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PROJECT REPORT

MCDA 5585 and MCDA 5586

Intelligent Automation and Reliability Engineering

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in partial fulfillment for the awarding of the degree:

Master of Science in

Computing and Data Analytics

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I extend my deepest gratitude to my manager and industry supervisor, **Mr. Kishor Deotale**, at Citco Technology Management (CTM). His mentorship gave me the opportunity to explore new technology stacks that had not previously been adopted for SRE at Citco — such as AWS-native tooling (CloudWatch, X-Ray, Application Signals) and Grafana as alternatives to Dynatrace — and to help lay the groundwork for a more modern observability approach. This involved assessing the benefits, limitations, and cost implications of each tool to shape a sustainable path forward: one that offers an “eagle-eye” view of the infrastructure, enabling both reactive detection of issues as they arise and proactive identification through warning and critical alerts.

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Finally, I would like to thank both **Saint Mary's University** and **Citco Technology Management (CTM)** for giving me this incredible opportunity to gain industry exposure and work on such a sophisticated real-world financial technology project.

CERTIFICATE

This is to certify that **Bhavik Kantilal Bhagat** (Student ID: **A00494758**), a student at Saint Mary's University in the Master of Science in Computing and Data Analytics program, has completed the projects in fulfillment of the course requirements for **MCDA 5585** and **MCDA 5586** on the project titled "**Intelligent Automation and Reliability Engineering for Financial Services**".

This project represents a total of **1020 hours** (900 internship hours + 120 academic hours) of technical and project work conducted from Mar 15 to Aug 31, 2026.

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1. ABOUT COMPANY & ORGANIZATION

1.1 The Citco Group of Companies

The Citco Group of Companies (Citco) is the world's largest independent hedge fund administrator, managing over \$1.8 trillion in assets under administration. Founded in 1948 as the *Curaçao International Trust Company* in Curaçao, Netherlands Antilles, the firm now operates from its headquarters in Tortola, British Virgin Islands. Over more than seven decades, Citco has grown from a civil law notary practice into a global financial services conglomerate with 50+ offices worldwide and approximately 16,000 employees across North America, Europe, Asia-Pacific, and the Caribbean.

Citco remains majority independently owned — a distinction that underpins its client-centric operating philosophy. CEO Christopher Smeets leads a management team that has overseen sustained organic growth, from \$250 billion in administered assets in 2005 to today's \$1.8 trillion benchmark. This growth reflects Citco's pioneering role in shaping the fund administration sector and its continuous investment in proprietary technology platforms.

1.1.1 Core Service Divisions

Citco's global operations are organized across several specialized divisions:

- **Citco Fund Services (CFS) & Investor Relations** — The flagship division and global leader in services for the alternative investment industry. CFS provides full-service fund administration, fund accounting, net asset value (NAV) calculations, investor relations, and reporting for over 2,000 funds. CFS has maintained SOC 1 Type 2 certification for 22 consecutive years and achieved its inaugural SOC 2 Type 2 certification in 2025, underscoring its commitment to operational controls and data security.
- **Citco Governance Services** — Provides corporate and global subsidiary governance, secretarial services, financial accounting and reporting, tax and audit liaison, entity incorporation, and market entry services across multiple jurisdictions.
- **Citco Banking** — Offers clients a comprehensive suite of banking services including investor accounts, operational and deposit bank accounts, foreign exchange, and credit services.
- **Citco Technology Management (CTM)** — The internal technology arm responsible for

developing and maintaining Citco's proprietary technology ecosystem, cloud infrastructure, data platforms, and enterprise applications that power fund administration and middle-office operations globally.

- **Middle Office Solutions** — Encompassing Treasury and Collateral Management, OTC settlements, Trade Matching, and Portfolio Finance. Citco's cloud-based *Æxeo*® Treasury platform provides centralized wire approval and fund movement visibility.

1.1.2 Industry Recognition

Citco consistently receives industry accolades, including the Hedgeweek European Award for Fund Administrator of the Year (Middle Office Services, 2026), Best Administrator over \$30 billion (APAC Hedge Fund Services Awards, 2026), and Fund Administrator of the Year for GPs over \$30 billion (Private Equity Wire US Awards, 2025). These awards reflect both depth of service and breadth of technological innovation.

1.2 Citco Technology Management — Halifax, Canada

My internship placement is within **Citco Technology Management (CTM)**, the technology division of the Citco Group. The Halifax, Nova Scotia office at 5151 George Street serves as a significant operational center for Citco Canada Inc., occupying multiple floors and functioning as a hub for technology development, fund administration services, and operational support for North American and global clients.

CTM designs, develops, deploys, and maintains Citco's internal technology platforms. This includes enterprise application development, cloud infrastructure management, DevOps and Site Reliability Engineering (SRE), data engineering pipelines, and intelligent automation initiatives. The Halifax CTM team collaborates with Citco colleagues across global offices to deliver technology solutions that underpin the firm's fund administration, investor relations, and governance services.

2. EXECUTIVE SUMMARY

This report documents my Mitacs Business Strategy Internship (BSI) at Citco Technology Management (CTM), Halifax, Nova Scotia, conducted between **Mar 15 and Aug 31, 2026** under the project title “*Intelligent Automation and Reliability Engineering for Financial Services*” (Mitacs reference IT49539).

Prior to my joining, the IA-IT-SRE team had no centralized observability platform for the IA-IT-SRE Infrastructure — a collection of six platforms (automationhub, bpm, citcoworks, ia-bpo, meridian, rpacfs1) spanning Lambda functions, ECS clusters, EC2 instances, MSK Kafka pipelines, and a range of database and messaging AWS services. Two documented production failures — a 29-day silent outage of the CAIS-Pricing GenAI pipeline and a 7.5-hour Meridian event processing delay — both went undetected until business teams escalated them, exposing the team’s reliance on reactive, manual incident detection. The research question framed for this engagement was: *can AWS-native services achieve Dynatrace-equivalent observability at a fraction of the cost?*

My approach was incremental across three phases. I first built CloudWatch dashboards targeting Lambda functions — the AWS service at the center of the silent outage — and documented each function’s trigger type and runtime across all six platforms. I then developed a registry-based IA Infrastructure inventory API that queries the AWS API dynamically and returns live, per-platform lists of all deployed AWS services. Finally, I built a centralized Amazon Managed Grafana (AMG) workspace as a proof-of-concept, covering all 13 AWS service categories across all six platforms via dynamic platform template variables. The RPA production support rotations ran in parallel throughout the co-op placement.

The key outcomes of this work are:

- Kafka consumer lag detection reduced from a 3.5-hour business-team escalation window to **under 30 seconds** via a CloudWatch MSK EstimatedMaxTimeLag alarm, directly addressing the Meridian Incident failure mode
- Dynamic CloudWatch alarms deployed across all monitored AWS services on all six platforms, where none or limited existed before
- Grafana AMG workspace with **multiple rows** covering every AWS service category in the IA-IT-SRE Infrastructure

- AWS service inventory across all six platforms, with **99% automated test coverage** on the inventory API
- AWS-native stack achieves **~60% Dynatrace feature parity** at \$77/month (CloudWatch + Grafana AMG + X-Ray), rising to **~80%** with Application Signals

The chapters that follow detail the tools and technologies used, the full project context and JIRA scope, the SRE methodology and requirements, the phased technical implementation, and the broader learning outcomes of the engagement.

3. TEAM COMPOSITION & RESPONSIBILITIES

My placement was within the **Intelligent Automation & Site Reliability Engineering (IA-SRE)** team at Citco Technology Management (CTM), Halifax. This chapter describes the team I worked with and the responsibilities I held during the co-op placement.

3.1 Team Structure

The IA-SRE team is co-located across Halifax (Canada) and Hyderabad (India), with stakeholders spanning Dublin and London. My direct supervisor was Mr. Kishor Deotale (Manager, CTM). The co-op period team comprised the following members:

- **Kishor Deotale** — Manager, CTM and Industry Supervisor. Kishor is responsible for all work streams, sets engineering priorities, and served as the final acceptance authority for all deliverables.
- **Ho Hoi Leung (“Horace”)** — IT Senior Developer, RPA Platform Lead. Horace is the primary knowledge holder for Blue Prism and UiPath platform operations, responsible for escalation triage in the Follow the Sun support model and the designated Halifax escalation point for production incidents.
- **Kristia Marie Labos (“Kri”)** — IT Developer, SRE Engineer. Kri leads the SRE practice for the team, including the four golden signals framework. She guided the observability platform design and validated alarm threshold selection throughout the dashboard build.
- **Sridhar Vadla and Soundarya Venkataraman** — Co-interns (SMU MCDA, Mitacs BSI IT49539). Both joined the Halifax team concurrently, each working on complementary workstreams of the IA monitoring and automation platform, enabling parallel progress across the observability roadmap.

3.2 My Role and Responsibilities

As a participant in the Mitacs Business Strategy Internship (BSI) program, I was embedded within the IA-SRE team at CTM Halifax. I collaborated closely with Kishor, Horace, Kri, co-interns Sridhar and Soundarya, and stakeholders across Halifax, Hyderabad, and Dublin. Weekly standups, sprint reviews, and direct Confluence and JIRA-based collaboration formed the day-to-day working

rhythm.

My responsibilities spanned the following areas:

- **IA Infrastructure Monitoring:** Developing per-platform CloudWatch dashboards for Lambda, ECS, EC2, SQS, EKS, and related AWS services across all six platforms, establishing the architectural patterns for dynamic resource discovery.
- **SRE Observability Platform Development:** Designing and building the centralized Amazon Managed Grafana (AMG) dashboard covering all AWS services across the six IA-IT-SRE Infrastructure platforms, including CloudFormation alarm provisioning and Infrastructure-as-Code delivery.
- **Inventory API Development:** Building the `ia-it-sre-infrastructure-inventory` Flask REST API for programmatic discovery of all deployed AWS services across the IA-IT-SRE Infrastructure, with a discoverer registry pattern that provides a live, per-platform inventory of all AWS services.
- **Cloud Infrastructure Engineering:** Authoring parameterized CloudFormation and AWS SAM templates, managing IAM roles for cross-account Grafana access, enabling CloudWatch Container Insights on ECS clusters, and navigating AWS service constraints encountered during delivery.
- **Scrum and Meeting Facilitation:** Leading daily standup calls on a rotation basis. Attending dashboard design reviews, knowledge-transfer sessions (RPA platforms, CitcoWorks, Event Store, Aexeo), security and compliance reviews, and stakeholder syncs with the wider CTM team.
- **JIRA Story Creation and Management:** Translating verbal requirements into structured JIRA stories and tasks with clear acceptance criteria and subtask decomposition. Raising defect tickets for issues identified through root cause analysis of production incidents.
- **RPA Production Support:** Providing operational support for Blue Prism and UiPath automation processes during dedicated support rotation weeks (11 AM–7 PM EST shift, once every four weeks). Responsibilities included monitoring bot queues, triaging service desk tickets, onboarding new clients for Corporate Actions workflows, retriggering failed processes, and escalating where required.

- **Incident Response and Hotfix Engineering:** Investigating and resolving production incidents including the CAIS Deadline credential failure, MESO month-end stuck tasks, and VPL pricing failures through root cause analysis and targeted code or configuration fixes.
- **AI-Augmented Engineering:** Leveraging the Kiro IDE, Amazon Q, and Claude Sonnet for spec-driven development, automated code scaffolding, CloudFormation template generation, and test suite creation.
- **Technical Documentation:** Maintaining Confluence knowledge-base pages for Lambda function inventories, runbooks, incident post-mortems, and architecture decisions.

These responsibilities required a blend of cloud engineering expertise, operational discipline, and cross-functional communication skills to deliver production-grade monitoring infrastructure while maintaining the reliability of live financial automation processes.

4. TOOLS & TECHNOLOGIES

During my co-op placement, the following tools, AWS services, and platforms were used to build the IA-IT-SRE Infrastructure monitoring deliverables, develop the inventory API, and support RPA production operations.

4.1 AWS Cloud Services

- **Amazon CloudWatch** — AWS's primary observability service, used for collecting metrics, logs, and alarms across all monitored AWS services. Metrics Insights SQL and Logs Insights were used for dashboard queries and log analysis.
- **AWS CloudFormation / SAM** — Infrastructure-as-Code framework used to provision CloudWatch alarms, Grafana workspaces, and Lambda functions via parameterized YAML templates.
- **AWS Lambda** — Serverless compute service hosting the automation workloads across all six platforms, and used as the runtime for the CloudFormation alarm provisioner and the resource inventory API.
- **Amazon ECS (Elastic Container Service)** — Container orchestration service used to run the Meridian platform services. CloudWatch Container Insights was enabled to surface task-level CPU and memory telemetry.
- **Amazon MSK (Managed Streaming for Apache Kafka)** — Managed Kafka service used as the Meridian event pipeline backbone. Consumer lag metrics from MSK were the primary observable for the Meridian Incident post-mortem monitoring gap.
- **Databricks** — Unified data analytics platform used within the Meridian event pipeline for ingesting and processing automated system events from Citco Recs. Events are received and processed in Databricks before being consumed downstream by the OpenSearch layer.
- **Amazon SQS (Simple Queue Service)** — Message queue service used as an inter-service trigger mechanism for Lambda functions. Queue depth and oldest message age were monitored as leading indicators of processing backlogs.
- **Amazon EC2** — Virtual machine service hosting RPA bot nodes and supporting infrastructure

across the platforms.

- **Amazon EKS (Elastic Kubernetes Service)** — Managed Kubernetes service used by the AutomationHub platform for container workloads.
- **Amazon ECR (Elastic Container Registry)** — Container image registry used to store and version Docker images for ECS and EKS deployments.
- **Amazon OpenSearch Service** — Search and analytics engine used by the Meridian platform as the event indexing layer queried by downstream services.
- **Amazon RDS Aurora** — Relational database service used by Meridian and other platforms for structured data storage.
- **Amazon DocumentDB** — MongoDB-compatible document database used by Meridian for document-oriented workloads.
- **Amazon Neptune** — Graph database used by the CitcoWorks platform for the Knowledge Graph hierarchy.
- **Amazon ElastiCache (Redis/Valkey)** — In-memory caching service used by Meridian's ODL layer for fast event state access.
- **Amazon S3** — Object storage used for CloudFormation template staging, Lambda deployment packages, and automation output artifacts.
- **AWS Secrets Manager** — Credential management service used to store and rotate API keys, database passwords, and service account tokens across all platforms.
- **AWS IAM** — Identity and access management service used to define least-privilege roles for the Grafana data source, Lambda provisioner, and inventory API.
- **AWS CodeCommit** — Git-based source control service used to host all automation and infrastructure repositories.

4.2 Observability & Monitoring

- **Amazon Managed Grafana (AMG)** — Fully managed Grafana service used to build the centralized IA-IT-SRE Infrastructure dashboard with multiple rows covering all monitored

AWS services across all six platforms.

- **CloudWatch Container Insights** — Managed telemetry extension for ECS and EKS that surfaces task-level CPU, memory, and network metrics, enabled on the Meridian clusters.
- **AWS X-Ray** — Distributed tracing service researched and partially evaluated for Lambda and ECS instrumentation; full deployment scoped as future work.
- **AWS Application Signals** — AWS-native SLO management and service health monitoring service evaluated as part of the Dynatrace parity analysis; identified as the path from 60% to ~80% AWS-native feature parity.
- **Dynatrace** — Enterprise APM platform already in use by the team for Blue Prism automation observability, used as the functional benchmark for the AWS-native observability PoC.

4.3 Development Tools

- **Python 3** — Primary programming language used for all automation scripts, the inventory API, and CloudFormation provisioner logic.
- **boto3** — AWS SDK for Python, used to interact with all AWS services programmatically in the inventory API and alarm provisioner.
- **Flask (async)** — Lightweight Python web framework used to build the `ia-it-sre-infrastructure-inventory` REST API with async route handlers.
- **FastAPI** — Python API framework used in the initial inventory API prototype before being migrated to Flask.
- **Pydantic** — Data validation library used for type-safe response models in the inventory API.
- **pytest + Hypothesis** — Testing frameworks used to build the automated test suite for the inventory API, achieving 99% code coverage with unit, integration, and property-based tests.
- **TypeScript / ReactJS / MUI** — Frontend technology stack used for the inventory dashboard UI, with ReactJS for component-based UI, MUI (Material UI) for design-system components, and TypeScript for type-safe frontend development.
- **Chart.js / Recharts** — JavaScript charting libraries proposed for the custom UI dashboard.

Chart.js for real-time streaming metrics with 30-second refresh intervals (canvas-based for smooth rendering); Recharts for static comparisons, bar charts, and gauges.

- **Cytoscape.js** — Graph visualization library proposed for X-Ray service map rendering in the custom UI dashboard, replacing Grafana's crash-prone nodeGraph plugin with a stable custom graph visualization implementation.
- **D3.js (hexbin)** — Data visualization library proposed for Dynatrace-style honeycomb health grids in the custom UI dashboard planning document.
- **Postman** — API testing tool used to validate and document the inventory REST API endpoints during development.
- **Apache JMeter** — Load testing tool used to validate API performance and throughput for the inventory REST API under simulated concurrent request loads.
- **DBeaver** — Universal database client used to inspect and query RDS, DocumentDB, and other database services across the platforms during incident investigation and development.
- **PowerShell / Shell Scripts** — Scripting environments used to automate deployment tasks, run AWS CLI commands, and execute operational scripts on Windows and Linux hosts.
- **Visual Studio Code (VS Code)** — Primary code editor used for Python, TypeScript, and YAML development, extended with the Kiro IDE plugin for AI-assisted engineering.
- **JIRA** — Project tracking tool used for sprint planning, story decomposition, and time logging across all co-op deliverables.
- **Confluence** — Internal wiki used for architecture documentation, Lambda function inventory pages, and runbooks.
- **Git / AWS CodeCommit / Bitbucket** — Version control system with two hosted repository services: AWS CodeCommit for Python and IaC repositories, and Bitbucket for UiPath automation repositories.
- **Remote Desktop Connection (RDC)** — Used to connect to remote Windows machines hosting RPA bot nodes and on-premises infrastructure for incident investigation and configuration changes.

4.4 AI-Native Development Tools

- **Kiro IDE** — AI-powered development environment built on VS Code, used for spec-driven development (`requirements.md` → `design.md` → `tasks.md`) and agentic code generation for the inventory API.
- **Amazon Q** — AWS integrated code assistant used for inline boto3 completions and project-level context retrieval.
- **Claude Sonnet & Opus (Anthropic)** — Large language model accessed via Kiro, used for architecture reasoning, CloudFormation template drafting, and test strategy design.

4.5 RPA Platforms

- **Blue Prism** — Legacy RPA platform hosting daily, monthly, and quarterly fund operations automation processes, monitored during production support rotations.
- **UiPath Cloud / On-Premises** — Target RPA platform for new automation development and Blue Prism migrations, operated via UiPath Orchestrator with SQS-triggered job invocations.

4.6 Communication & Collaboration

- **Microsoft Teams** — Primary communication platform used for daily standups, sprint planning meetings, technical discussions with the SRE team, and screensharing during incident investigation and pair debugging sessions.
- **Microsoft Outlook** — Email client used for formal communications, meeting scheduling, calendar management, and receiving production alert notifications from CloudWatch SNS topics.

5. PROJECT OVERVIEW

5.1 Project Background and the Motivating Incident

5.1.1 Pre-Co-op Placement Monitoring Posture

Before my co-op placement began in mid-Mar 2026, the IA-SRE team operated without a centralized observability stack. Monitoring relied entirely on an *ad hoc* model: business teams in Dublin, London, and Halifax would notice anomalies in fund processing outputs and raise tickets. There were no CloudWatch alarms, no dashboards aggregating health across the six platforms, and no mechanism for the SRE team to receive proactive notification of degrading conditions — the team was blind to *saturation* and *error rate* signals until they had already caused downstream business impact.

5.1.2 CAIS-Pricing Silent Outage (Jan–Feb 2026)

Between **Jan 20 and Feb 17, 2026**, the CAIS-Pricing Extract GenAI pipeline under the `rpacfs1` platform stopped delivering output for **29 consecutive days** without a single alert firing. The root cause was gradual exhaustion of the external CTI AI model API token, which controls the async SQS callback that triggers the downstream Lambda stages; without that callback, three downstream functions dropped to zero invocations while two upstream functions kept running, making the pipeline appear healthy. The outage was detected only when business users reported missing pricing data — 29 days after onset. The post-mortem identified the absence of throughput alarms, SQS queue depth monitoring, and end-to-end output verification as the root causes of the failure going unnoticed. A sub-task was assigned directly to me as the immediate remediation and became the starting point for the full monitoring programme.

5.1.3 The Meridian Incident (Apr 2026)

The Meridian platform processes daily statements, reconciliations, and event notifications across fund administration operations. A surge of automated system events from Citco Recs overwhelmed Meridian's OpenSearch consumption layer, causing Kafka consumer lag to reach approximately **9 million messages** and halting daily statement processing for 7.5 hours. The SRE team received no alert; the Dublin business team escalated over three and a half hours after the first failure signal. The post-mortem recorded: *"No proactive monitoring/alerting in place on the*

SRE side for Meridian event volume anomalies.” This became the primary driver for the centralized dashboard project.

5.1.4 From Incident to Programme

Both incidents shared the same root cause: no proactive monitoring. The post-mortems established two specific gaps — no Kafka consumer lag alarm, and no centralized visibility across the IA-IT-SRE Infrastructure — that directly motivated the six work streams described in the following section. Kishor elevated the dashboard project to **Critical** priority on May 6, 2026, and my co-op placement pivoted from exploratory dashboard research to a full-scale observability platform build.

5.2 Work Streams Overview

The two motivating incidents established a clear engineering mandate: build proactive observability across the entire IA-IT-SRE Infrastructure. Six work streams were pursued, each directly motivated by the findings and limitations of the previous. Full technical scope and acceptance criteria for each deliverable are documented in Chapter 6.

5.2.1 Work Stream 1: CloudWatch Dashboards and IA Infrastructure Monitoring

Following the CAIS-Pricing Silent Outage, the objective was to build unified CloudWatch dashboards starting with Lambda — the AWS service at the center of that incident. The scope extended incrementally to SQS queues, EC2 instances, ECS clusters, EKS nodes, and database services across all six platforms, establishing the tag-based resource discovery patterns that all later work built upon.

A fundamental limitation became apparent: CloudWatch dashboards are siloed per AWS service and do not support cross-service filtering across platforms. This directly motivated Work Stream 2. *Key deliverables: IISS-435, IISS-455, IISS-461, IISS-469, IISS-482–486.*

5.2.2 Work Stream 2: Grafana AMG Centralized Dashboard

Following the Meridian Incident, the project was elevated to **Critical** priority. The objective was to build a centralized observability platform using Amazon Managed Grafana (AMG) — addressing what CloudWatch dashboards could not: a unified, cross-platform view of the entire IA-IT-SRE

Infrastructure in a single workspace. The broader research objective was to evaluate whether this AWS-native stack could achieve Dynatrace-equivalent observability. *Key deliverable: IISS-487.*

5.2.3 Work Stream 3: IA Infrastructure Inventory API

A critical limitation surfaced during the Grafana build: CloudWatch Metric Insights SQL does not support pattern-based filtering on dimension values. To solve this, the Lambda discovery scripts were extended into a full Flask REST API to discover AWS services dynamically across all platforms, enabling the Grafana dashboard variables to be populated from live infrastructure rather than static configuration. *Key deliverable: IISS-507.*

5.2.4 Work Stream 4: RPA Production Support

In parallel with the engineering work streams, dedicated RPA production support rotations were carried out as part of the team's Follow the Sun model. The North America slot (11 AM–7 PM EST) was covered once every four weeks, serving as the primary contact for RPA incidents escalated through the SD Portal, Innovation IT email inbox, IA_INCIDENTS distribution list, and Teams channels.

Each shift began with a review of the Confluence handover log from the Manila shift and a queue scan to identify open items or approaching SLA breaches. Incidents were triaged, investigated, and either resolved or escalated to the development team with documented findings. *Key incident tickets: IISS-706, IISS-538, IISS-741, IISS-876.*

5.2.5 Work Stream 5: RPA LLM Agent — Ask Citco RPA

This work stream involved a production quality and reliability improvement programme for the **Ask Citco RPA** chatbot — a RAG pipeline using Amazon Bedrock, OpenSearch Serverless, and 11 Lambda functions. The scope covered production bug resolution, code quality improvements, structured logging migration, and X-Ray distributed tracing instrumentation. *Key deliverable: IISS-824.*

5.2.6 Work Stream 6: Citco Enterprise Custom UI Dashboards

While the Grafana AMG dashboard addressed the primary observability gap, it remained constrained by fixed panel types and no drag-and-drop layout. This work stream focused on

designing and building a Citco-owned enterprise monitoring UI by extending the existing ia-config-portal application (built by Soundarya and Sridhar) with two new dashboard pages. *Key deliverables: IISS-509, IISS-919. Status: In progress; implementation continuing in Sep–Dec 2026 extension.*

6. PROJECT GOALS & DELIVERABLES

I set the following engineering objectives at the outset of the co-op placement, guided by the two motivating incidents and the requirements sessions with Kishor and Kri. Each objective maps directly to one of the six work streams described in Chapter 5.

6.1 Project Goals

The primary goals of the co-op placement project were as follows:

- **Eliminate reactive-only incident detection.** Replace the dependency on business-team escalation as the primary detection mechanism with proactive CloudWatch alarms that fire within seconds of a threshold being crossed, directly addressing the failure modes exposed by the CAIS-Pricing Silent Outage and the Meridian Incident.
- **Build a centralized observability platform for the IA-IT-SRE Infrastructure.** Provide a single-pane-of-glass view across all six platforms and all 13 AWS service categories, enabling the SRE team to assess platform health, drill down to individual AWS services, and respond to incidents without navigating six separate CloudWatch consoles.
- **Automate AWS service discovery and inventory.** Replace manual Confluence-maintained Lambda lists with a programmatic, always-current inventory API that enumerates all deployed AWS services per platform and exposes them via REST endpoints for use by the dashboard and other downstream tooling.
- **Evaluate AWS-native observability as a Dynatrace alternative.** Conduct a structured feature-parity analysis comparing the AWS-native stack (CloudWatch, Grafana AMG, X-Ray, Application Signals) against Dynatrace capabilities, quantifying the achievable parity percentage and the incremental monthly cost at each tier.
- **Deliver Infrastructure-as-Code for all monitoring components.** Ensure all alarms, dashboards, IAM roles, and supporting infrastructure are provisioned via parameterized CloudFormation templates and AWS SAM, making the platform reproducible, version-controlled, and environment-portable.
- **Establish engineering quality standards.** Produce a comprehensive test suite for the inventory API, document all architectural decisions and limitations, and maintain a

Confluence knowledge base for the Lambda inventory, runbooks, and incident post-mortems.

6.2 Deliverables and Technical Scope

This section documents the technical scope and stack for each primary deliverable. Detailed implementation decisions are covered in Chapter 10.

Table 6.1: Project Deliverables — JIRA Scope

JIRA	Title / Scope
IISS-435, 455, 461, 469	CloudWatch Lambda Monitoring Dashboard and inventory
IISS-482–486	IA Infrastructure Monitoring Platform (CloudWatch dashboards extended to all platforms)
IISS-487	Centralized Grafana AMG SRE Dashboard (primary deliverable)
IISS-507–510	IA Infrastructure Inventory API (Flask REST API)
IISS-707, 538, 741, 753, 876	RPA Production Support Incidents
IISS-509, IISS-919	Citco Enterprise Custom UI Dashboards (Infrastructure Inventory + Monitoring)
IISS-824	RPA LLM Agent — Ask Citco RPA (bug fixes, KB sync, quality improvements)

6.2.1 IA Infrastructure Monitoring Platform

Current State:

No CloudWatch dashboards existed for any of the six IA platforms. Lambda function health was monitored only through the AWS Console on an ad-hoc basis. No tag-based resource discovery was in place and the Lambda inventory was maintained manually on Confluence.

Scope:

Build unified CloudWatch dashboards covering all six platforms — automationhub, bpm, citcoworks, ia-bpo, meridian, and rpacfs1 — across Lambda, EC2, ECS, EKS, SQS, and supporting AWS services. Implement a tag-based resource discovery pipeline using BudgetCode as the primary filter and AppName as the mandatory fallback discriminator for the meridian/citcoworks shared BudgetCode ambiguity.

Deliverables:

CloudWatch dashboards for all six platforms; a resilient three-tier SQS registry pipeline (live API → S3 cache → bundled inventory); a Python-based dashboard builder framework deployed via CloudFormation.

6.2.2 Centralized Grafana SRE Dashboard**Current State:**

No centralized observability workspace existed. Platform health required navigating six separate CloudWatch consoles. No Kafka consumer lag alarm existed for the Meridian platform, which was the direct cause of the April 2026 incident going undetected for over three hours.

Scope:

Build a centralized Amazon Managed Grafana (AMG) workspace providing a unified cross-platform view across all six platforms and all monitored AWS service categories. Implement proactive CloudWatch alarms across all services, an SNS notification pipeline to the SRE team, and Grafana template variables backed by the Inventory API for dynamic resource filtering.

Deliverables:

Grafana AMG workspace with 13+ dashboard rows covering all service categories; CloudWatch alarms deployed across all six platforms; SNS alarm notification pipeline; Infrastructure-as-Code (AWS SAM / CloudFormation) for reproducible deployment.

6.2.3 IA Infrastructure Inventory API**Current State:**

No programmatic AWS resource inventory existed. Lambda function lists were manually maintained on Confluence and became stale quickly. CloudWatch Metric Insights SQL lacked support for pattern-based dimension filtering, making dynamic Grafana template variable population impractical.

Scope:

Build a Flask REST API that dynamically discovers all deployed AWS services per platform across all six IA platforms. Implement a discoverer registry pattern for extensibility, a two-stage tag filter to resolve the meridian/citcoworks BudgetCode ambiguity, and an S3-backed cache layer to keep enumeration latency within the API Gateway 30-second timeout.

Deliverables:

REST endpoints for Lambda, ECS, EC2, SQS, S3, and health resources; per-platform filtering support; S3 cache with CloudWatch metrics publishing; Lambda deployment via API Gateway.

6.2.4 RPA Production Support**Current State:**

The IA-SRE team operates a Follow the Sun support model with shifts across Manila (APAC), Dublin (EMEA), and Halifax (NA). The NA slot required a dedicated engineer covering 11 AM–7 PM EST, one week in every four.

Scope:

Cover the North America RPA production support rotation, handling incidents escalated through the SD Portal, IA_INCIDENTS distribution list, and Teams channels. Triage, investigate, and resolve or escalate RPA process failures within the defined SLA, and maintain daily Confluence shift handover logs.

Deliverables:

Five completed support rotations (W8, W12, W16, W20, W24); incident triage and resolution for all escalated tickets; Confluence handover logs maintained for each shift day.

6.2.5 RPA LLM Agent — Ask Citco RPA**Current State:**

The **Ask Citco RPA** chatbot was a production RAG pipeline using Amazon Bedrock (Claude), OpenSearch Serverless (1024-dim FAISS/HNSW vectors), and 11 Lambda functions ingesting from SharePoint, Databricks, and the BPM API. The system had multiple known reliability issues including a recurring KB sync timeout and several production bugs identified via CloudWatch log analysis.

Scope:

Resolve production reliability issues, enrich the Knowledge Base, and improve code quality across all Lambda functions per the IA Best Practice checklist. Evaluate and propose a Hybrid Graph-RAG architecture extension using Amazon Neptune.

Deliverables:

Production bug fixes deployed; NavChecklist PDFs added to Knowledge Base; modular code

structure with structured logging, Pydantic validation, tenacity retry, kill switch, and X-Ray tracing; comprehensive unit test suite; architecture proposal for Graph-RAG extension.

6.2.6 Citco Enterprise Custom UI Dashboards

Current State:

The Grafana AMG dashboard provided approximately 60% Dynatrace feature parity but was constrained by fixed panel types, no drag-and-drop layout, and limited custom visualization options. A Citco-owned enterprise UI was identified as the next step to close the remaining observability gap.

Scope:

Extend the existing `ia-config-portal` (React 19/TypeScript, ECS Fargate, built by Soundarya and Sridhar) with two new dashboard pages — an Infrastructure Inventory dashboard (IISS-509) and an Infrastructure Monitoring dashboard (IISS-919) — using a config-driven panel schema and CloudWatch data integration.

Deliverables:

Architectural implementation plan covering gap analysis, technology stack decisions, API design, and task breakdown; initial frontend scaffolding with AG Grid resource tables and config-driven panel components; continuing in Sep–Dec 2026 extension.

6.3 Acceptance Criteria

The following criteria defined whether each deliverable met its stated objectives.

6.3.1 IA Infrastructure Monitoring Platform

- **Platform coverage:** CloudWatch dashboards exist for all six platforms covering the key AWS services (Lambda, ECS, EC2, SQS, EKS) with tag-based resource filtering.
- **Resilient SQS monitoring:** The SQS registry pipeline remains functional during AWS service degradation via the three-tier fallback model.

6.3.2 SRE Observability Platform

- **Proactive incident detection:** The team can be alerted to Kafka consumer lag, Lambda error rate spikes, ECS CPU saturation, and SQS queue backlogs before business teams escalate — directly addressing the failure modes exposed by the CAIS-Pricing Silent Outage and the Meridian Incident.
- **Cross-platform visibility:** The SRE team can assess the health of all six platforms and all monitored AWS services from a single Grafana workspace without navigating individual CloudWatch consoles.
- **Alarm coverage:** CloudWatch alarms are deployed and active across all monitored AWS services on all six platforms, with SNS notifications routed to the SRE team.
- **Container-level telemetry:** Task-level CPU and memory metrics are available for the Meridian ECS clusters, enabling resource saturation to be detected before service degradation occurs.
- **Reproducible infrastructure:** All monitoring infrastructure — Grafana workspace, alarms, IAM roles, S3 storage — is provisioned as Infrastructure-as-Code and can be redeployed consistently across environments.

6.3.3 IA Infrastructure Inventory API

- **Dynamic discovery:** The API returns a current, accurate list of deployed AWS services per platform at query time, enabling Grafana template variables to reflect the live fleet without manual updates.
- **Tag disambiguation:** Resources from the meridian and citcoworks platforms are correctly separated despite sharing the same BudgetCode tag value.
- **Extensibility:** New platforms or AWS service types can be onboarded by implementing a single discoverer class with no changes to the API routing or test infrastructure.

6.3.4 RPA Production Support

- **Incident response:** Production incidents are triaged, root-caused, and either resolved within the support shift or escalated with a documented RCA and proposed fix.

- **SLA compliance:** Tickets are acknowledged within the L2 response SLA and carried over with complete handover notes per the IA Support Model.

6.3.5 RPA LLM Agent

- **KB sync reliability:** The IRCOEKBExtractLambda completes within the 300-second timeout on every daily run, with steady-state duration under 60 seconds for unchanged content.
- **Production stability:** All identified bugs (OpenSearch silent delete, fire-and-forget pattern, Athena race condition, SSL verification, encryption inconsistency) are resolved and deployed.
- **KB content:** NavChecklist documents are indexed and retrievable via the chatbot.

6.3.6 Citco Enterprise Custom UI Dashboards

- **Architecture plan:** A comprehensive implementation plan covering gap analysis, technology decisions, API design, and task breakdown is produced before implementation begins.
- **Backend API design:** New Lambda handlers designed with request/response schemas and IAM permissions documented.
- **Component specifications:** SRE-specific UI components (honeycomb, service map, trace waterfall) specified with library selections and integration patterns.

7. LEARNING GOALS

The following learning goals were established at the start of my co-op placement in consultation with my academic and industry supervisors. They are organized into technical and non-technical categories to reflect the dual competency development expected of an MCDA industry placement.

7.1 Technical Learning Goals

- **AWS Cloud Services and Infrastructure.** Gain production-level proficiency with the AWS services central to the IA-IT-SRE Infrastructure. During the part-time onboarding phase, I focused on understanding the full AWS service landscape in use at Citco: Amazon CloudWatch (metrics, alarms, Logs Insights, Metric Insights SQL), AWS Lambda (serverless execution, cold-start behavior, concurrency), Amazon ECS and EKS (containerized workloads), Amazon EC2 (virtual machine infrastructure), Amazon MSK (managed Kafka, consumer lag), Amazon SQS (queue-based messaging), Amazon SNS (alerting), Amazon S3, AWS CloudFormation and SAM (Infrastructure-as-Code), AWS CodePipeline (CI/CD), and AWS Secrets Manager (credential management). A deliberate goal was exposure to the full spectrum of database categories deployed at Citco: Amazon DynamoDB (key-value), Amazon DocumentDB (document), Amazon Neptune (graph database — CitcoWorks Knowledge Graph), Amazon RDS Aurora (relational), Amazon ElastiCache for Redis and Valkey (in-memory caching — Meridian ODL tier), and Amazon OpenSearch Service (search and analytics). Additional infrastructure covered includes Amazon ECR, AWS CodeCommit, and the AWS IAM permission model (including cross-account role chaining for Grafana data source access).
- **Observability, Monitoring, and the Four Golden Signals.** Develop a working understanding of modern observability principles and apply them to a production financial services platform. This goal was developed through discussions with Kri (SRE mentor) on the four golden signals framework — latency, traffic, errors, and saturation — and their application to the IA platform AWS service fleet. Full details are covered in Chapter 9.

Beyond the four golden signals, learning goals included understanding key observability metrics for each AWS service on the dashboard — MSK consumer lag, SQS queue depth and oldest message age, ECS task CPU and memory, Lambda duration and concurrent executions, EC2 CPU and network, OpenSearch JVM heap pressure, and Redis memory

utilization — as well as evaluating advanced observability tooling including AWS X-Ray (distributed tracing), AWS Application Signals (SLO management), Grafana Trace Maps, and improvement opportunities for the existing IA ML/LLM RPA Support Agent.

- **Infrastructure-as-Code and CI/CD.** Author parameterized CloudFormation and AWS SAM templates for alarm and infrastructure deployment; develop a Lambda-backed dynamic alarm provisioner; and understand the AWS CodePipeline CI/CD model used for automation deployments across the platforms.
- **Grafana AMG Platform.** Build and operate a production-grade Amazon Managed Grafana workspace: configure CloudWatch as a data source, design collapsible dashboard rows for each AWS service, implement platform template variables for dynamic filtering, and learn Grafana panel types (time series, stat, bar gauge, heatmap, top-K list, table with deep-link URLs).
- **Backend API Development.** Write production-quality Python APIs using Flask and FastAPI with async route handlers, boto3 for AWS service interaction, Pydantic for response validation, and pytest with Hypothesis for property-based testing.
- **Software Engineering Practices.** Practice spec-driven development via the Kiro IDE workflow (.kiro/specs/ with requirements.md, design.md, tasks.md), and apply AI-augmented engineering using Kiro (Claude Sonnet) and Amazon Q to accelerate design documentation, code scaffolding, and test generation.
- **Proactive vs. Reactive Reliability Engineering.** Transition from a reactive support model (incident detection by business escalation) to a proactive one (alarm-backed detection within 30 seconds) by applying SLI/SLO targets and error budget concepts to real production workloads.

7.2 Non-Technical Learning Goals

- **Scrum and Engineering Leadership.** Lead daily standup calls on rotation, facilitate team coordination, translate verbal requirements from Kishor into structured JIRA stories with clear acceptance criteria and subtask decomposition, and develop task estimation and prioritisation skills in a production engineering environment.
- **Cross-Functional and Cross-Timezone Collaboration.** Build working relationships across

Halifax IA-SRE, Innovation IT Hyderabad (IST), Dublin and London business operations teams, and the RPA support tier. Participate in structured knowledge-transfer sessions, design reviews, security and compliance meetings, and stakeholder syncs across time zones.

- **Production Incident Response and RCA.** Develop the discipline of structured root cause analysis: follow the IA Support Model escalation tiers, document findings in SD tickets and Confluence, communicate clearly with business and platform teams under time pressure, and propose fixes or raise JIRA defects with actionable resolution steps.
- **Technical Communication and Documentation.** Produce engineering-grade written deliverables: incident RCA reports, architecture design documents, Confluence knowledge-base pages, runbooks, and this academic report.
- **Financial Domain Knowledge.** Develop working familiarity with fund administration operations — VPL queue (5 PM EST daily deadline), Corporate Actions workflows (Mandatory, Voluntary, Ops Transfer), MESO month-end cycles, and CAIS regulatory filing deadlines — to make informed monitoring and incident response decisions.

8. REQUIREMENTS ELICITATION

During my co-op placement I worked with a small set of internal stakeholders whose needs shaped the functional and non-functional requirements of the IA-IT-SRE Infrastructure monitoring platform.

8.1 Stakeholder Identification

Table 8.1 enumerates the key stakeholders, their roles, and their specific interests in the project.

Table 8.1: Stakeholder Identification

Stakeholder	Role	Interest / Influence
Kishor	Manager, CTM; Industry Supervisor	Primary sponsor; set engineering priorities; escalated IISS-487 to Critical; final acceptance authority
Horace	IT Senior Developer, RPA Platform Lead	RPA subject matter expert; delivered Blue Prism and UiPath KT; defined RPA monitoring scope
Kri	IT Developer, SRE Engineer	Delivered golden signals and alarm threshold training; reviewed dashboard design
IA IT SRE Team	Internal consumers	Primary users of the Grafana dashboard and CloudWatch alarm notifications
Meridian Team	Kafka/ECS architecture owners	Provided MSK consumer group structure, ECS cluster topology, and OpenSearch integration context
Sridhar & Soundarya	Co-interns	Parallel development of related monitoring features; design review and test validation

8.2 Requirements Elicitation

The elicitation methods used were appropriate for an in-house engineering context: structured knowledge-transfer sessions, incident post-mortems, and JIRA story refinement, rather than formal user interviews or focus groups.

8.2.1 Elicitation Methods

Knowledge-Transfer Sessions (Training Week, Apr 13–18, 2026). The first week of full-time engagement was structured as an intensive knowledge-transfer programme covering the full breadth of the IA platform. Sessions with Horace on the Blue Prism and UiPath architectures, with Bhanu Teja Murari on CitcoWorks and the Event Store platform, and with the Meridian team on the Kafka event pipeline (comprising the Event Routing Service, Action Processor, and OpenSearch consumer) provided the architectural grounding needed to define meaningful SLIs. Key requirements insight from training week: the Æxeo Treasury platform (cloud-based wire approval and fund movement) and the MESO platform (high-memory autoscale issue encountered later) each had distinct resource profiles requiring separate monitoring strategies.

Incident Post-Mortem (Apr 17, 2026). The Meridian production incident yielded the single most important requirements insight of the entire engagement: MSK consumer lag is the primary leading indicator of Meridian health. The post-mortem made concrete the latency between signal onset and detection under the reactive model, and directly specified the target for the lag alarm evaluation period (NFR2: ≤ 1 minute).

Golden Signals Session (May 25, 2026). Kri's dedicated SRE methodology session translated the theoretical four golden signals into concrete metric choices for the IA platform. This session produced the initial SLI/SLO target table and established the design constraint that all metrics must be queryable through Amazon CloudWatch — ruling out external time-series databases such as Prometheus and anchoring the architecture to CloudWatch Metric Insights as the sole data source.

JIRA Story Refinement. The dashboard project went through multiple refinement cycles between May 6 and May 11 (infrastructure deployment date). These sessions with Kishor and Kri surfaced the infrastructure-as-code requirement (FR7), the multi-environment reuse requirement (NFR5), and the credential management constraint (NFR7). The elevation of the dashboard to Critical priority on May 6 also functionally set NFR1 (dashboard refresh latency ≤ 30 seconds) as a hard constraint rather than an aspirational target.

8.2.2 Key Insights

Three insights from the elicitation process materially shaped the resulting design:

1. **Tag-based resource discovery is non-trivial at Citco scale.** The training week revealed that the citcoworks and meridian environments share the **Shared Cloud Artifacts** BudgetCode tag, meaning a simple BudgetCode lookup conflates resources from two operationally distinct environments. This directly motivated FR9 (tag-based discovery with AppName fallback).
2. **The primary audience for the dashboard is the SRE team, not business operators.** Business operations teams would continue to use their existing domain-specific tools; the dashboard needed to be useful for the SRE first-responder, not designed as a business reporting tool. This shaped the panel selection (technical metrics over business KPIs) and the deep-link design (panels link directly to the relevant AWS Console page).
3. **Infrastructure-as-Code is a first-class requirement, not an after-thought.** The team had experienced the cost of manually provisioned monitoring infrastructure in previous projects (configuration drift, undocumented alarms). The JIRA refinement sessions made FR7 an explicit acceptance criterion rather than a **nice-to-have**.

8.3 Functional and Non-Functional Requirements

The following requirements were derived from the elicitation activities described above. They governed the design and acceptance criteria for all six work streams of my co-op placement.

8.3.1 Functional Requirements

Table 8.2 enumerates the nine functional requirements, each with the primary deliverable that satisfies it.

8.3.2 Non-Functional Requirements

Table 8.3 enumerates the seven non-functional requirements, each with the measurable target and the achieved value where applicable.

8.3.3 Requirements Traceability

All nine functional requirements and all seven non-functional requirements were satisfied by my delivered work streams. The test suite for the inventory API provides quantitative evidence for

Table 8.2: Functional Requirements

ID	Requirement	Satisfied By
FR1	Dashboard displays real-time Lambda metrics (invocations, errors, duration, throttles) per platform	Grafana AMG, Lambda rows (IISS-487)
FR2	Dashboard displays ECS CPU and memory utilization per cluster and service	Container Insights + ECS rows (IISS-487, IISS-483)
FR3	Dashboard displays MSK consumer lag per consumer group	MSK rows; EstimatedMaxTimeLag alarm (IISS-487)
FR4	Dashboard displays SQS queue depth and oldest message age	SQS rows in Grafana; SQS monitoring (IISS-482)
FR5	CloudWatch alarms provisioned for all monitored resource types with SNS routing to SRE team	Dynamic alarms via CloudFormation (IISS-487)
FR6	Dashboard supports variable-based filtering by platform and resource	Grafana template variables (IISS-487)
FR7	All monitoring infrastructure deployed as Infrastructure-as-Code	4 CloudFormation stacks; Lambda-backed IaC provisioner (IISS-487)
FR8	Inventory API enumerates Lambda, ECS, EC2, and S3 resources per platform via REST	Flask REST API, discoverer registry (IISS-507)
FR9	Inventory API supports two-stage tag-based discovery for shared BudgetCode platforms	Two-stage AppName filter in citcoworks/meridian discoverers (IISS-507)

NFR3 and NFR4. NFR6 was validated in practice when the Grafana workspace was upgraded from version 10.4 to version 12.4 on May 13, 2026 — a major version migration that required updating CloudWatch Metric Insights query syntax across all panels but did not require any architectural change to the underlying monitoring design.

The most technically demanding requirement was FR9, which exposed a fundamental ambiguity in the AWS tagging strategy for the IA platform. My discovery that citcoworks and meridian share the same BudgetCode tag value (**Shared Cloud Artifacts**) was not documented anywhere before I built the inventory API. The solution — designating AppName as a mandatory secondary discriminator for these two environments — became the standard pattern for all resource discovery going forward and was documented in the inventory API's README as a platform-wide

Table 8.3: Non-Functional Requirements

ID	Requirement	Target	Achieved
NFR1	Dashboard refresh enables sub-minute incident detection	≤ 30 sec	≤ 30 sec
NFR2	CloudWatch alarm evaluation period for MSK lag and Lambda error rate	≤ 1 min	1 min
NFR3	Inventory API response time for full resource enumeration	≤ 10 sec	< 10 sec
NFR4	Inventory API automated test coverage	$\geq 95\%$	99%
NFR5	CloudFormation templates parameterized for multi-environment reuse	Parameterized YAML	Achieved
NFR6	Dashboard functional after Grafana version upgrades without manual rework	Standard plugin queries	Achieved (10.4→12.4)
NFR7	Zero hardcoded credentials — all secrets via AWS Secrets Manager	No plaintext secrets	Achieved

architectural note.

NFR7 received an unexpected real-world validation during the CAIS Deadline hotfix on Jun 8, 2026. The incident revealed that the ECS container’s credential loading pattern (reading from Secrets Manager once at module import time via `constants.py`) was fragile under secret rotation. The hotfix I delivered introduced fail-fast credential validation — a design improvement that reinforced NFR7 by ensuring that absent or expired credentials from Secrets Manager produce an immediate, visible failure rather than a silent downstream authentication error.

8.3.4 Out-of-Scope Requirements

The following were evaluated and explicitly descope:

- **Automated error budget burn-rate alerting.** Computing burn rate requires Metric Math expressions that divide the rolling error count by the rolling SLO budget. While the SLO targets and SLI metrics are now in place, configuring burn-rate alarms requires additional CloudFormation work that I descope from the dashboard project.
- **Prometheus as a data source.** A self-managed Prometheus server with a CloudWatch

Exporter sidecar was evaluated as an alternative data layer. It was ruled out because Amazon Managed Grafana does not support a self-managed Prometheus endpoint without Amazon Managed Service for Prometheus (AMP), which would add infrastructure cost and operational overhead. CloudWatch Metric Insights SQL provides equivalent multi-resource aggregation capability without the additional stack. Documented as Decision 3 in Section 10.8.3.

- **Pyroscope (continuous profiling).** Pyroscope is a Grafana-ecosystem continuous profiling tool that would surface CPU flame graphs and memory allocation traces at the function level. While it would complement the golden signals dashboard, it requires an instrumentation agent deployed alongside each monitored service. Given the scope of the current engagement and the constraint that the Grafana workspace uses only CloudWatch as its data source, Pyroscope was identified as a post-placement enhancement.

9. METHODOLOGIES

During my co-op placement I adopted Site Reliability Engineering as the primary methodology. I applied the four golden signals framework and SLI/SLO discipline to address the observability gap exposed by the CAIS-Pricing Silent Outage and the Meridian Incident. This chapter covers three areas: the SRE principles I applied, the gap analysis that established the starting posture, and the Dynatrace parity evaluation that framed the research question.

9.1 SRE Principles

Site Reliability Engineering is an operational discipline that applies software engineering thinking to keeping production systems reliable at scale. Its core concepts — golden signals, Service Level Indicators (SLIs), Service Level Objectives (SLOs), and error budgets — provided the theoretical backbone for every design decision I made during this co-op placement.

9.1.1 The Four Golden Signals

I discussed the four golden signals framework with Kri during a dedicated knowledge-transfer session. These four signals define the minimum instrumentation required to characterize service health:

1. **Latency** — *How long does a request take?* For the IA platform, latency shows as Lambda function duration (P50, P95, P99 percentiles) and ECS service response time. Sustained P99 increases warn of resource saturation before errors appear.
2. **Traffic** — *How much demand is on the system?* Measured as Lambda invocation count, SQS message ingestion rate, and MSK throughput (bytes in/out per second). Traffic anomalies often precede error and saturation signals.
3. **Errors** — *What is the rate of failed requests?* Tracked as Lambda error and throttle counts, ECS task failures, and NumberOfMessagesFailed on SQS queues. An error rate crossing the defined threshold indicates a defect requiring immediate attention.
4. **Saturation** — *How full is the service?* Saturation is the signal most relevant to the Apr 17, 2026 Meridian incident. It is measured as Lambda concurrent execution count relative to the account concurrency limit, ECS task CPU and memory utilization, MSK

consumer lag (the number of messages awaiting processing on a Kafka topic), and SQS queue depth with oldest-message age. A service approaching saturation degrades before it fails — making this the most actionable of the four signals.

Figure 9.1 illustrates how a request flows through the monitored services, exposing each of the four golden signals at different stages of the request lifecycle.

The Cast

AWS Component	Restaurant Equivalent	What it does
ALB (Application Load Balancer)	Host / Receptionist	Greets customers at the door, checks their reservation, directs them to the right section
Target Group	A section of tables	All tables assigned to the same team of waiters — customers with similar needs go here
ECS Service	Kitchen staff roster	"I need 3 cooks on shift" — defines how many workers should be running
ECS Task	One individual cook	A single worker doing their job right now
Health Check	Food safety inspection	The host periodically asks each cook "are you okay to serve?"

How a Request Flows

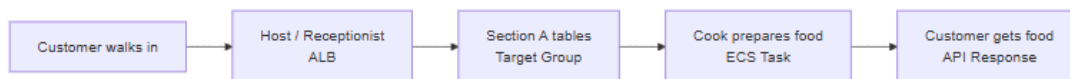


Figure 9.1: Request Flow Through Monitored Services — Golden Signals at Each Stage

9.1.2 Service Level Indicators and Objectives

A **Service Level Indicator (SLI)** is the specific metric chosen to represent one of the four golden signals for a service — the measurement. A **Service Level Objective (SLO)** is the target threshold set on that measurement — the acceptable limit.

For the IA platform, I selected the following SLIs and set corresponding SLO thresholds, implemented as CloudWatch alarm trigger values:

- **Lambda error rate:** SLI = Errors / Invocations per function. SLO threshold = fewer than 1% of invocations result in an error over any 24-hour window. Implemented as a CloudWatch alarm with Metric Math expression $\text{ErrorRate} = \text{Errors} / \text{Invocations}$.

- **MSK consumer lag:** SLI = EstimatedMaxTimeLag per Meridian consumer group. SLO threshold = lag must remain below **10,000 messages** at any point. Derived from the Apr 17 post-mortem: lag reached 9 million messages before detection; a 10,000-message threshold would have fired within minutes of onset.
- **ECS CPU utilization:** SLI = task-level CPU utilization per cluster via Container Insights. SLO threshold = no service sustains above **80%** over a 5-minute evaluation period.
- **SQS queue depth:** SLI = ApproximateNumberOfMessagesVisible per queue. SLO threshold = configurable per-queue maximum based on expected processing throughput.

These thresholds were encoded directly as CloudWatch alarm trigger values — each alarm fires when its SLI crosses the SLO boundary and routes an SNS notification to the SRE team. This constitutes **SLO-aligned alerting**: the alarms enforce the objectives but do not compute running error budget consumption or burn rate. This distinction is elaborated in the following subsection.

9.1.3 Error Budgets

An **error budget** is the complement of the SLO: the permitted amount of unreliability before the objective is breached. For example, if the Lambda error rate SLO is below 1% over 24 hours, the error budget is that 1% of invocations permitted to fail.

During this co-op placement I did not implement automated error budget tracking. The Grafana dashboard surfaces current SLI values and the CloudWatch alarms fire at SLO breach — which provides a reactive signal that the budget has been consumed, but does not calculate how quickly it is being drawn down. Full burn-rate alerting (requiring Metric Math expressions dividing rolling error counts against the rolling SLO budget) was identified as a post-placement enhancement and is scoped in Chapter 15.

9.2 Observability Gap Analysis

A prerequisite for building any monitoring platform is understanding the current state of observability. Before I wrote a single line of code for the centralized dashboard, I performed a structured gap analysis to characterize what the IA platform could and could not see about itself.

9.2.1 Pre-Co-op Placement Monitoring Posture

At the commencement of my co-op placement in Mar 2026, the IA-SRE team's monitoring posture exhibited the following deficiencies:

- **No CloudWatch alarms on any Meridian service.** The Meridian Kafka-backed event pipeline — the most business-critical component of the IA platform — had zero proactive alerting. No alarms existed on MSK consumer lag, ECS task failure counts, Lambda error rates, or SQS queue depth for any Meridian-owned resource.
- **No centralized dashboard.** Monitoring was entirely fragmented. Operators who wished to check the health of a service were required to navigate to that service's individual CloudWatch console page. There was no eagle-eye view from which to assess the overall health of the platform.
- **Incomplete Container Insights coverage.** Of the six Meridian ECS clusters, only one (enabled in the DEV environment since Jul 2024) had Amazon CloudWatch Container Insights activated. The remaining five clusters were invisible to task-level CPU and memory telemetry.
- **No MSK consumer lag monitoring.** Despite Kafka consumer lag being the primary leading indicator of Meridian processing health — as demonstrated conclusively by the Meridian Incident — no metric or alarm existed for this signal. The `EstimatedMaxTimeLag` and `SumOffsetLag` metrics available natively from Amazon MSK were not surfaced anywhere.
- **No SLO tracking or error budget calculation.** The team had no defined SLIs, no published SLO targets, and therefore no basis for calculating error budgets or determining when the platform was in an unacceptable reliability state.
- **No automated resource inventory.** The inventory of Lambda functions, ECS clusters, and EC2 instances was maintained manually via Confluence pages and spreadsheets. This made it impossible to know, at any given moment, exactly which resources existed across all six platforms.
- **RPA bot monitoring limited to orchestrator UIs.** Blue Prism and UiPath bots were monitored exclusively through their respective orchestrator platforms (Blue Prism Orchestrator and UiPath Cloud). No CloudWatch integration existed, and there was no mechanism for the

SRE team to receive proactive notification of bot failures or queue backlogs outside the orchestrator UI.

- **Reactive operational posture.** The cumulative effect of the above gaps was that the team's primary incident detection mechanism was business-team escalation. When a fund processing automation failed silently, the first signal received by the SRE team was a ticket or message from operations staff in Dublin, London, or Halifax who had noticed anomalous fund processing outputs. This reactive posture introduced structural latency between failure onset and engineering response.

9.2.2 Gap Analysis Summary

Table 9.1 summarizes the pre-placement gaps and the post-placement resolutions for each monitoring domain.

Table 9.1: *Observability Gap Analysis — Pre- vs. Post-Co-op Placement*

Domain	Pre-Co-op Placement State	Post-Co-op Placement State
CloudWatch Alarms	0 alarms on any IA resource	Dynamic alarms across 7 AWS service categories
Centralized Dashboard	None	Multiple rows and panels (Grafana AMG)
Container Insights	1 of 6 Meridian clusters	5 of 6 Meridian clusters
MSK Consumer Lag	No monitoring	CloudWatch alarm on EstimatedMaxTimeLag (<30 sec detection)
SLO Tracking	None	SLI/SLO targets defined; dashboard exposes current values
Resource Inventory	Manual (Confluence)	Automated REST API (Lambda, ECS, EC2 across all platforms)
Incident Detection	Business team escalation	Proactive alarm with SNS routing

9.3 Dynatrace Parity Analysis

A key framing question for my co-op placement — introduced in Chapter 2 and developed here in methodological detail — is how closely an AWS-native observability stack can replicate the capabilities of Dynatrace, and at what incremental cost. The PoC investigation evaluated 25 Dynatrace features against five AWS tool combinations to quantify achievable parity.

Table 9.2 summarizes the parity analysis for the ten most operationally relevant Dynatrace capabilities to the IA platform.

Table 9.2: *Dynatrace Capability Parity — AWS-Native Stack vs Dynatrace (selected features)*

Dynatrace Feature	CW	+Grafana	+X-Ray	+App Signals	+Tempo
Alerting	✓	✓	—	—	—
Infrastructure Metrics	✓	✓	—	—	—
Log Analytics	✓	✓	—	—	—
Container Monitoring	✓	✓	—	✓	—
Error Rate Tracking	✓	✓	✓	✓	—
Latency Percentiles	—	✓	✓	✓	✓
Distributed Trace Waterfall	—	—	✓	—	✓
Service Map / Topology	—	—	✓	✓	—
SLO Management	—	—	—	✓	—
Dependency Mapping	—	—	✓	✓	—
Parity (% of 25 features)	30%	33%	60%	80%	83%
Monthly Cost (est.)	\$49	\$68	\$77	\$232	\$269

Three capabilities remain **permanently outside the AWS-native reach**: Davis AI causal analysis (proprietary ML, no AWS equivalent); automatic baselining (Dynatrace learns per-service baselines over weeks; AWS alarms require manual thresholds); and session replay (full DOM capture, not available in any OSS tool). These account for approximately 9–10% of the functional gap that no combination of AWS tools can close.

Adopted stack. The PoC I delivered in this co-op placement corresponds to the **CloudWatch + Grafana + X-Ray** column: 60% parity at approximately \$77/month. Application Signals (the path to 80% parity) requires deploying an ADOT sidecar on each ECS service and is scoped as post-placement work. The gap-to-tooling mapping is used throughout Chapter 10 and Chapter 13 to contextualize what the PoC does and does not cover.

10. IMPLEMENTATION & ARCHITECTURE

10.1 Platform Overview: Four-Layer Architecture

The three deliverables of my co-op placement form a coherent four-layer observability stack. Figure 10.1 shows how the layers relate to each other and to the underlying IA platform infrastructure.

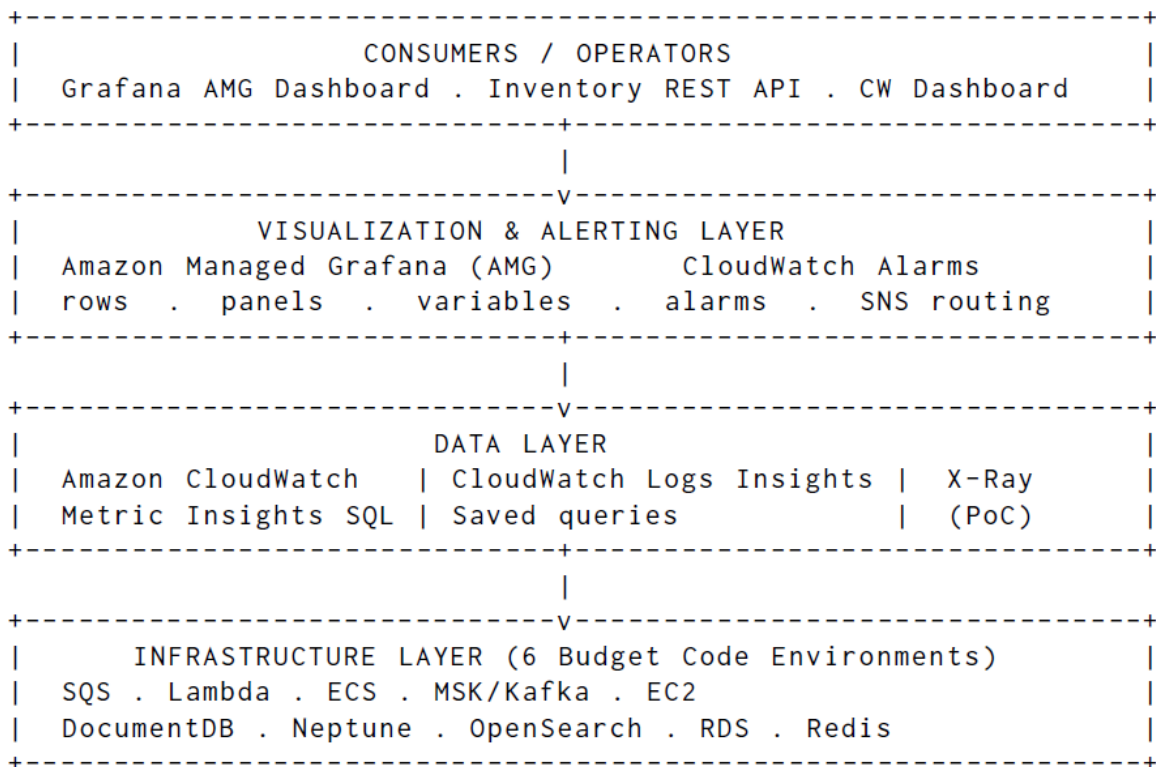


Figure 10.1: IA-SRE Observability Platform — High-Level Architecture (four layers)

Figure 10.2 shows the full system architecture of the Grafana AMG observability platform, detailing how each monitored AWS service feeds CloudWatch, how the Lambda provisioner reads alarm templates from S3, and how the SNS routing delivers email notifications.

Figure 10.3 provides a rendered view of the same architecture from the project documentation.

The **infrastructure layer** is the monitored subject: six budget platforms housing Lambda functions, ECS clusters, EC2 instances, and the full spectrum of AWS database and messaging services used by the Meridian and Citco Works platforms.

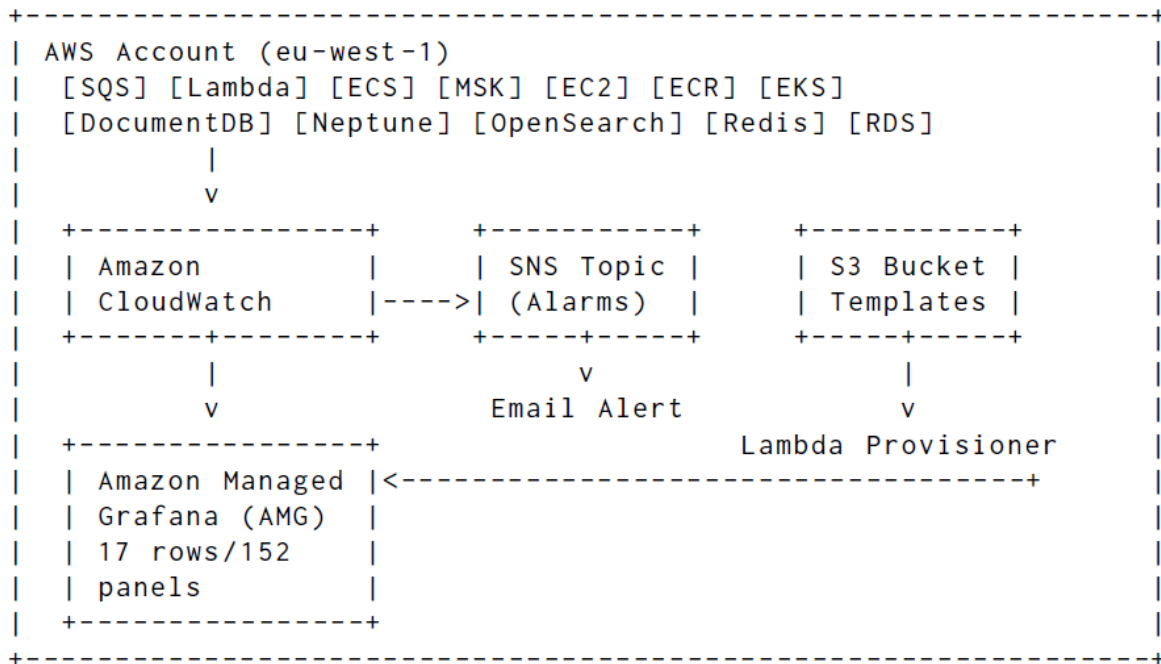


Figure 10.2: AMG Observability Platform — System Architecture

The **data layer** ingests, stores, and queries observability signals from the infrastructure layer. Amazon CloudWatch serves as the sole data source. This was a deliberate architectural decision (see Section 10.8.3) that eliminates the operational overhead of a self-managed Prometheus stack. CloudWatch Metric Insights SQL provides per-resource aggregation, enabling multi-service panels without requiring one panel per resource.

The **visualization and alerting layer** splits between two components. Amazon Managed Grafana (AMG) renders the multi-row dashboard for human operators. CloudWatch Alarms with SNS routing fires notifications when metrics breach their SLO-aligned thresholds.

The **consumers layer** includes three groups: operators who use the Grafana dashboard during incident response, the IA Resources Inventory REST API that exposes the AWS service fleet programmatically, and the SRE team’s email distribution list that receives SNS alarm notifications.

10.2 CloudWatch Dashboard Platform

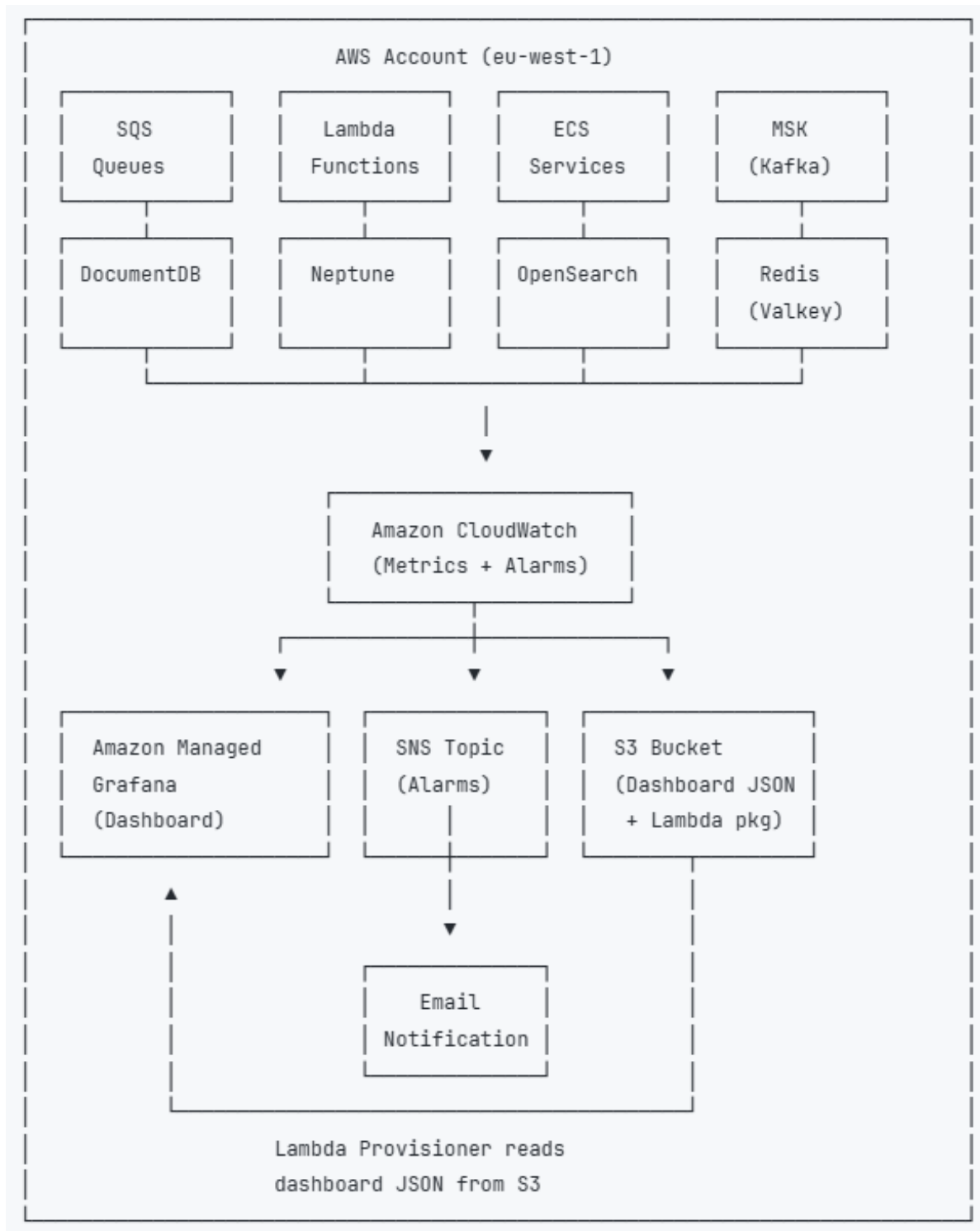


Figure 10.3: AMG Observability Platform — Architecture Diagram (from documentation)

10.2.1 Initial Lambda Dashboard

My earliest monitoring work, completed in Apr 2026, produced a native Amazon CloudWatch dashboard covering the `rpacfs1` and `ia-bpo` Lambda function fleets — 51 functions total. The dashboard displayed per-function panels for invocation counts, error rates, duration averages, and throttle events. I also created a Confluence inventory page mapping each function.

The architectural pattern for this early work was straightforward: CloudWatch native widgets queried the `AWS/Lambda` namespace using `SEARCH()` expressions to enumerate all functions matching a `FunctionName` prefix. This approach worked adequately for small, homogeneous fleets but surfaced two structural limitations that drove the later Metric Insights SQL migration: the **`SEARCH()` visibility gap** (idle functions disappear from panels) and **no cross-environment aggregation** (a single expression cannot span multiple tag dimensions simultaneously). Both are documented in full in Chapter 14.

Figure 10.4 shows a live screenshot of the CloudWatch Lambda dashboard displaying invocation and error metrics.

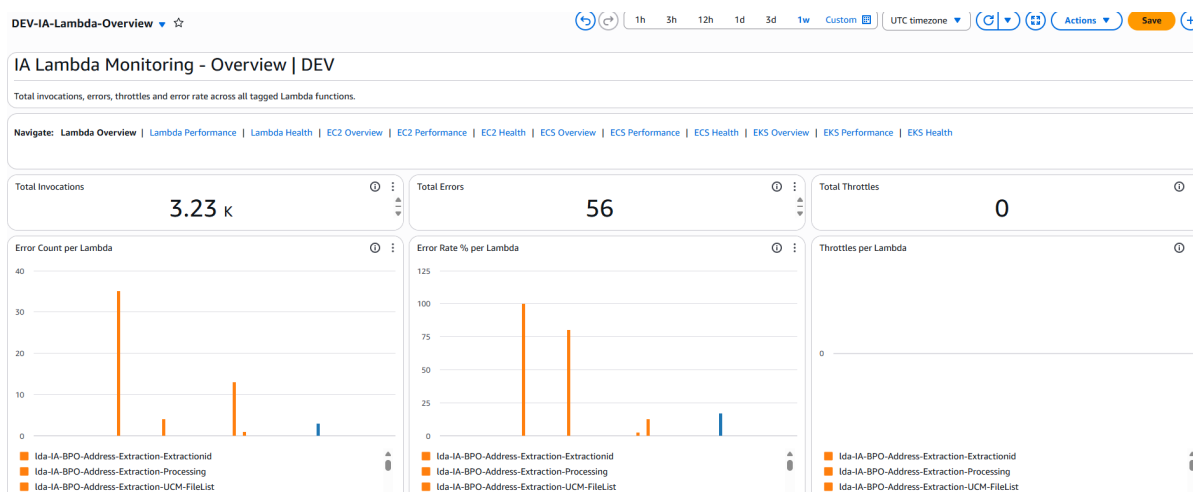


Figure 10.4: CloudWatch Lambda Dashboard — Live Screenshot (*ia-resources-monitoring*)

10.2.2 Extended Dashboard and Architecture Analysis

I extended the initial dashboard with additional tag dimensions covering `automationhub`, `bpm`, `citcoworks`, and `meridian` budget codes, and introduced CloudWatch Logs Insights widgets for error log surfacing. Rather than relying solely on Errors metrics from the `AWS/Lambda` namespace, the Logs Insights widgets queried the `/aws/lambda/` log group prefix directly,

enabling pattern-matched error extraction regardless of whether Lambda had published a corresponding metric.

A significant architectural investigation within this task evaluated the feasibility of a fully centralized log aggregation approach using CloudWatch metric subscription filters. My analysis quantified the infrastructure cost of this design: routing log events from all IA Lambda functions into a centralized processing layer would require **106 metric subscription filters**—one per function per metric dimension—across the six platforms. While technically feasible from a CloudWatch service quota perspective, this approach was ultimately blocked by an IAM permission boundary constraint: the `logs:PutSubscriptionFilter` action was not granted within the IA team’s scoped permission set, preventing programmatic creation of subscription filters at the scale required.

This investigation also evaluated five distinct approaches to out-of-memory (OOM) detection for Lambda functions: (1) parsing REPORT log lines from CloudWatch Logs Insights; (2) comparing MaxMemoryUsed against MemorySize via the Lambda enhanced monitoring extension; (3) CloudWatch Lambda Insights (paid tier); (4) custom metric emission on memory threshold breach; and (5) structured log parsing via a metric filter pattern. I documented each approach with its detection latency, cost, and permission requirements. This analysis informed the Container Insights cost evaluation and the alarm design for the dashboard.

10.2.3 IA Infrastructure Monitoring Sub-Stories

EC2 Inventory and Tag Mismatch Discovery

My gap analysis of the EC2 inventory scanned **83 instances** across the six platforms. This scan revealed a critical tagging inconsistency: the `rpacfs1` environment’s EC2 instances were tagged with `BudgetCode="CFS/RPA"` rather than the expected **"rpacfs1"** value, making them invisible to tag-based `BudgetCode` lookups. This discovery led me to introduce the `AppName` fallback pattern: when a primary `BudgetCode` lookup returns no results for `rpacfs1`, the discovery logic falls back to filtering by `AppName="rpacfs1"`. This fallback became the standard for all resource discovery in the inventory API (see Section 10.4).

Container Insights Cost Analysis

With ECS task-level telemetry absent from five of the six Meridian ECS clusters, I evaluated four Container Insights deployment options:

1. **No Container Insights** — zero cost, zero task-level visibility.
2. **Enhanced Container Insights** — full task and pod metrics, but carries per-metric ingestion cost that scales with task count.
3. **Standard Container Insights** — cluster- and service-level CPU/memory metrics via the ECS/ContainerInsights namespace, charged at CloudWatch standard metric pricing.
4. **CloudWatch Agent on ECS (self-managed)** — maximum flexibility, highest operational overhead.

My analysis recommended **Standard Container Insights** at an estimated cost of **\$10.44/month** for full coverage across all 22 ECS clusters. This option provided the SLI data required for FR2 (ECS CPU/memory dashboard) at negligible incremental cost relative to the existing CloudWatch spend. I enabled Standard Container Insights on five of the six Meridian clusters on May 11–12, 2026.

SQS Registry Architecture

The SQS monitoring feature I added on introduced a **registry-based architecture with a three-tier fallback model** for resilient SQS queue data retrieval. The three tiers operated as follows:

1. **Tier 1 — Live API**: Query CloudWatch AWS/SQS metrics directly via the CloudWatch API. Fast and fresh, but requires network connectivity and API quota.
2. **Tier 2 — S3 Cache**: If the API call fails (rate limit, transient error), retrieve the last-known queue snapshot from an S3 bucket. Stale by up to one cache TTL window, but always available.
3. **Tier 3 — Local File**: If both API and S3 fail, fall back to a locally bundled queue inventory file. Guaranteed to return data; snapshot reflects deployment-time inventory.

This three-tier model ensured the SQS monitoring pipeline remained functional even during AWS service degradation events. The feature was validated with **automated tests** covering the registry dispatch logic, fallback transitions, and queue metadata serialization.

10.3 Grafana AMG Observability Platform

This was my primary deliverable over significant duration. It delivered a centralized Amazon Managed Grafana workspace with various dashboard rows and panels, dynamic CloudWatch alarms, and a Lambda-backed IaC provisioner — all built on a four-layer architecture described below.

10.3.1 Architectural Layers

Layer 1: Data Layer

Amazon CloudWatch serves as the **sole data source** for the entire Grafana workspace. My deliberate choice to use only the native Grafana CloudWatch plugin — rather than a self-managed Prometheus/CloudWatch exporter stack — eliminated all additional infrastructure requirements. The data layer is characterized by:

- **CloudWatch plugin:** The Grafana AMG workspace connects to CloudWatch via the built-in plugin, authenticated through an IAM role. No additional data collectors, agents, or scrapers are required.
- **IAM role model:** A **16-role deployment** (one role per AWS account, covering all six IA platform accounts) built on a **three-role IAM chaining architecture** provides the Grafana service account with cross-account read access across all IA AWS accounts. Each role grants the minimum permissions required: `cloudwatch:GetMetricData`, `cloudwatch:ListMetrics`, `cloudwatch:DescribeAlarms`, `logs:StartQuery`, `logs:GetQueryResults`, and `ec2:DescribeTags` (for tag-based resource enumeration). Figure 10.5 shows the three-role IAM chaining architecture required because CTM-Developer cannot create IAM roles directly and the shared CITCO-CloudFormationBuild role lacks `grafana:*` permissions.
- **Metric Insights SQL:** The primary query language for multi-resource panels. CloudWatch Metric Insights enables SQL-like aggregation across all resources in a namespace with a single query, replacing the brittle `SEARCH()` expressions used in the earlier CloudWatch dashboard.

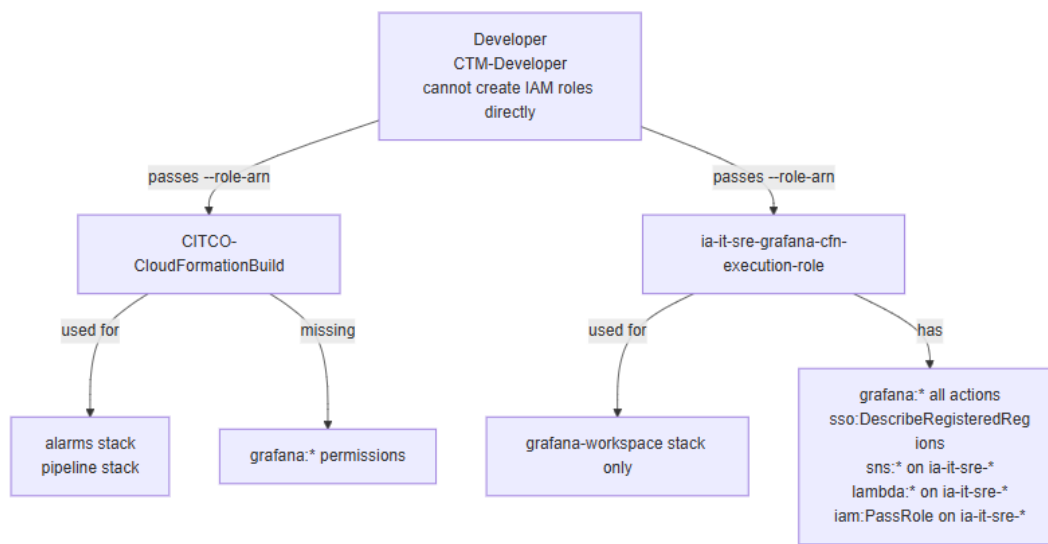


Figure 10.5: IAM Role Architecture — Three-Role Model for Grafana AMG Deployment

Layer 2: Visualization Layer

The Grafana AMG workspace hosts **17 dashboard rows** organized by monitored service category. Table 10.1 summarizes the dashboard structure.

Figures 10.6–10.8 show live screenshots from the deployed Grafana AMG dashboard, demonstrating the Health Overview, Lambda monitoring, and ECS monitoring rows.

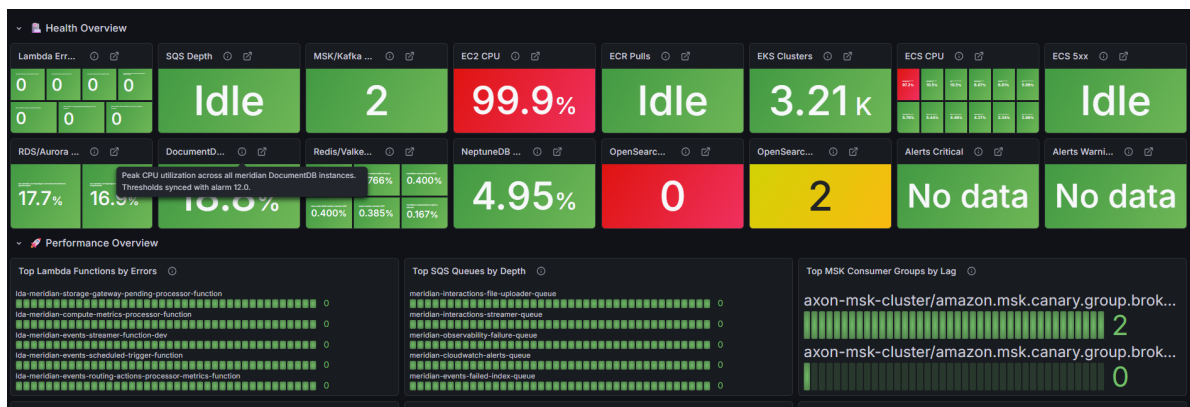


Figure 10.6: Grafana AMG Dashboard — Health Overview Row (Live Screenshot)

The Lambda Monitoring Row (Figure 10.7) provides real-time visibility into all Lambda functions across the selected environment. The top table displays function names, trigger types, runtime versions, memory allocations, and timeout configurations — enabling operators to identify

Table 10.1: Grafana AMG Dashboard — Row Structure (Rows 1–15 shown; Rows 16–17: Alarms & Notifications, X-Ray Traces)

Row	Service	Panels and Metrics
1	Health Overview	Color-coded stat tile per service (Lambda, SQS, MSK, EC2, ECR, EKS, ECS, RDS, DocumentDB, Redis, Neptune, OpenSearch); critical/warning alarm counts
2	Performance Overview	Top-10 bargauge comparisons per service — highest queue depth, Lambda errors, ECS CPU, Kafka consumer lag, Redis memory, etc.
3	Lambda	Invocations, errors, throttles, concurrent executions, duration P50/P99; Logs Insights error panel
4	SQS	Queue depth (ApproximateNumberOfMessagesVisible), oldest message age, DLQ depth
5	MSK/Kafka	Consumer lag: SumOffsetLag and EstimatedMaxTimeLag; broker CPU, disk, messages in/out per second
6	EC2	CPU (avg/max), memory (CloudWatch Agent), network I/O, disk I/O, status checks
7	ECR	Container image pull activity
8	EKS	Kubernetes node and pod metrics
9	ECS Services	CPU and memory utilization per cluster and service; Container Insights task-level metrics
10	ECS Target Groups	ALB request throughput, response time P50/P95/P99, HTTP 5xx/4xx error counts
11	RDS / Aurora	CPU, database connections, read/write latency, IOPS, freeable memory
12	DocumentDB	CPU, disk queue depth, active connections, read/write latency, IOPS
13	Redis / Valkey	Engine CPU (avg/max), memory, cache hit rate, replication lag, connections, latency
14	Neptune DB	CPU, Gremlin/SPARQL errors, freeable memory, queue depth, network throughput
15	OpenSearch	Cluster status (green/yellow/red), JVM heap pressure, indexing rate, search latency



Figure 10.7: Grafana AMG Dashboard — Lambda Monitoring Row (Live Screenshot)

misconfigured functions at a glance. Below the table, four time-series panels track the four golden signals: **Total Errors** (count of failed invocations), **Invocations** (throughput), **Concurrent Executions** (saturation indicator), and **Throttles** (capacity limit breaches). The legend below each panel enables drill-down to individual functions, with color-coding to identify anomalous patterns.



Figure 10.8: Grafana AMG Dashboard — ECS Monitoring Row (Live Screenshot)

The ECS Monitoring Row (Figure 10.8) surfaces Container Insights metrics for all ECS services in the selected cluster. The service table shows cluster assignment, desired count, and running count — highlighting any discrepancies that indicate failed deployments or scaling issues. The four time-series panels display CPU and memory utilization (both average and maximum) across all tasks, enabling operators to identify resource saturation before it impacts service availability. This row was particularly valuable for monitoring the Meridian ECS services, where memory pressure had previously gone undetected until task failures occurred.

Grafana Template Variables

The dashboard supports environment-level filtering through three Grafana template variables backed by CloudWatch dimension-value queries:

- `$budget_code` — filters all panels to a specific environment (rpacfs1, ia-bpo, meridian, citcoworks, automationhub). Backed by a CloudWatch `ListMetrics` call enumerating distinct `BudgetCode` tag values.
- `$ecs_cluster` — filters ECS panels to a specific cluster. Backed by a `dimension_values()` query on the ECS/ContainerInsights namespace.
- `$lambda_function` — filters Lambda panels to a specific function. Backed by a `dimension_values()` query on the AWS/Lambda namespace.

When a user selects a value from any variable, all panels that reference that variable re-execute their queries with the selected filter applied, effectively scoping the entire dashboard to a single environment or resource. This eliminates the need to maintain separate dashboards per platform.

CloudWatch Metric Insights SQL: Architecture Pattern

CloudWatch Metric Insights SQL is the key architectural pattern that enables per-resource drill-down across large fleets within a single query. The standard form uses `SELECT ... FROM ... GROUP BY` to return metrics for each resource as a separate time series, covering all resources without requiring one panel per function. The `WHERE` clause can filter by tag or dimension to scope results to specific platforms.

Percentile Limitation and Workaround. A fundamental constraint of Metric Insights SQL is that the `GROUP BY` mode only supports the aggregate functions `AVG`, `SUM`, `MIN`, and `MAX`. Percentile statistics (p50, p95, p99) are *not available* in `GROUP BY` mode. This is a CloudWatch API limitation, not a Grafana limitation.

For panels requiring true latency percentiles, the workaround is to use the *standard CloudWatch metrics* query mode with a wildcard dimension value (e.g., `FunctionName: [ "*"]`). This returns individual time series for each function that has emitted metrics within the selected time window. The tradeoff is that only recently active functions appear — the same “visibility gap” limitation noted for `SEARCH()`. For the Lambda duration P50/P95 panels, I accepted this tradeoff: a function

that has not been invoked recently has no meaningful latency distribution to display.

Layer 3: Alerting Layer

I provisioned **dynamic CloudWatch alarms** across all monitored platforms, each connected to an SNS topic that routes notifications to the IA-SRE team's email distribution list. Alarm categories include:

- **Lambda alarms:** Error rate exceeding 1% over a 5-minute rolling window; throttle count exceeding threshold; concurrent execution count approaching account limit.
- **ECS alarms:** CPU utilization sustained above 80% over a 5-minute evaluation period; task desired count $\neq$ running count (task failure detection).
- **MSK alarms:** EstimatedMaxTimeLag exceeding 10,000 messages on any consumer group. This alarm directly addresses the Meridian Incident: it would have fired within minutes of the lag beginning to build, compared to the 3.5-hour detection gap that occurred under the reactive model.
- **SQS alarms:** Queue depth exceeding configurable threshold per queue; ApproximateAgeOfOldestMessage exceeding 1 hour (configurable per SLA).
- **Composite alarms:** Platform-level health signals that aggregate per-service alarm states into a single OK/ALARM indicator per budget code environment.

Layer 4: Infrastructure-as-Code Provisioner

The IaC layer is a **Lambda-backed CloudFormation provisioner** deployed via AWS SAM. Its architecture addresses a specific scaling constraint: with 119 Lambda functions, a static CloudFormation template defining one alarm per function would contain $119 \times (\text{alarm types per function})$ resources, rapidly approaching the CloudFormation 500-resource-per-stack limit.

The provisioner resolves this by *dynamically discovering* the current function fleet at deploy time and synthesizing alarm resources programmatically:

1. The provisioner Lambda function is invoked by a CloudFormation custom resource on every `samconfig.toml` deploy.

2. It reads parameterized CloudFormation YAML templates from an S3 bucket (see Section 10.8.1).
3. For each template (Lambda alarms, ECS alarms, SQS alarms, MSK alarms), it calls the relevant AWS discovery API to enumerate all current resources in the specified platform.
4. It synthesizes CloudFormation `AWS::CloudWatch::Alarm` resource blocks for each discovered resource and passes the generated template to CloudFormation for stack deployment.
5. Four CloudFormation stacks were deployed on **May 11, 2026**: `ia-sre-lambda-alarms`, `ia-sre-ecs-alarms`, `ia-sre-sqs-alarms`, and `ia-sre-msk-alarms`.

A `samconfig.toml` with environment-specific parameter files enables reuse across DEV, UAT, and PROD without code changes. The budget code, alarm thresholds, and SNS topic ARN are all externalised as SAM parameters, satisfying NFR5 (multi-environment reuse).

Figure 10.9 illustrates the alarm notification flow from CloudWatch alarm state change through SNS topic fan-out to the IA-SRE team's email distribution list.

Figure 10.10 shows the CodePipeline CI/CD deployment flow that delivers all stacks across the three environments.

10.3.2 Grafana Version Upgrade: Architectural Resilience Test

On **May 13, 2026** — two days after the initial infrastructure deployment — the Grafana AMG workspace was upgraded from version **10.4 to 12.4**. This was a major version migration that constituted an unplanned but highly informative architectural resilience test.

The upgrade introduced breaking changes in two areas:

- **CloudWatch Metric Insights SQL editor:** The query syntax editor in Grafana 12.4 changed the way SQL expressions were associated with panel targets. All SQL panels required migration to the updated CloudWatch query model.
- **Variable interpolation:** Template variable syntax in panel queries changed between versions, causing several `$budget_code` and `$ecs_cluster` references to fail to resolve correctly after the upgrade.

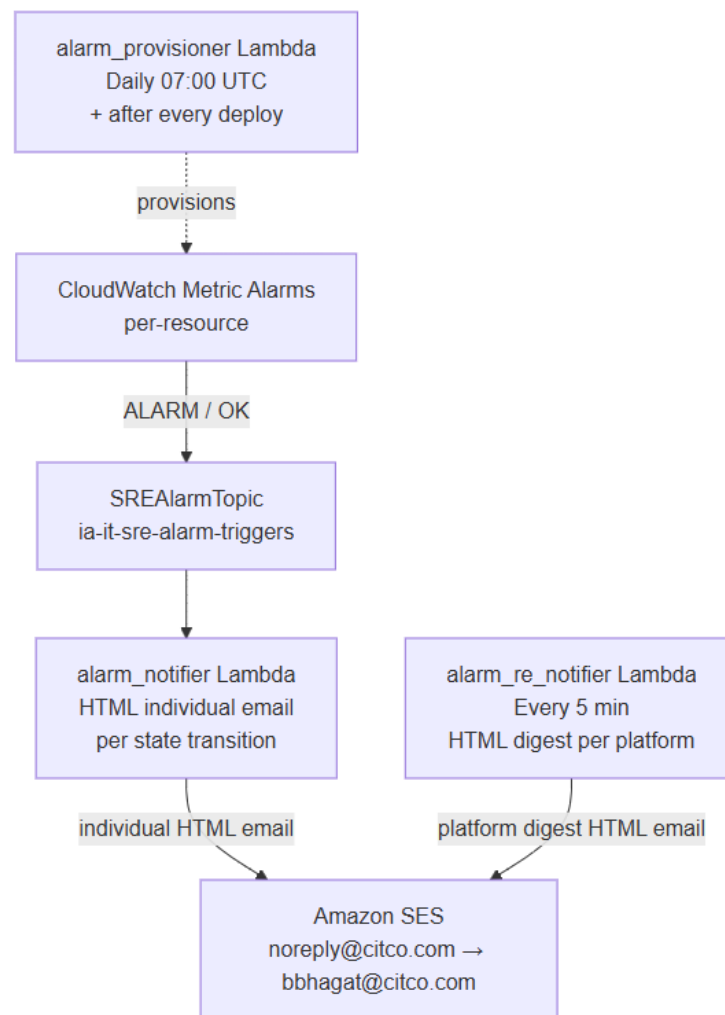


Figure 10.9: Alarm Notification Flow — CloudWatch to SNS to Email Distribution

I resolved this through a systematic audit of all panels to identify affected queries. The migration reinforced a key design principle documented as NFR6: **use only standard CloudWatch plugin queries and avoid AMG-version-specific syntax**. I migrated all panel queries to the stable Grafana 12.4 CloudWatch query model, and the workspace has remained stable through subsequent Grafana patch releases.

10.3.3 X-Ray Distributed Tracing (PoC)

As part of the observability stack, I integrated AWS X-Ray tracing to provide distributed trace visibility across the IA platform. The X-Ray integration enables service map visualization and trace waterfall analysis — capabilities that contribute to the 60% Dynatrace parity achieved by the

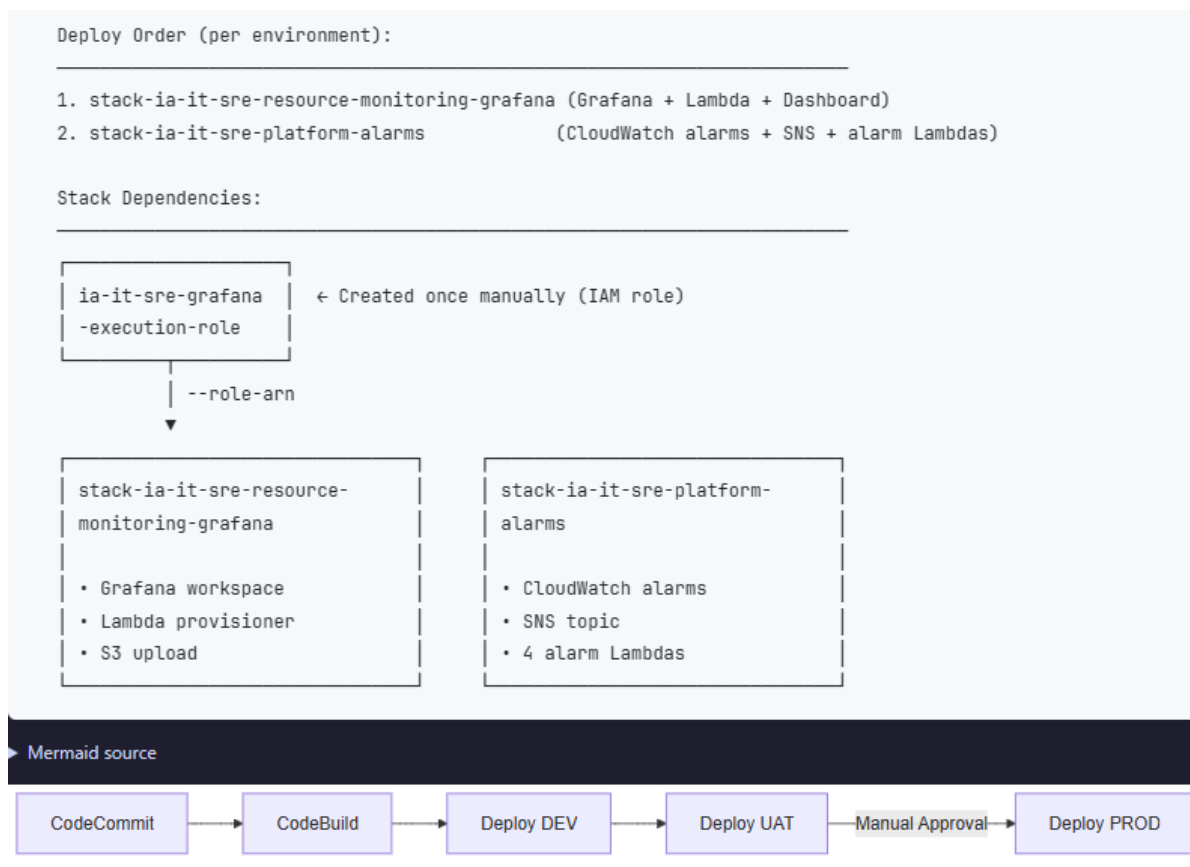


Figure 10.10: CI/CD Deployment Pipeline — CodeCommit to AMG Deployment Flow

CloudWatch + Grafana + X-Ray stack (see Section 9.3).

Figure 10.11 shows the X-Ray service map displaying the trace topology for the RPA-LLM platform, illustrating how requests flow through Lambda functions, external APIs, and AWS services.

The X-Ray integration delivers three key capabilities:

- **Service Map:** Visual representation of service dependencies and request flow, enabling rapid identification of bottlenecks and failure points.
- **Trace Waterfall:** Detailed timing breakdown for individual requests, showing latency contribution from each service in the call chain.
- **Error Analysis:** Automatic capture of exceptions and error responses with stack traces, correlated to specific trace segments.



Figure 10.11: AWS X-Ray Service Map — Distributed Trace Topology (Live Screenshot)

X-Ray tracing is currently enabled in DEV environment only; UAT and PROD enablement is pending senior approval due to the additional CloudWatch costs associated with trace segment storage.

10.4 IA Resources Inventory API

The inventory API delivered a **Flask REST API** that I built for programmatic discovery of all IA platform cloud resources. I housed the project in the `ia-sre-resources-inventory` repository and scaffolded it on Jun 11, 2026; it reached production-ready state with 562 automated tests and 99% code coverage.

Figure 10.12 shows the high-level architecture of the Inventory API, illustrating how discoverers query AWS services and return unified resource models.

10.4.1 Technology Stack

- **Language:** Python 3, flask[async] for async route handlers
- **AWS SDK:** boto3 for all resource discovery
- **Validation:** Pydantic response models (type-safe Resource dataclass with fields: name,

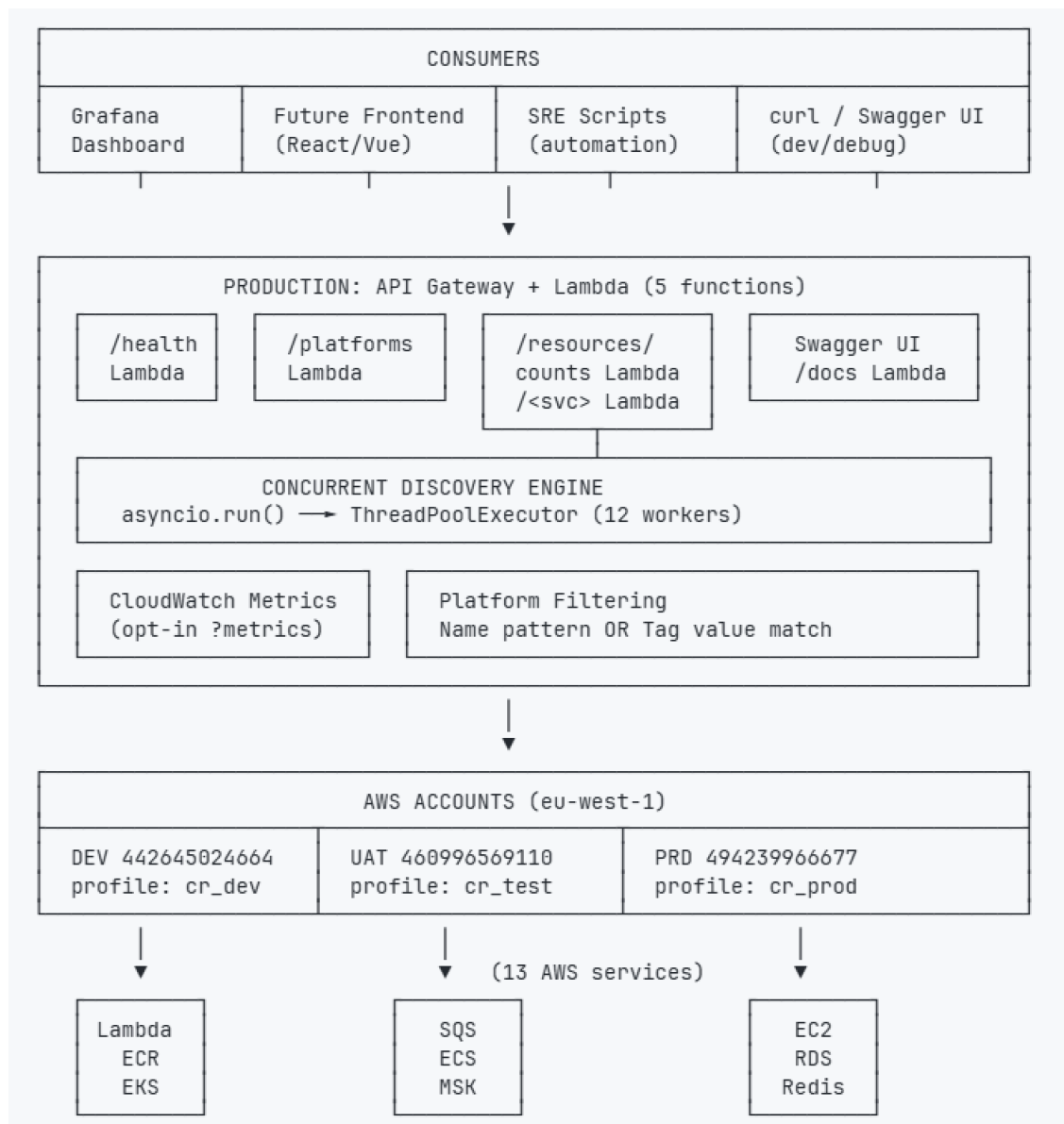


Figure 10.12: IA Resources Inventory API — Architecture Overview

type, budget_code, tags, arn)

- **Deployment:** AWS Lambda via serverless-wsgi adapter
- **Testing:** pytest + Hypothesis (property-based tests)

10.4.2 Discoverer Registry Pattern

My core architectural contribution in the inventory API is the **Discoverer Registry pattern**. Rather than routing resource discovery logic through a monolithic handler, the API maintains a registry that maps (resource_type, budget_code) tuples to concrete discoverer classes. Each discoverer implements a single abstract method `async def discover() -> List[Resource]`. Adding support for a new platform or resource type requires implementing one class and registering one entry — zero changes to routing, serialization, or test infrastructure.

10.4.3 API Endpoints

Table 10.2: IA Resources Inventory API — Endpoint Summary

Endpoint	Description
GET /api/v1/lambda	All Lambda functions; optional ?budget_code=rpacfs1 filter
GET /api/v1/ecs	All ECS clusters across all platforms
GET /api/v1/ec2	All EC2 instances (tag-based discovery)
GET /api/v1/secrets	Secrets Manager secrets inventory
GET /api/v1/s3	S3 buckets per budget code
GET /api/v1/health	Health check — returns 200 OK

10.4.4 Tag-Based Discovery and the Two-Stage Filter

The most technically interesting discovery challenge was the citcoworks and meridian tag ambiguity. Both environments share the AWS tag value **"Shared Cloud Artifacts"** for the BudgetCode dimension — a legacy of the two platforms being onboarded before the IA team established per-platform BudgetCode uniqueness requirements. A naive BudgetCode lookup would conflate resources from two operationally distinct environments.

Figure 10.13 shows how the Inventory API integrates with the Grafana AMG dashboard — the cache-refresh Lambda writes S3 snapshots and publishes CloudWatch metrics, which Grafana

reads via template variables and Metric Insights SQL panels.

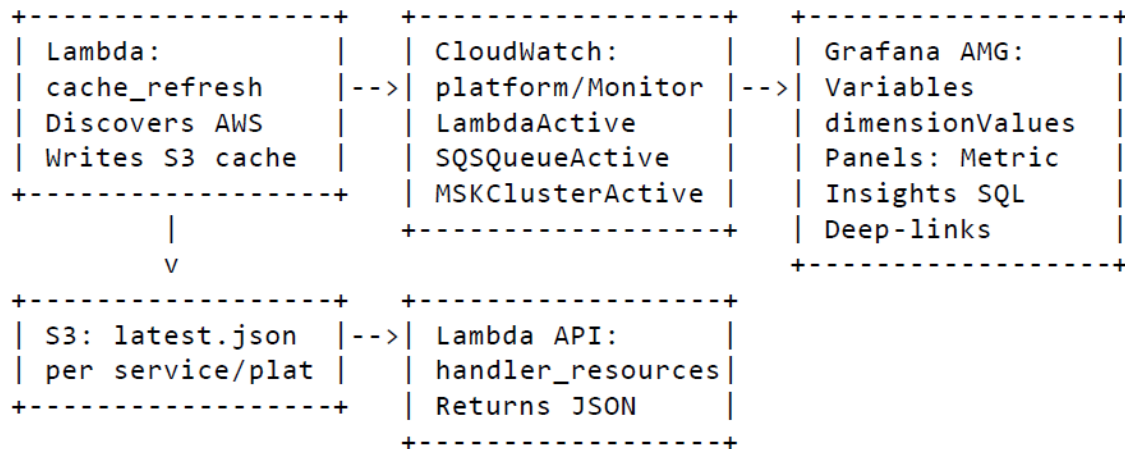


Figure 10.13: Inventory API — Grafana AMG Integration

My solution is a **two-stage filter** implemented in both the CitcoWorksECSDiscoverer and MeridianECSDiscoverer. Stage 1 filters by BudgetCode (which is shared between both platforms), then Stage 2 uses the AppName tag to discriminate Meridian resources from CitcoWorks resources.

The AppName tag is now designated the **mandatory secondary discriminator** for all resources in the **"Shared Cloud Artifacts"** BudgetCode environments. Any new resource added to either environment without the correct AppName tag will be invisible to the inventory API. This risk is mitigated by a tag compliance CloudWatch alarm that fires if new ECS services or Lambda functions lack the required tags.

Figure 10.14 illustrates the complete resource discovery flow, showing how the API routes requests through the discoverer registry and applies the two-stage filter for ambiguous budget codes.

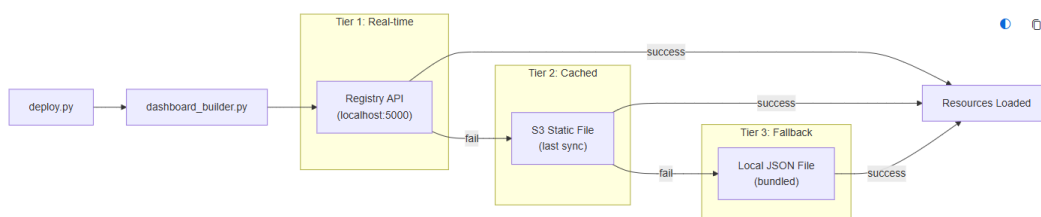


Figure 10.14: Resource Discovery Flow — Discoverer Registry and Two-Stage Filter

10.4.5 Async Architecture

The API uses `flask[async]` with async route handlers. Each handler aggregates boto3 calls concurrently using `asyncio.gather()`, reducing total API latency by parallelizing the AWS API calls within a single request. All matching discoverers run concurrently rather than sequentially, reducing API response time from the sum of all boto3 call latencies to the latency of the single slowest call — consistently keeping the API within the 10-second NFR3 target.

10.4.6 Test Strategy

My test suite for the inventory API achieves **99% code coverage**:

- **Unit tests:** Each discoverer class tested in isolation with mocked boto3 clients. Verifies correct tag filtering, pagination handling, and Resource object construction.
- **Integration tests:** Full Flask test client hitting all REST endpoints. Verifies routing, serialization, and error response format.
- **Property-based tests (Hypothesis):** Tag-parsing logic, registry dispatch, and Pydantic response serialization tested with randomly generated inputs. Verifies that the two-stage filter never returns a meridian resource when `budget_code=citcoworks` is specified, regardless of what combination of tag values is present.
- **Edge case tests:** Malformed tags, empty budget code environments, boto3 pagination edge cases (empty pages, single-page results, maximum page sizes).

10.5 Work Stream 4: RPA Production Support

Throughout the co-op placement I participated in dedicated RPA production support rotations as part of the IA team's **Follow-the-Sun model**. I held the North America slot (11 AM–7 PM EST) once every four weeks, serving as the primary contact for RPA incidents during that window. This was not project work with JIRA tickets — it was ongoing operational responsibility with Service Desk tickets, email triage, and ad-hoc troubleshooting.

10.5.1 Support Rotation Structure

Each support week followed a consistent daily rhythm:

- **Shift start (11 AM EST):** Review the Confluence handover log from the Manila shift to pick up carried-over items or approaching SLA breaches. Check open tickets in the SD Portal, Innovation IT email inbox, IA_INCIDENTS distribution list, and Teams notifications.
- **During shift:** Monitor VPL queue health (5 PM EST daily checkpoint), respond to escalations from Dublin/London business teams, triage incoming tickets, and coordinate with developers when code-level fixes are required.
- **Shift end (7 PM EST):** Update the Confluence handover log with ticket status, any pending actions, and context for the next shift (Manila).

10.5.2 Support Weeks Completed

I completed four full support rotations during the co-op placement:

- **Week 8** (May 4–8): First rotation. Familiarization with SD Portal, VPL queue monitoring, and the escalation ladder. Shadowed Horace on Corporate Actions ticket triage.
- **Week 12** (Jun 1–5): MESO hotfix week. Encountered IISS-538 (ECS autoscaling gap) during routine monitoring — this led to a code fix, not just ticket resolution.
- **Week 20** (Jul 27–31): IISS-876 (VPL Pricing GTL booking error). Coordinated with the business team on NAV impact assessment. Also handled routine queue monitoring and ticket triage.
- **Week 23** (Aug 17–21): Final rotation. Lighter ticket volume; used remaining time for report writing and KT preparation.

10.5.3 Notable Incidents Handled

While most support work involved routine ticket triage and queue monitoring, several incidents required deeper investigation or code-level fixes:

- **IISS-706 (CAIS Deadline):** UCM upload failure with `SMBAuthenticationError: SpnegoError`. Seven-hour investigation traced to Secrets Manager credential rotation leaving empty values cached at module import time. Hotfix added fail-fast validation. Filing completed before regulatory deadline.
- **IISS-538 (MESO ECS):** Tasks stuck, not autoscaling. Root cause: ECS service autoscaling policy

was unconfigured. Fixed the scaling policy and added an ECS `SaturationAlarm` to the IISS-487 alarm set.

- **IISS-741 (MESO Critical)**: INNOCAP and SAMLMOSE funds stuck in queue with no Lambda heartbeat. Manually retriggered the `ia_meso_process` pipeline. Processing completed approximately 2 hours after escalation.
- **IISS-753 (VPL IPV)**: Æxeo pricing window `CheckpointException` in UiPath — bot exhausted its retry budget waiting for a dynamic UI element. Estimated 16 hours for selector resilience fix; escalated to the development team.

10.5.4 Value of Support Rotation for SRE Work

The support rotation was not separate from the SRE observability work — it was *complementary*. Several incidents I encountered during support weeks directly influenced the dashboard and alarm design:

- The MESO ECS incident (IISS-538) led to adding the `ECS_CPUSaturation` alarm to the Grafana dashboard.
- The CAIS credential failure (IISS-706) reinforced the importance of fail-fast validation patterns documented in the Challenges chapter.
- VPL queue monitoring during support shifts provided real-world validation of the SQS dashboard panels and queue depth alarms.

When ticket volume was low, I used remaining support shift time for development work on the dashboard, inventory API, or documentation — maintaining productivity while remaining available for escalations.

10.6 Work Stream 5: RPA LLM Agent — Ask Citco RPA

During Week 21 I engaged with a parallel work stream outside the SRE monitoring scope: the **Ask Citco RPA** support chatbot, implemented across two CodeCommit repositories (`m1-rpa-status-llm-agent` and `core-smart-rpa-llm-ecr-image`). My engagement began as a research and documentation task but expanded to production bug-fixing after log analysis revealed critical reliability failures.

10.6.1 System Architecture

The chatbot is a **Retrieval-Augmented Generation (RAG)** pipeline built on Amazon Bedrock, designed to answer RPA operator queries about process status, procedures, and Legal Entity assignments without requiring developers to be on-call for routine questions.

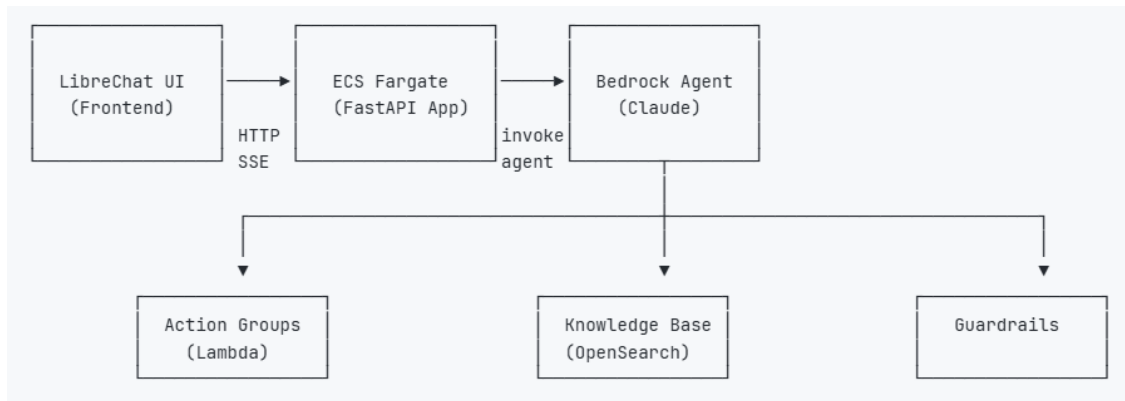


Figure 10.15: Ask Citco RPA — RAG Architecture (IISS-824)

The **ECS Fargate** layer runs the FastAPI container, authenticates users via LDAP (checking `rpa_support_agent_user` AD group), invokes the Bedrock agent, and streams responses back in OpenAI SSE format. The **Bedrock Agent** (Claude) orchestrates between action groups and the Knowledge Base depending on the query type: process status queries hit the Event Service Lambda; procedural questions retrieve chunks from OpenSearch via hybrid vector and keyword search.

10.6.2 Knowledge Base Pipeline

The KB ingestion pipeline runs daily in two stages:

1. **12:00 UTC — IRCOEKBExtractLambda:** Syncs 49 SharePoint IRCOEKB wiki libraries (~5,000 pages) to S3 as plain-text files with SharePoint metadata preserved.
2. **14:00 UTC — RPALLMKBIngestionFunction:** Triggers Bedrock KB ingestion, which chunks (hierarchical: 1,200 token parent / 300 token child), embeds (1024-dim), and indexes into OpenSearch Serverless (FAISS/HNSW, L2 distance).

Static documents (PDFs, process guides) are uploaded separately via an idempotent upload script (`scripts/upload_kb_content.py`) that compares content hashes before writing to S3.

10.6.3 Production Reliability Issues Identified and Fixed

My engagement began with CloudWatch log analysis. Analysis of **93 days of production logs** (May 5 – Aug 5, 2026; 148 Lambda runs) revealed a critical reliability failure in the KB sync:

- **52% timeout rate on IRCOEKBExtractLambda.** 77 of 148 production runs hit the 300-second Lambda timeout. June 2026 was the worst month at 71% timeout rate. Average run duration was 291 seconds — every run within 36 seconds of the limit. When the Lambda timed out, the Knowledge Base was incomplete for that day: pages not yet uploaded when the timeout fired were missing from the KB until the next successful run.
- **Root cause: unconditional full re-upload.** The original code re-uploaded all ~980 pages on every daily run regardless of whether content had changed. The KB Articles library alone (1,266 pages at ~220ms per S3 head_object call) required 278 seconds just for change detection I/O.

I implemented a two-phase hybrid sync (**v1.1.x + v1.1.18**) that reduced steady-state run time from 291 seconds average to approximately **25 seconds**:

1. **Phase 1 — Library-level change detection (~25s for all 49 libraries).** One lightweight ALB call per library using a SharePoint filter=Modified gt {last_synced_at} query. Zero results means the entire library is unchanged and is skipped with no further I/O.
2. **Phase 2 — Conditional sync.** Only changed libraries are processed. Sequential if five or fewer changed; parallel via ThreadPoolExecutor(max_workers=3) if more than five changed.
3. **Per-page idempotency.** Within each synced library, each page's SharePoint Modified timestamp is compared to the sp-modified field stored in S3 object metadata. Unchanged pages skip HTML extraction and S3 upload entirely.

Beyond the KB sync, log analysis also surfaced seven additional production bugs fixed across versions v1.1.1–v1.1.32:

- **OpenSearch silent delete** (v1.1.1): index creation failure silently deleted the index and reported CloudFormation SUCCESS. Fixed with proper exception handling and idempotency check.
- **Fire-and-forget Lambda pattern:** EventStoreExtractLambda and

ClientFundMappingLambda always reported SUCCESS regardless of actual outcome due to incorrect exception class references (`boto3.exceptions.BotoError` does not exist).

- **Athena SSE encryption inconsistency:** queries used SSE_S3 instead of SSE_KMS in some Lambdas.
- **Athena race condition:** MSCK REPAIR TABLE executed immediately after CREATE TABLE without waiting for completion.
- **SSL verification disabled:** BPM API calls used `verify=False`.

10.6.4 Knowledge Base Enrichment: NavChecklist Documents

As part of the RPA LLM Agent work, I uploaded the NavChecklist process documentation PDFs (*Architecture – NAV CHECKLIST.pdf*, *FAQ – NAV CHECKLIST.pdf*) to the KB S3 bucket under RPA Processes/NAV Checklist/ using the idempotent upload script. These documents — describing the NAV calculation checklist process flow — are now indexed in the production Knowledge Base and searchable by the chatbot.

10.6.5 Code Quality and Engineering Improvements

Alongside the bug fixes, I delivered the following engineering quality improvements to bring the codebase in line with IA Best Practice guidelines (pending dev approval gate):

- **Module refactor:** the 1,140-line monolithic `function.py` split into 8 focused modules (`handler.py`, `event_service_api.py`, `secrets.py`, `project_lookup.py`, etc.)
- **structlog migration:** replaced `print()` and root logger across all 11 Lambda functions with structlog structured logging
- **Type hints and docstrings:** PEP 484 type hints and Google-style docstrings added to all functions and classes
- **Test suite:** Comprehensive unit tests at 99% code coverage (v1.1.x series)
- **Pydantic models:** Databricks API response and BPM API response wrapped in Pydantic validation models
- **Kill switch:** SSM Parameter Store kill switch pattern

(/Env/rpa-llm/ircoekb/process-run-status: RUN/STOP) implemented across IRCOEKB, Event Store, and Mapping Lambdas

- **X-Ray tracing:** AWS Lambda Powertools Tracer instrumented on all functions (enabled in DEV; pending approval for UAT/PROD)

Versions v1.1.1 through v1.1.32 were released during Aug 3–20, 2026.

10.6.6 Proposed: Hybrid Graph-RAG Architecture

During the codebase analysis I identified a structural limitation of the current flat-vector RAG approach for the Citco RPA domain. The entities in this domain are highly relational: clients have funds, funds run specific processes, processes execute on either Blue Prism or UiPath, and each platform has its own operational knowledge base. A standard vector similarity search treats all these entities as independent text chunks, losing the relational structure.

I propose a **Hybrid Graph-RAG** enhancement: adding Amazon Neptune as a graph layer alongside the existing OpenSearch vector store. The graph would model the entity relationship network (client–fund–process–platform), enabling the Bedrock agent to traverse relationships for multi-hop queries (*“Which Blue Prism processes handle FX Closeouts for client X?”*), then retrieve dense context from the vector KB for the identified process entities. The dual-path retrieval would improve answer accuracy for complex operational queries while maintaining the existing RAG pipeline for straightforward procedural lookups. This proposal will be submitted as a formal IISS follow-on ticket.

10.7 Work Stream 6: Citco Enterprise Custom UI Dashboards

Work Stream 6 is in active development, extending the existing `ia-config-portal` application with two new dashboard pages. During this co-op placement period, I completed an **architectural implementation plan** (IISS-919-IMPLEMENTATION-PLAN.md) and began initial implementation. Development is continuing during the co-op placement extension period (Sep–Dec 2026).

10.7.1 Planning Document Deliverable

The primary deliverable for this work stream is a comprehensive 865-line implementation plan that documents:

- **Gap analysis:** 45 documented Grafana AMG limitations with severity ratings, identifying which gaps justify a custom UI solution.
- **Technology stack decisions:** Evaluation of charting libraries (Chart.js vs Recharts), service map rendering (Cytoscape.js for X-Ray topology), and component framework integration with Citco's internal @citco/core and @citco/theme packages.
- **Backend API design:** Four new Lambda handlers (/xray/service-map, /xray/traces, /xray/traces/{traceId}, /alarms) with request/response schemas and IAM permission requirements.
- **Task breakdown:** 12 implementation tasks covering backend handlers, workspace scaffolding, component development, and portal integration.

10.7.2 Motivation: Grafana AMG Limitations

The Grafana AMG dashboard built in Work Stream 2 achieved approximately 60% Dynatrace feature parity, but 45 architectural limitations were identified. The most critical gaps driving the custom UI proposal:

Table 10.3: Critical Grafana AMG Limitations Identified

ID	Limitation	Severity
L1	CloudWatch 10 TPS hard throttle — panels error on every load	Critical
L9	X-Ray service map crashes on load — requires Edit mode workaround	Critical
L18	Table filter does not cross-filter other panels	Critical
L21	No topology / honeycomb / Dynatrace-style health view	Critical
L23	No alarm acknowledgement or snooze	Critical
L31	Cannot embed AMG in other portals (iframe blocked)	Medium

10.7.3 Planned Solution Architecture

The planning document proposes two new dashboard pages within the existing ia-config-portal (built by Soundarya Venkataraman and Sridhar Vadla):

- **Infrastructure Inventory Dashboard (IISS-509):** Frontend for the Inventory API built in Work Stream 3, displaying resources across all platforms with filtering and Excel export capabilities.

- **Infrastructure Monitoring Dashboard (IISS-919):** Config-driven monitoring interface with drag-and-drop panel layout, Dynatrace-style honeycomb health visualizations, and X-Ray service map integration.

Key planned components include:

- **CitcoHoneycomb:** D3 hexbin-based health grid where each hexagon represents one resource instance, colored by real-time health status.
- **CitcoServiceMap:** Cytoscape.js topology visualization replacing the crash-prone Grafana X-Ray plugin.
- **CitcoTraceWaterfall:** Custom Recharts component for X-Ray trace span visualization.

10.7.4 Status and Timeline

Table 10.4: Work Stream 6 — Status Summary

Item	Description	Status
Planning Document	865-line implementation plan with gap analysis, tech stack, task breakdown	Complete
Backend Handlers	4 new Lambda API endpoints for X-Ray and Alarms	Planned
Component Library	SRE-specific wrappers on @citco/core	Planned
Inventory Dashboard	IISS-509 frontend for resource inventory	In Progress
Monitoring Dashboard	IISS-919 config-driven monitoring UI	In Progress

Implementation is deferred to the internship extension period (Sep–Dec 2026), allowing the Grafana AMG dashboard to stabilize in production and gather user feedback on which limitations are most impactful to address first.

10.8 Design Decisions & Tradeoffs

Four design decisions during my co-op placement warranted explicit documentation because each involved a meaningful tradeoff between competing engineering constraints. This section records the decision, the alternatives considered, and the rationale for the chosen approach.

10.8.1 Decision 1: Secrets Manager 64 KB Limit → S3 Migration

Problem. The CloudFormation IaC provisioner Lambda function stores its CloudFormation YAML templates as a payload retrieved at deploy time. Initially, this payload was stored in AWS Secrets Manager as a single JSON-encoded string. As the monitoring scope expanded to cover all seven AWS service categories, the generated YAML grew to **14,720 lines** at version v728 — exceeding the **64 KB hard limit** on Secrets Manager secret values. The Lambda function began failing with `SecretSizeExceeded` on the `PutSecretValue` call.

Options Considered.

1. **Split into multiple secrets:** Partition the YAML into n secrets, each under 64 KB, and have the Lambda function reassemble them. This preserves Secrets Manager as the storage layer but introduces assembly logic and a hard dependency on secret ordering.
2. **Compress the payload:** gzip-compress the YAML before storing it as a binary secret. Achieves $\approx 70\%$ compression on YAML, bringing the payload well under 64 KB. Adds a decompression step in the Lambda function.
3. **Migrate to S3:** Store the YAML templates as S3 objects, versioned using S3 object versioning. The Lambda function retrieves templates via `s3:GetObject` at deploy time.

Decision. I chose **S3 migration**. The rationale:

- No size limit — templates can grow arbitrarily as additional AWS service categories are onboarded.
- Native versioning via S3 object versioning, providing a rollback mechanism for template changes at no additional cost.
- No reassembly or decompression complexity in the Lambda function.
- Cost negligible: a single `GetObject` call per deploy adds less than \$0.001 to operational cost.

Tradeoff. The Lambda execution role now requires `s3:GetObject` on the template bucket, broadening its IAM surface slightly. The `GetObject` call adds approximately 50 ms of latency to the provisioner bootstrap — acceptable for a deploy-time operation that runs infrequently.

10.8.2 Decision 2: AppName Tag as Mandatory Secondary Discriminator

Problem. As described in Section 10.4, the citcoworks and meridian environments share BudgetCode="Shared Cloud Artifacts". A single-stage BudgetCode lookup cannot distinguish resources from the two environments, causing inventory contamination.

Root Cause. Both platforms were onboarded before the IA team established the policy requiring each platform to have a unique BudgetCode tag value. Retroactively changing the tag across all existing resources in two production environments would require a coordinated tagging campaign with risk of operational disruption.

Decision. I designated the AppName tag as the **mandatory secondary discriminator** for all resources in the **"Shared Cloud Artifacts"** BudgetCode. Discoverers for citcoworks and meridian must implement the two-stage filter described in Section 10.4.

Tradeoff. This decision creates a tagging governance obligation: all new resources added to either environment *must* carry the correct AppName tag. A resource missing this tag will be invisible to the inventory API, creating a silent coverage gap. This risk is mitigated by a CloudWatch tag compliance alarm and documented in the ia-sre-resources-inventory README as a platform-wide architectural constraint.

10.8.3 Decision 3: CloudWatch as Sole Data Source (No Prometheus)

Problem. A production-grade Grafana deployment typically uses Prometheus as the time-series database, with CloudWatch Exporter bridging AWS metrics. This is the pattern recommended by the Grafana community for large-scale AWS environments.

Alternative. Deploy a self-managed Prometheus server (on ECS or EC2) with the CloudWatch Exporter sidecar, and configure Grafana to use Prometheus as its data source.

Decision. I used the **native Grafana CloudWatch plugin** exclusively.

Rationale.

- **No additional infrastructure:** A Prometheus server and exporter fleet would introduce new ECS tasks, IAM roles, configuration management, and an additional failure domain.

- **Amazon Managed Grafana constraint:** AMG does not support self-managed Prometheus as a built-in data source without a Prometheus-compatible remote write endpoint. Connecting AMG to a self-managed Prometheus would require Amazon Managed Service for Prometheus (AMP), adding further cost and complexity.
- **Direct IAM authentication:** The native CloudWatch plugin authenticates directly via IAM roles, requiring zero credential management beyond the initial role configuration.
- **Cost:** CloudWatch metrics are already being published by AWS services at no marginal cost. Prometheus would require scraping, storage, and retention management.

Tradeoff. The accepted limitation is that CloudWatch Metric Insights SQL does not support p50/p99 percentiles in GROUP BY mode (see Section 10.3). Latency distribution dashboards must use AVG and MAX as proxies, or fall back to the standard metrics wildcard mode which only shows recently active functions. For the current monitoring maturity of the IA platform, this tradeoff is acceptable — the P50/P99 gap can be addressed in a future phase by adding custom Lambda Duration metric emission with explicit percentile dimensions.

10.8.4 Decision 4: Lambda-Backed IaC Provisioner vs. Static CloudFormation

Problem. Provisioning one CloudWatch alarm per Lambda function for all monitored metric types would generate over 100 AWS::CloudWatch::Alarm resources per stack. Approaching the **CloudFormation 500-resource per-stack limit** was a realistic concern as the fleet grew. A static template encoding all alarms explicitly would also require manual updates every time the Lambda fleet changed.

Alternative. Static CloudFormation templates with each alarm resource hardcoded — one YAML block per function per alarm type, updated manually when functions are added or removed.

Decision. I chose a **Lambda-backed dynamic provisioner** that discovers the function fleet at deploy time and synthesizes alarm resources programmatically.

Rationale.

- **Fleet-scale coverage:** The provisioner ensures every Lambda function in every platform has alarms configured, regardless of when it was deployed. No manual update is needed when

a new function is added.

- **Avoids static YAML maintenance:** A static template for 119 functions $\times$ 3 alarm types = 357 resources would be a 500+ line YAML file requiring hand-editing for every fleet change.
- **Parameterization:** Thresholds, evaluation periods, and SNS topic ARNs are all externalised as SAM parameters, enabling multi-environment reuse without code changes.

Tradeoff. The Lambda provisioner introduces pipeline ordering complexity: the provisioner Lambda must complete successfully before the CloudFormation stack update can proceed, creating a sequential dependency. A Lambda execution failure (timeout, permissions error) would halt the deployment pipeline without deploying updated alarms. This is mitigated by comprehensive error logging and a manual re-invocation pathway via `python main.py deploy --phase 3`.

Table 10.5: Design Decision Summary

Decision	Chosen Approach	Key Tradeoff
Template storage	S3 (from Secrets Manager)	+s3:GetObject permission; ~50 ms latency
Tag disambiguation	AppName mandatory secondary key	Tags governance obligation; missing tag = invisible resource
Metrics data source	CloudWatch native plugin only	No p50/p99 in GROUP BY mode; AVG/MAX as proxies
Alarm provisioning	Dynamic Lambda provisioner	Sequential deploy dependency; Lambda failure halts pipeline

Taken together, these four decisions shaped an architecture that favors operational simplicity and zero-maintenance fleet coverage over strict metric precision, while accepting bounded and well-understood tradeoffs at each decision point. I documented the design in the respective repository README and JIRA acceptance criteria so that future team members can revisit any decision as the platform's monitoring maturity evolves.

11. WEEKLY IMPLEMENTATION BREAKDOWN

This chapter documents the work delivered each week across the 24-week co-op placement (Mar 15 – Aug 31, 2026). Weeks 1–4 operated at part-time intensity (~15–20 h/week) while balancing university coursework; full-time engagement (~40 h/week) began in Week 5.

11.1 Week 1: Environment Setup & Access Onboarding (Mar 16–20)

Week 1 marked the formal start of the co-op placement with Day-1 orientation on March 16. I met with manager Kishor Deotale for a comprehensive overview of the team structure, project scope, and co-op placement expectations. The team introduction call connected me with co-interns Sridhar Vadla and Soundarya Venkataraman, as well as senior mentors Horace (Ho Hoi Leung) and Kri (Kristia Marie Labos).

The primary focus was establishing access to Citco’s development infrastructure. I raised access request tickets for Confluence (#14193911), JIRA (#14193674, #14193943), and BitBucket (#14196961). AWS Core Dev access was initiated through the IDM portal. Local development tools were installed: VSCode with Python and AWS Toolkit extensions, Git with SSH key configuration, PyCharm, Postman, and the Amazon Q plugin for AI-assisted coding.

On March 20, I attended a security meeting with Kishor, Ivan Carlo, and Maria covering the BPM Control Questionnaire. This session provided foundational knowledge of Citco’s security posture: Active Directory integration for authorization, Ping authentication (migrated from SiteMinder), penetration testing processes, and access logging infrastructure including ELK stack, Kibana, and Elasticsearch with 7–14 day retention policies. I also learned about Dynatrace application monitoring for anomaly detection and alerting.

11.2 Week 2: AWS Services Deep Dive (Mar 23–26)

With access requests progressing, Week 2 shifted to intensive AWS services learning. I completed a structured study of AWS database offerings: relational options (RDS with Single/Multi-AZ deployments, Aurora auto-scaling, Redshift for OLAP workloads), NoSQL options (DynamoDB, DocumentDB), and specialized databases (Neptune for graph queries, TimeStream for time-series, QLDB for immutable ledgers).

The Neptune study proved directly relevant—the CitcoWorks Knowledge Graph uses Neptune to store L1–L5 hierarchy relationships. I learned both Gremlin (property graph traversal) and SPARQL (RDF pattern matching) query languages, understanding when each is appropriate: Gremlin for application-level traversal, SPARQL for semantic data integration.

CloudWatch and CloudFormation consumed the remainder of Week 2. I studied the CloudWatch CMAA cycle (Collect, Monitor, Act, Analyze), metric types (EC2 1-minute detailed vs 5-minute basic), log retention policies, alarm configuration (metric alarms, composite alarms), and EventBridge integration. CloudFormation IaC fundamentals covered templates (YAML/JSON), stacks, and change sets—knowledge immediately applied in Week 3’s dashboard work.

11.3 Week 3: CloudWatch Pipeline Dashboard (Mar 30 – Apr 1)

Week 3 transitioned from learning to delivery with the first JIRA ticket: IISS-435, a CloudWatch pipeline health dashboard for the `rpacfs1-cais-pricing-extract-genai-infrastructure` repository.

On March 30, I added CloudWatch metrics instrumentation to track CTI (CAIS Pricing Extract) submission activities within the pipeline. The Pipeline Health Dashboard was created to provide operational visibility into system performance. I refactored the metrics code into a dedicated module for maintainability and added comprehensive unit tests to prevent regressions. Repository hygiene tasks included updating `.gitignore` to exclude virtual environments, pytest cache, and test artifacts.

March 31 focused on the technical dashboard with a CloudWatch Logs Insights widget for deeper system analysis. This widget enabled error pattern extraction from Lambda execution logs, providing root cause analysis capabilities that the basic metrics dashboard lacked. I configured appropriate time ranges and query syntax for CTI pipeline logs.

April 1 completed the testing phase: unit tests for Logs Insights query functionality, CFN template validation tests, and a full test suite run. Code quality improvements included moving imports to file tops following Python best practices and adding type hints for IDE support.

11.4 Week 4: Lambda Monitoring Dashboard & Documentation (Apr 6–8)

In Week 4, the focus was on completing the IA Lambda Monitoring Dashboard, addressing PR review feedback, and delivering structured Lambda inventory documentation on Confluence.

Merging IISS-435 and Pivoting to IISS-455:

The week opened on April 6 with the successful merge of IISS-435 (CTI CloudWatch Dashboard) into the release/SRE branch, closing the first JIRA ticket of the co-op placement. I immediately transitioned to IISS-455, which required building a monitoring solution covering all 51 Lambda functions across the rpacfs1 and ia-bpo platforms. Unlike the CTI dashboard, which targeted a single pipeline, IISS-455 needed to serve the entire IA Lambda fleet with a single unified view.

Dashboard Architecture and Widget Design:

I designed the dashboard using a modular YAML widget architecture where each widget type lives in its own file, allowing independent updates without touching surrounding logic. Widgets covered five monitoring domains: filter info with inline usage instructions, overview metrics (Total Invocations, Errors, Throttles across all functions), error analysis (bar charts and a CloudWatch Logs Insights investigation widget), duration metrics (Average, Max, and P99 latency), and operational health metrics (cold starts, concurrency, throttling, and memory utilization). The unit test suite reached 21 tests covering dashboard builder validation and widget contract enforcement.

PR Review and Refinement:

On April 8, Horace conducted a detailed PR review. Feedback focused on widget layout usability and filter propagation behavior. I addressed all comments, refined the widget grid, and added test cases for edge cases identified during the review. This review interaction introduced me to the team's standard for dashboard quality: functional correctness was necessary but not sufficient—the operational user experience mattered equally.

IISS-461 Confluence Lambda Inventory:

In parallel with the PR review, I created IISS-461 as a subtask to document the Lambda inventory on Confluence. The published page cataloged all 51 functions with trigger mappings (S3, SQS, EventBridge, API Gateway), memory configurations, timeout settings, runtime versions, and budget code allocations for cost tracking. This page became the authoritative reference for the team during subsequent dashboard and support work.

By the end of Week 4, IISS-435 was production-merged, IISS-455 delivered a 51-function Lambda

dashboard with a passing test suite, and the Lambda inventory documentation was live on Confluence—completing the part-time phase and setting the stage for full-time engagement.

11.5 Week 5: Training Week & Meridian Incident (Apr 13–17)

Week 5 marked the beginning of full-time engagement and was structured primarily as a formal knowledge-transfer week, with four dedicated sessions covering the platforms I would be monitoring and supporting throughout the co-op placement. It also included a critical production incident that would shape the entire direction of IISS-487.

RPA Support Model Knowledge Transfer (Horace):

Horace (Ho Hoi Leung) conducted a comprehensive session on the RPA operational model. I learned the Blue Prism and UiPath platform architectures, pool and queue mechanics, and the VPL queue monitoring cadence (checked at 5:00 PM EST daily). The session covered the Corporate Actions three-track workflow (Mandatory, Voluntary, and Ops Transfer), and the Follow-the-Sun escalation ladder that governs how incidents are handed between APAC, EMEA, and NA teams. This session directly prepared me for the RPA support rotations beginning in Week 8.

CitcoWorks and Event Store (Bhanu Teja Murari):

Bhanu introduced the CitcoWorks Knowledge Graph—a Neptune graph database storing the full L1-to-L5 entity hierarchy used across all Citco fund services. He walked through the Event Routing Service and Action Processor deployment model, and provided hands-on guidance on the Event Store structured logging library. Understanding the mandatory fields (subject, process_identifier, run_id) proved essential when validating log events in later dashboard and LLM agent work.

Meridian and Æxeo Platform Sessions:

The Meridian platform session covered the three-stage Kafka event pipeline (Event Routing Service to MSK topics, MSK to OpenSearch Consumer, OpenSearch to AXI), the six ECS clusters hosting Meridian services, and consumer lag as the primary health signal. The Æxeo session explained cloud-based wire approval and fund movement via COMAPI and the Æxeo Reporter Server REST API—workloads central to MESO and E-Binder reconciliation processes.

Dev Work and Meridian Production Incident (Apr 13 and Apr 17):

Despite the training schedule, I completed development on April 13 by refactoring the Lambda monitoring dashboard to a multi-dashboard architecture with explicit Lambda list filtering and

adding error investigation capabilities. On April 17 (Friday afternoon), the Meridian Incident occurred while I was completing the final training session: a surge of automated events from Citco Recs congested the OpenSearch consumption layer, causing approximately nine million Kafka messages to queue and halting daily statement processing for 7.5 hours. The detection latency alone was 3.5 hours because no MSK consumer lag alarm existed. Observing the post-incident discussion in real time crystallized the monitoring gap I would spend the next ten weeks closing. IISS-469 was completed this week and IISS-487 scope was formally defined.

By the end of Week 5, I had completed the formal KT programme, delivered a multi-dashboard architecture refactor, and directly witnessed the production incident that established `EstimatedMaxTimeLag` as the primary MSK SLI and drove the creation of IISS-487.

11.6 Week 6: Platform Extension & Grafana AMG Story Creation (Apr 20–24)

In Week 6, the focus was on extending the CloudWatch monitoring platform beyond Lambda to EC2, ECS, and EKS resources, creating the centralized `ia-resources-monitoring` repository, and formally creating the IISS-487 Grafana AMG story.

IISS-469 Dashboard Extension and API Refactoring:

The week opened with IISS-469 work: adding a metric filters generator, implementing dashboard export functionality with per-widget CSV files and a combined XLSX report, and converting the Lambda registry from a FastAPI service to a Flask application with Swagger documentation. I renamed `lambda_registry_api` to `resources_monitoring_api` to reflect its broader scope and added an S3 export endpoint with versioned Lambda list saving. Trigger source detection was added, identifying whether each Lambda function is invoked by SQS, API Gateway, or EventBridge. The test suite reached 262 passing tests with a clean pyright status.

EC2, ECS, and EKS Resource Discovery (IISS-482, IISS-483):

I extended the API to cover EC2 instances, ECS services, and EKS clusters. During EC2 discovery I identified the `rpacfs1` tagging inconsistency—the fleet uses `BudgetCode="CFS/RPA"` rather than the expected `"rpacfs1"` value—and introduced the `AppName` fallback pattern as the standard workaround. EC2 response models were enriched with `availability_zone` and `tags` fields. The registry package was refactored into per-resource sub-modules for long-term maintainability.

ia-resources-monitoring Repository Creation (IISS-455):

On April 24—the most commit-dense day of the co-op placement with 39 commits across two repositories—I set up the centralized `ia-resources-monitoring` repository. All three monitoring packages (Lambda, EC2, and the API) were migrated from `rpacfs1-cais-pricing-extract-genai-infrastructure` into this dedicated repository. Code quality tooling was established: Ruff linting, mypy type checking, deptry dependency hygiene, and pre-commit hooks. The `ia_paths.py` module was introduced to eliminate all `sys.path.insert` calls and a unified `main.py` CLI entry point provided access to all deployment and dashboard generation commands. The `release/SRE` branch was merged to master, making this the first production deployment from the new repository.

IISS-487 Story Creation:

Following the Week 5 Meridian Incident and the emerging need for a platform-wide observability solution, I met with Kishor to formally scope and create IISS-487—the Centralized Grafana AMG Infrastructure Monitoring Dashboard. The story was created with requirements drafted to cover all Meridian-platform AWS services and address the MSK consumer lag gap that had caused the incident.

By the end of Week 6, the CloudWatch monitoring platform covered four resource types, the centralized repository was established and production-deployed, and IISS-487 was formally created with an agreed scope and priority.

Technical Challenges Resolved:

Two infrastructure issues were resolved during the migration day. The `moto[all]` dependency triggered SSL certificate download failures on Windows; switching to specific service extras (`moto[s3,lambda,ec2]`) resolved this. CloudFormation YAML files using `!Sub` and `!Ref` tags failed the `check-yaml` pre-commit hook; CFN files were excluded from that hook while retaining validation via `cfn-lint`. Both resolutions were documented in the `CONTRIBUTING.md` for future contributors.

11.7 Week 7: CloudWatch Platform Completion (Apr 27 – May 1)

Week 7 was dedicated to completing the CloudWatch monitoring platform by delivering ECS and EKS dashboards, implementing structured logging across all packages, and pushing the overall test coverage above the 80% target.

ECS Monitoring Dashboard (IISS-484):

The `ia_ecs_monitoring_dashboard` package was built from scratch on April 27, including health and performance dashboards, YAML widget definitions, CloudFormation templates, export functionality, and deploy support scripts. A key technical decision was replacing SQL arithmetic operations in health widgets with native `PendingTaskCount` metrics after discovering that CloudWatch Metrics Insights does not support arithmetic in certain widget modes. On April 30, CPU and Memory reservation widgets were added alongside a service inventory table, giving operational teams both capacity and inventory views in a single dashboard.

EKS Monitoring Dashboard (IISS-485):

The EKS dashboard was completed on April 28, then substantially revised on May 1 to switch from custom-namespace queries to the official AWS/EKS control plane metrics. The EKS discovery scope was subsequently narrowed: only one active EKS cluster (`uipath-eks-cluster-01`, AUTOMATIONHUB budget code) exists and it was idle at the time. The cluster was noted for future activation and the dashboard placed in a ready-but-deferred state. EKS widget YAMLs were split to one file per row for consistency with the ECS structure, and test coverage for the EKS package reached 81%.

Structured Logging and Test Coverage Milestone (IISS-482):

On May 1 I replaced all `print` statements and `disable_warnings` calls across the API, EKS, ECS, EC2, and Lambda packages with `structlog` structured logging, providing consistent log output across all components. The test suite was then pushed from 67% to 81% coverage (830 tests) in three incremental PRs, crossing the 80% target committed in the ISS-482 acceptance criteria.

Dashboard Header Standardisation and EC2 Polish (IISS-483):

Dashboard headers were standardized across all four resource types, enforcing a consistent visual language. EC2 dashboard improvements included Fleet Status, EBS metrics, the `AppName` fallback for discovery, correct header height, Credit Balance widget on the health dashboard, and annotations with AVG line on the performance dashboard. Trailing spaces were added to markdown headers to force CloudWatch's text widget line-break rendering.

By the end of Week 7, the CloudWatch monitoring platform was complete across Lambda, EC2, ECS, and EKS; 830 tests passed at 81% coverage; structured logging was consistent across all packages; and the repository was production-merged for the second time, closing the ISS-482–486 family of JIRAs.

Technical Decisions and Constraints:

A key CloudWatch Metrics Insights constraint was confirmed during ECS duration panel work: the GROUP BY aggregation mode supports only AVG, SUM, MIN, and MAX — percentile functions (p50, p99) are not available in SQL mode. The Metrics Insights query parser also required cluster names containing hyphens to be substituted *before* JSON serialisation; applying substitution post-serialisation resulted in double-escaped output. Both constraints were documented in a technical decisions log to prevent re-investigation in later weeks. Additionally, CloudWatch Logs Insights `put_query_definition` enforces a hard limit of 20 saved queries per account, which constrained the saved-query strategy to a Intelligent-Automation folder with errors-only scope.

11.8 Week 8: Support Rotation #1 & Grafana AMG Repository Scaffold (May 4–8)

In Week 8, the focus shifted to the first RPA production support rotation while simultaneously scaffolding the Grafana AMG repository and completing final CloudWatch dashboard polish.

RPA Production Support Rotation (May 4–8):

The first support rotation ran Monday to Friday, covering the North America shift from 11:00 AM to 7:00 PM EST. Each morning I reviewed the Confluence shift handover log from the APAC team, then monitored the SD Portal queue and Microsoft Teams channels for incoming incidents. Tickets handled across the week included: MESO process status inquiries (traced via CloudWatch logs using Run ID), FX Closeout retrigger requests (verifying the Daily565 folder and archiving old reports), Corporate Actions file mask updates for new client onboarding, an Auto Rec Download configuration request (backing up and editing `gridview.py` and `config.ini`), and a VPL process exception traced through UiPath Orchestrator transaction details. End-of-shift handover notes were posted to Confluence each evening. Following the Week 5 training and this first live rotation, the RPA support model became familiar and manageable alongside development work.

AUTOMATIONHUB Budget Code and Dynamic Tag Discovery (IISS-504, IISS-508):

During quieter periods in the support queue I completed two CloudWatch platform JIRAs. IISS-504 added the AUTOMATIONHUB budget code across all discovery packages, enabling dashboards to surface UiPath-platform resources alongside the existing Meridian and IA-BPO resources. IISS-508 followed with dynamic tag discovery using `budget_code` filtering, BudgetCode-first discovery with AppName fallback, citcoworks budget code support, multi-region and multi-account query parameters, and cluster-based ECS fallback. These changes made the resource API significantly more resilient to infrastructure naming inconsistencies across the five Citco AWS accounts.

Grafana AMG Repository Scaffold and Agentic-AI Architecture:

On May 6, following Kishor's decision to elevate IISS-487 to **Critical** priority in the post-incident review, I created and scaffolded the `ia-it-sre-infrastructure-monitoring-amg` repository. The initial commit established a multi-agent project structure: enforced LF line endings via `.gitattributes`, pre-commit hooks with code quality tooling, and a `structlog` structured logging and agent audit trail. On May 7, agent skill definitions and role scaffolds were implemented for the multi-agent orchestration layer. On May 8, the Python version was upgraded from 3.11 to 3.14, documenting GIL and free-thread status for the new runtime.

By the end of Week 8, the first support rotation was complete, IISS-504 and IISS-508 closed the CloudWatch platform feature set, and the IISS-487 repository was scaffolded and ready for the full infrastructure build beginning in Week 9.

Key Architectural Decision:

The discovery of the CITCOWORKS and MERIDIAN platforms sharing the **"Shared Cloud Artifacts"** BudgetCode was the most significant architectural finding of the week. A naive single-stage tag filter would conflate operationally unrelated resources from two distinct platforms. The AppName discriminator was established as a mandatory second filter stage for these two platforms, a pattern that became the standard two-stage discovery architecture used throughout the inventory API in subsequent months. Additionally, five of the six Meridian ECS clusters had no resource tags at all, requiring a cluster-name-based fallback strategy to ensure complete fleet coverage.

11.9 Week 9: Infrastructure Deployment Day (May 11–15)

Week 9 was the most infrastructure-dense week of the co-op placement, involving the full deployment of the Grafana AMG stack, the first live CloudWatch alarms, a forced Grafana version migration, and the construction of the first three dashboard rows.

May 11 Infrastructure Deployment:

On Monday May 11 I deployed the full IISS-487 infrastructure in a single sustained session spanning 49 commits from 9:59 AM to 9:05 PM. The work began by enforcing coding standards (uv, JavaDoc-style docstrings, strict type hints) and implementing the Human-In-the-Loop gate with trigger rules and audit trail for agent oversight. I then built the CloudFormation templates in sequence: Grafana AMG workspace with IAM role, Lambda custom resource for Grafana API provisioning, the Grafana dashboard JSON covering all ten rows, a CloudWatch alarms template

with all 17 alerting rules, and the multi-environment CodePipeline with ChangeSet deployment and EventBridge trigger. A dedicated Grafana CFN execution role was created to resolve IAM bootstrap permission issues, and the `IamPermissionResolver` agent was added to automatically diagnose and fix permission errors in future deployments. The day concluded by importing the Meridian dashboard via the Lambda provisioner and storing its JSON in Secrets Manager. With this deployment complete, the MSK consumer lag alarm was live against production traffic — the alarm that would have detected the April Meridian Incident was now operational.

Documentation Sprint and 100% Test Coverage (May 12):

May 12 delivered the complete documentation suite: release notes, installation guide, architecture document (PNG diagrams replacing Mermaid blocks for CodeCommit rendering compatibility), process flow document, contribution guide, permission fixes guide, and a detailed user guide for all ten dashboard rows. The test suite simultaneously reached 99% coverage on the agentic-AI subsystem, then 100% coverage on `trigger_rules` and `orchestrator`—a milestone that provided confidence in the agent dispatch layer.

SQS Panel Fixes and Grafana 12.4 Upgrade (May 13–14):

On May 13, the SQS row panels (Queue Depth, Messages In-Flight, Age of Oldest Message, DLQ Depth) were fixed with per-queue breakdowns and `QueueName` legends, achieving 100% test coverage on the Row 2 SQS panel tests. An unplanned Grafana 10.4 to 12.4 upgrade was then required after the AMG workspace was automatically updated by Amazon. The breaking changes in the Metric Insights SQL editor and template variable syntax consumed approximately one full working day on May 13–14, migrating all existing panels to the stable 12.4 query model and updating the schema version to 41. May 14 additionally delivered the full 23-agent architecture (including `cloudwatch_alarms_expert`, `amg_deployment_expert`, and `amg_observability_expert`), resolved the Grafana API key bootstrap chicken-and-egg issue, and implemented the SQS Health gauge with composite critical logic.

Lambda Row and ECS Row Completion (May 15):

On Friday May 15 I overhauled the Lambda monitoring row: per-function breakdown for all panels using the Metric Insights SQL `GROUP BY` format, split Duration into p50 and p99 panels, paired Throttles with Concurrency, and introduced the `$function` variable for interactive function-level drill-down. The `lda-` naming prefix was applied to Lambda-specific resources and the ECS Container Insights Lambda was removed after confirming native CloudWatch Container Insights was sufficient.

By the end of Week 9, the full Grafana AMG infrastructure was deployed and live, the MSK consumer lag alarm was operational, dashboard Rows 1–3 were built, and the 23-agent architecture was established with 100% test coverage across the core orchestration layer.

Grafana API Bootstrap and SQS Thresholds:

An important infrastructure challenge was the Grafana API key chicken-and-egg problem: the provisioner Lambda needed an API key to provision resources, but the key had to be created in Grafana which did not yet exist. A hotfix series resolved this by bootstrapping the key separately as a one-time setup step before the full deploy. The SQS Health gauge composite logic was configured with precise thresholds reflecting the Meridian queue operating range: 0–34,999 messages (OK, green), 35,000–49,999 (WARNING, amber), and 50,000+ (CRITICAL, red). These thresholds were derived from the April Meridian Incident peak queue depth of approximately nine million messages, setting the warning level well below the incident threshold while avoiding false positives during normal processing. A three-hour knowledge transfer workshop with Kishor, Nathaniel, and two additional team members (May 13, 188 minutes) provided deep context on the Meridian operational model that informed dashboard panel design decisions for the remainder of the co-op placement.

11.10 Week 10: Dashboard Build Intensive (May 18–22)

In Week 10, the focus was on building the remaining middle rows of the Grafana dashboard, resolving CloudWatch Metrics Insights SQL limitations, and migrating dashboard storage from Secrets Manager to S3 to overcome payload size constraints.

ECS, MSK, DocumentDB, Neptune, and OpenSearch Rows (May 19–20):

The week opened with Row 5 (ECS APIs, ALB metrics) and Row 6 (MSK/Kafka) on May 19. Row 5 required a multi-iteration approach to the `$platform` variable in `filterFieldsByName`: ALB resource names do not contain the platform string, requiring a split between Row 4 (ECS container health) and Row 5 (ALB traffic metrics). Row 6 MSK was migrated to Metric Insights SQL with per-broker and per-consumer-group breakdown, and a meridian cluster filter was added. On May 20, Rows 7–9 (DocumentDB, Neptune, and OpenSearch) were completed with SQL migration and per-instance breakdowns. A new Aurora/RDS Row 8 was added and the existing rows renumbered to accommodate it. The SQS Inventory table panel with clickable AWS Console links was introduced on May 20 to reduce incident response navigation time.

Metric Insights SQL Percentile Limitation:

During Lambda and ECS duration panel work, I confirmed a fundamental Metric Insights constraint: GROUP BY mode supports only AVG, SUM, MIN, and MAX aggregation functions—p50 and p99 percentiles are not available via SQL. I implemented a wildcard dimension workaround (FunctionName: ["*"]) for Lambda duration percentile panels, accepting the tradeoff that idle functions would appear in the list. This constraint was documented as a known limitation to prevent future re-investigation.

Secrets Manager 64 KB Limit and S3 Migration (May 25):

By late Week 10 the CloudFormation provisioner payload grew to approximately 14,720 lines as additional alarm templates were added, triggering SecretSizeExceededException from Secrets Manager. I migrated all dashboard JSON and alarm templates to S3 object storage with AES256 server-side encryption to comply with the CITCO-S3ForceEncryptNoPublic bucket policy. The provisioner Lambda was updated to fetch content via s3:GetObject at deploy time, removing the 64 KB ceiling entirely.

Agentic AI System Refactor and 597-Test Milestone (May 22):

The agentic AI subsystem underwent a major refactor on May 22: agents grouped into five capability tiers (infra, grafana, development, workflow, safety), the aws_config_troubleshooter role added, and all subprocess-dependent tests fixed to prevent hanging in CI. The test suite reached **597 tests at 100% coverage**—the largest single-day test milestone of the co-op placement. ProvisionerVersion was bumped to 355 after two sequential deploys validating the refactored architecture.

By the end of Week 10, Rows 1–9 of the dashboard were built and deployed, the Secrets Manager storage constraint was resolved via S3 migration, and the percentile limitation was documented.

Dashboard Variable Architecture:

The \$platform variable was implemented as a cross-row filter using filterFieldsByName with a pipe-formatted multi-value pattern. This allowed a single variable selection to filter data across 13 rows simultaneously, reducing the operational burden of context-switching between platform-specific views. The SQS inventory table was introduced with clickable AWS Console links using Grafana data links, enabling one-click navigation from queue depth panels to the SQS console page for that specific queue. The Aurora/RDS row was added on May 20 as an additional row, requiring all subsequent row numbers to be incremented by one — the renumbering.md migration document established a convention for future row additions.

11.11 Week 11: Database Monitoring Rows & Alarm Infrastructure (May 25–29)

Week 11 was dedicated to completing the database monitoring rows, delivering the alarm and notification infrastructure, and bringing the dashboard to a state ready for stakeholder review.

Health Panel Completion for All Database Services (May 25):

On Monday May 25, health panels were added for DocumentDB (CPU utilization with 60/80% thresholds), Neptune DB (same threshold structure), RDS/Aurora (CPU thresholds with row reordering), and OpenSearch (Green/Degraded status derived from `ClusterStatus.green` metric). The OpenSearch health panel was particularly significant: OpenSearch was the bottleneck service in the April Meridian Incident, and having a dedicated health indicator for it addressed the primary observability gap identified in the post-mortem. ECS metrics were enhanced to show CPU and Memory usage MAX per service ordered by `MAX() DESC`, making the busiest services immediately visible at the top of each panel.

Redis/Valkey Row and Dashboard Visual Polish (May 26):

The Redis/Valkey row was completed on May 26 with 11 panels covering the four golden signals (latency, traffic, errors, saturation), fully compliant with the AMG 1-query-per-panel constraint. Row files were renumbered to sequential order `row_01` through `row_13` for consistency. The CLI was refactored to replace the `--dry-run/--validate-only` flags with a unified `--validate` flag and a new `--smoke-test` option. Visual polish across the dashboard included Retro LCD bar gauges, consistent series colors across all timeseries panels, and `renameByRegex` transformations for clean legend names. MSK Row 6 was fixed to replace Bytes In with Messages Out per second for more meaningful throughput visibility.

Datasource Analysis and CloudWatch Logs Integration (May 26–27):

I completed a datasource analysis to determine the full set of Grafana datasources required, verifying the conclusions via Amazon Q. CloudWatch Logs and OpenSearch were added as new datasources alongside the existing CloudWatch Metrics datasource. This enabled Log-derived panels to be added to the dashboard in subsequent weeks.

Alarm and Notification Infrastructure (May 27–29):

The alarms and notifications layer was delivered across three days. On May 27, I created a comprehensive alarms and notifications guide, performed an alerts gap analysis documenting existing versus required coverage, parameterized the `Platform` variable in CloudWatch alarm names, added a `python main.py alarms` CLI command for SNS subscription management,

and secured both SNS topics with `KmsMasterKeyId alias/aws/sns` encryption. On May 28, I implemented EventBridge + Lambda integration (Option A) for real-time CloudWatch alarm state in Grafana, replacing a polling approach after evaluating the two options against latency and cost criteria. On May 29, an SNS Re-Notifier Lambda was deployed for ongoing alarm state reminders, composite alarms were configured to aggregate per-service states into platform-level OK/ALARM indicators, and the alarm state history LCD panel was built with severity-based colors spanning the full dashboard time range.

By the end of Week 11, all 13 dashboard rows were built and deployed, the OpenSearch JVM heap pressure alarm was operational, the SNS Re-Notifier Lambda was live, and the dashboard was ready for the Week 14 stakeholder review and final sign-off.

Alarm Architecture Decision (EventBridge vs Scheduled Lambda):

On May 28, I evaluated two approaches for real-time alarm state delivery to Grafana panels: Option A (EventBridge + Lambda) and Option B (Scheduled Lambda polling at 1-minute intervals). Option A was chosen for its event-driven nature—alarm state changes are delivered within seconds of the CloudWatch alarm transition, versus up to 60 seconds of lag with the polling approach. The cost difference was negligible: EventBridge charges per rule invocation (~\$0.01/1M events) versus Lambda invocations at 1-minute intervals (~\$0.02/month). The EventBridge approach also aligned with the existing CodePipeline trigger pattern already used in the repository. Both SNS topics were secured with KMS encryption using `alias/aws/sns` to comply with Citco encryption-at-rest requirements. The May 26 first major stakeholder review confirmed the dashboard visual language and row organization were acceptable to the team before the final build sprint began.

11.12 Week 12: Support Rotation #2 & Alarm Tuning (Jun 1–5)

In Week 12, the focus was the second RPA production support rotation alongside finalising the IISS-487 alarm infrastructure and completing EC2 dashboard rows.

RPA Production Support Rotation (Jun 1–5):

The second rotation followed the same North America shift pattern (11:00 AM–7:00 PM EST). Tickets handled across the week included: MESO process status inquiries traced through CloudWatch logs using Run IDs, FX Closeout retrigger coordination (verifying the Daily565 folder, archiving stale reports before triggering the UiPath job), the AVCAB date format exception (investigated and confirmed an incorrect input date supplied by the requester, who was asked to

retrigger after correction), and an Auto Rec Download CAPULA2 configuration update (backed up `gridview.py`, added the ASPECT client configuration block). Horace conducted knowledge transfer sessions spanning 35 minutes on Monday, 27 minutes on Tuesday, and 88 minutes on Friday, covering RPA support procedures and post-incident RCA walkthroughs. Stale tickets exceeding 14 days without end-user response were closed per the IA Production Support guide. End-of-shift handover notes were posted to Confluence each evening.

IISS-487 Alarm Tuning and CloudFormation Improvements:

During quieter support periods, CloudWatch alarm thresholds were calibrated against actual DEV baseline metrics for DocumentDB and Neptune: CPU utilization 60%/80% thresholds were confirmed correct. The CloudFormation deploy script was improved with `--preview` mode and `changeset` status reporting. All 14 dashboard rows and 17 CloudWatch alarms were validated in DEV, and no-op `changeset` detection logic was confirmed working correctly to prevent false redeployes.

EC2 Dashboard Rows (IISS-483):

EC2 monitoring Rows 7–8 were completed with the `AppName` fallback pattern applied across all 83 instances. Fleet Status, EBS metric corrections (switching from `DiskReadOps` to the correct `EBSReadOps` namespace), and per-instance drill-down panels were delivered. Test coverage for the EC2 dashboard package reached 96%.

By the end of Week 12, the second support rotation was complete, IISS-487 alarm infrastructure was validated, and EC2 dashboard rows were merged to release.

11.13 Week 13: CAIS Hotfix & Per-Panel Folder Refactor (Jun 8–12)

Week 13 was the most incident-dense week of the co-op placement, beginning with an emergency hotfix for a regulatory deadline and continuing with significant IISS-487 structural improvements.

IISS-706: CAIS Deadline SpnegoError Hotfix:

The week opened with a critical failure in the CAIS Deadline Lambda — `SMBAuthenticationError: SpnegoError` during UCM upload, blocking a regulatory CAIS filing. A seven-hour investigation on June 8 identified the root cause: `constants.py` was caching credentials at module import time, meaning a Secrets Manager rotation that had left empty values during container startup caused all subsequent authentication attempts to fail silently with the pre-loaded empty credential. Three targeted fixes were applied: fail-fast UCM

credential validation at startup and before each SMB connection attempt; a guard preventing empty `deadline_ids` from generating an invalid SQL `IN()` clause; and a zip file existence check before attempting the UCM upload. Comprehensive unit tests were added for all three fixes, including mock patches for `trigger_process` and `email_function` that had been preventing test isolation. The CAIS filing completed before the regulatory deadline and a full RCA was documented on Confluence.

IISS-487: Per-Panel Folder Refactor and SQS Dashboard:

Following the hotfix, I completed a structural refactor of the IISS-487 repository: 110+ dashboard panels were extracted into per-panel YAML folders under `rows/row_02/` through `row_14/`, enabling independent panel updates without touching surrounding files. An EC2 mapper Lambda was deployed with EventBridge trigger, and the SQS monitoring dashboards went live (**960 tests passing**). The IISS-482 SQS feature was formally closed.

IISS-507: Inventory API Scaffold:

On June 11, a Kiro spec-driven workflow was used to scaffold the `ia-it-sre-infrastructure-inventory` repository in 2.5 hours: Flask async API structure, discoverer registry pattern, Pydantic response models, and alignment of `service_type` names with the dashboard rows.

By the end of Week 13, the CAIS regulatory filing was unblocked, IISS-487 reached 357 tests on the refactored architecture, and the inventory API foundation was established.

CloudFormation Template Size Constraint:

A second infrastructure size constraint was encountered on June 10 when the full CloudFormation alarm template could not be deployed inline — the 51,200-byte inline template limit was exceeded. Rather than splitting into nested stacks, the decision was made to upload templates to S3 and use `TemplateURL` references, keeping the deployment model flat and avoiding the CFN nested stack dependency chain that had caused rollback issues in earlier test environments. The `s3:ListBucket` permission was required alongside `s3:GetObject` to produce meaningful error messages on `NoSuchKey` failures during template retrieval.

11.14 Week 14: Final Dashboard Deployment v728 (Jun 15–19)

Week 14 delivered the final Grafana AMG dashboard deployment and formally closed IISS-487.

Deep-Link Panels for All Service Rows:

Deep-link panels were added to all nine monitored service rows (Lambda, SQS, ECS, MSK, DocumentDB, Neptune, RDS, Redis, OpenSearch), connecting each resource panel directly to the relevant AWS Console page. This addressed a recurring friction point observed during the Week 8 and Week 12 support rotations: incident responders were manually navigating to the console to cross-reference alarm states with resource metrics. Fixing this reduced context-switching during on-call investigation.

Row Restructuring and Visual Consistency:

DocumentDB, Redis, and OpenSearch rows were restructured to a consistent 4+4 (w=6) layout, standardizing panel widths and eliminating asymmetric gaps. ECS bargauge legend names and deep-link URL formats were corrected. An EBS metric correction replaced the incorrect `DiskReadOps` dimension with `EBSReadOps` from the AWS/EBS namespace. Grafana transformation ordering for the alarm state history panel was documented to prevent future regression. Ten new dashboard template variables were added completing row-level interactive filtering across all nine services: `lambda_function`, `sqs_queue`, `ecs_service`, `ecs_target_group`, `msk_consumer_group`, `docdb_instance`, `rds_instance`, `redis_cluster`, `neptune_instance`, and `opensearch_domain`. The `ProvisionerVersion` was incremented from 698 to 726 across five sequential deploys during the June 17 restructuring session alone, reflecting the iterative nature of the final polish work.

Dashboard v728 Merge and IISS-487 Closure:

On June 18, the dashboard was finalized as **version v728**. All four acceptance criteria were verified: MSK consumer lag alarm live against production traffic, all 14 rows rendering correctly in the AMG workspace, deep-links functional for all nine service types, and alert emails delivering to the configured SNS distribution list. The release PR was merged to main and IISS-487 was closed — the largest single JIRA of the co-op placement.

11.15 Week 15: Post-Release Improvements & Resource Mappers (Jun 22–26)

Following the IISS-487 closure, Week 15 focused on post-release improvements to the MSK row and the delivery of six new resource mapper Lambdas.

MSK Consumer Lag Refinements:

MSK consumer lag panels were enhanced with per-topic breakdown (previously only

per-consumer-group), and a GROUP BY ClusterName + ConsumerGroup + Topic triple-axis structure was implemented to enable topic-level drill-down. The MSK cluster ARN chain regex was corrected from Python to JavaScript syntax after discovering it was preventing the `$msk_cluster_arn` variable from resolving in Grafana's template engine. MSK performance panel topic deep-link URL format was fixed to use the correct AWS Console path structure, and stable panel IDs were enforced to prevent deep-link breakage across dashboard reddeploys.

Six Resource Mapper Lambdas:

Six new mapper Lambda functions were deployed to the resource-mappers CloudFormation stack, each responsible for discovering and caching one resource class: ECS target groups, DocumentDB clusters, RDS instances, Redis cache nodes, Neptune clusters, and OpenSearch domains. These mappers replaced the previous SQS Queue Mapper's names-pattern approach with a tag-based discovery strategy, enabling the Grafana `$variable` dropdowns to be populated dynamically from live AWS inventory rather than from static configuration lists.

Knowledge Transfer Sessions (Horace):

Horace provided a 60-minute session on the FX Closeout process exception handling procedure, covering how to identify settled-vs-live position mismatches and the correct retrigger sequence. A follow-up session on FX Settled exception handling was held the following day. Both sessions were documented in Confluence for the next support rotation team member.

By the end of Week 15, the MSK row was production-quality, six mapper Lambdas were live and populating dashboard variables, and support procedures for FX Closeout were formally documented.

SQS Platform Coverage Resolution:

The SQS case-insensitive regex fix on June 22 resolved a critical coverage gap: the `ia-bpo` platform was returning only 2 of its 55 queues due to an uppercase/lowercase mismatch in the queue name pattern. An attempted Infinity plugin approach (direct S3 variable population) was abandoned after discovering an unresolvable Go HTTP client timeout in the plugin's transport layer. The SQS Queue Mapper Lambda approach replaced a 5-minute EventBridge schedule with CloudTrail-triggered invocations on `CreateQueue` and `DeleteQueue` API calls, reducing Lambda invocations from approximately 8,640 per month to 2–5 per month while ensuring the queue inventory stayed current within minutes of infrastructure changes. A content-hash S3 key scheme was implemented for Lambda deployment zips so CloudFormation correctly detected code changes between deploys — the prior filename-only approach caused silent no-op deploys of

updated Lambda code.

11.16 Week 16: Support Rotation #3 & Inventory API Discoverers (Jun 29 – Jul 3)

Week 16 was the third RPA support rotation, coinciding with Canada Day (July 1), alongside completion of ECS cluster discoverers for the inventory API.

RPA Production Support Rotation (Jun 29 – Jul 3):

Week 16 support handled several notable tickets. Two Corporate Actions new client onboarding requests were completed: SD#15103842 for Leaside (LEESIDE_Trades_HFXCA added to BPSHare Mandatory/Voluntary/Ops) and SD#15113393 for Orchid Tree (OrchidTree_trades_HFXCA added across all three tracks). An OTC process inquiry for Lumyna Equity Funds (SD#15102495) identified a dependency on the MNL team for configuration, and coordination was established with Miko and Daniel for knowledge transfer. On Canada Day (Wednesday), an INNOCAP SAMLMOS MESO retrigger was coordinated with Horace: the process had failed overnight, was manually marked failed, and retrigged to completion. Horace conducted two extended knowledge transfer sessions: 80 minutes on Thursday and a joint 96-minute session with Kishor on Friday covering the full RPA support handover model.

IISS-487: ECS Variable and Health Panel Optimisation:

During quieter periods, an `ecs_cluster` dashboard variable was added for cluster-level filtering, and health stat panel queries were optimised by removing the `seriesToRows` transform and replacing it with `ORDER BY MAX() DESC`. This change made the busiest services appear at the top of each panel without requiring a client-side sort transform.

IISS-507: ECS and S3 Cache Layer:

ECS cluster discoverers were implemented for all 22 clusters with the two-stage filter (BudgetCode then AppName discriminator) established in Week 6. A bi-weekly EventBridge- triggered S3 cache layer was added to the inventory API, reducing per-request latency from the 10-second boto3 enumeration to a sub-second cache read. A four-part IISS-845 observability solution was drafted covering log group discovery, API endpoint, Grafana panel, and saved query generation.

By the end of Week 16, the third support rotation was complete with new client onboarding delivered, ECS discoverers were in production, and the inventory API S3 cache was operational.

Canada Day Operational Coverage:

Working through the Canada Day public holiday on July 1 was necessary to maintain the INNOCAP SAMLMOS MESO process continuity. The process had failed overnight and required admin-level intervention to mark the queued items as failed before retriggering. Horace provided admin access and live KT on the marking procedure during a remote session. This experience directly motivated adding ECS task health monitoring to the IISS-487 alarm set — specifically the “desired vs running task count” discrepancy alarm that would surface this class of silent failure automatically. The two-stage BudgetCode + AppName filter pattern proved essential again this week: 22 ECS clusters were discovered, but the MERIDIAN and CITCOWORKS clusters sharing a BudgetCode prefix required the AppName discriminator to be classified correctly for the dashboard rows.

11.17 Week 17: Inventory API Milestone & X-Ray Architecture (Jul 6–10)

Week 17 delivered the inventory API to production readiness and established the X-Ray distributed tracing architecture plan.

IISS-508: Inventory API v1.5.0 and CI/CD Pipeline:

The inventory API was migrated from a Flask monolith to four dedicated Lambda handlers behind API Gateway (v1.5.0). A /docs endpoint was added with embedded Swagger UI, and an async background refresh mechanism was implemented to avoid the 30-second API Gateway timeout on full inventory scans. The /refresh endpoint was introduced supporting per-platform, per-service, and full refresh targeting. A full CI/CD pipeline for the citcodevops01 account was added with a deployment guide. All AWS resources were renamed to the ia-it-sre-resources-inventory-* naming convention for platform consistency. A VPN-graceful SSL degradation fix (v1.1.1/v1.1.2) was applied after discovering the S3 cache crashed when run outside the Citco network.

X-Ray Distributed Tracing Architecture (IISS-837–844):

BPM Flight Control meetings (Monday and Friday) and Blue Prism Troubleshooting sessions (Thursday) informed the X-Ray architecture approach. I drafted the full tracing plan across seven IISS tickets: IISS-838 (Lambda Powertools Tracer), IISS-839 (ECS ADOT sidecar), IISS-840 (EKS DaemonSet), IISS-841 (API Gateway active tracing), IISS-842 (SQS trace header propagation), IISS-843 (MSK trace propagation), and IISS-844 (EC2 standalone). The unified team-requests.md document compared structured vs unstructured logging patterns and documented the recommended approach for the ia_meso_process codebase.

VPL IPV CheckpointException (IISS-753) Follow-up:

The VPL IPV Æxeo pricing window CheckpointException pattern was identified as systemic across multiple VPL workflows targeting dynamically loading Æxeo screens. The fix was estimated at 16 hours for selector resilience work and escalated to the development team due to BitBucket repository access restrictions.

By the end of Week 17, the inventory API was production-deployed at v1.5.0 with CI/CD, the X-Ray architecture was formally planned across seven IISS tickets, and all 83 EC2 instances were confirmed in the live inventory.

Infinity Datasource Migration:

Fourteen dashboard variables were migrated from the deprecated mapper Lambda approach to the Infinity datasource via CloudFront, eliminating the SigV4 authentication complexity that had prevented direct S3 access from the AMG workspace. CloudFront served as an unsigned proxy with S3 origin access control, making the inventory JSON available to Grafana without any client-side AWS credential management. The X-Ray datasource UID was set to the stable value `xray-ia-it-sre-dev` and the CloudWatch provisioner UID fixed to `cloudwatch` to match the 100+ panel references already embedded in the assembled dashboard JSON. The inventory API was refactored into four sub-packages (`core`, `api`, `infra`, `metrics`) and bumped to v1.4.0 on July 8, with seven S3-lifecycle expiration rules added per platform/service combination in v1.4.3 to manage storage costs.

11.18 Week 18: Alarm Email Deep-Links & Log Group Observability (Jul 13–17)

Week 18 delivered the alarm email log group deep-link system and completed the five-phase log group observability specification.

Alarm Email Log Group Deep-Links (IISS-845 Phase 2):

The alarm formatter Lambda was extended to auto-construct CloudWatch Logs Console deep-link URLs from the cached log group inventory. When an alarm fires, the formatted email now includes a direct link to the relevant log group filtered to the alarm timestamp, eliminating the need to manually navigate to Logs Insights and reconstruct the query. The log group URL encoding utility was tested for all seven platform types, handling the differences in log group naming conventions between Lambda, ECS, EC2, and MSK resources.

Grafana CloudWatch Logs Panel:

A placeholder Row 17 panel was implemented in the Grafana dashboard using the CloudWatch Logs datasource, with a `$platform` variable filter. This provided a foundation for the full X-Ray tracing row planned for delivery in later weeks.

Five-Phase Log Group Observability Specification:

The IISS-845 specification was finalised covering: Phase 1 (log group discoverer Lambda), Phase 2 (inventory API `/api/log-groups` endpoint), Phase 3 (alarm email deep-links), Phase 4 (Grafana log panel row), and Phase 5 (CloudWatch Insights saved queries). All five phases were subsequently tracked as individual sub-tasks on the JIRA board.

IISS-507: Async Inventory and Pydantic Models:

The inventory API async route handler architecture was validated: `asyncio.gather()` dispatching all discoverers concurrently via `ThreadPoolExecutor` kept full-fleet enumeration under the 10-second non-functional requirement. Twelve service-specific Pydantic response models were added with versioned S3 keys (`latest.json` plus timestamped history entries).

By the end of Week 18, alarm emails included actionable log group deep-links, the inventory API reached **99% code coverage** with a 453-test suite, and the log group observability specification was fully defined.

Dynamic Alarm Architecture and X-Ray Instrumentation:

The dynamic alarm provisioner Lambda replaced 17 static alarm definitions with a data-driven approach: on each deploy it reads the full resource inventory from S3 and creates or updates per-resource CloudWatch alarms automatically. Nine alarm types absent from the original static set were added, including Lambda error rate (using a CloudWatch math expression rather than a metric alarm), MSK consumer lag (`SumOffsetLag` with thresholds 1,000/5,000), and EC2 instance and system status checks. Concurrent alarm creation was throttled to two `ThreadPoolExecutor` workers with an adaptive retry strategy of up to 10 attempts, resolving `PutMetricAlarm` throttling that had previously caused partial alarm set deployments. Lambda deployment zip sizes grew from 9 KB to 170 KB after bundling `aws-xray-sdk` with `patch_all()` auto-instrumentation. The X-Ray gap analysis established the baseline: CloudWatch monitoring alone provided approximately 30% parity with Dynatrace; adding X-Ray, Application Signals, and the full Grafana stack raised this to 88%.

11.19 Week 19: CloudWatch Logs Layer & Saved Queries (Jul 20–24)

Week 19 delivered the CloudWatch Logs observability layer including 30 saved queries across all monitored platforms.

Saved Queries Generation Script (IISS-845 Phase 5):

`create_saved_queries.py` was written to programmatically create 30 CloudWatch Logs Insights saved queries, covering six query types (recent errors, Lambda cold starts, request tracing, performance analysis, audit events, and resource utilization) across seven platforms. The script queries the live log group inventory to resolve group names at creation time, ensuring queries target the correct log groups even if naming conventions change. Running the script in DEV confirmed 748 matched log groups from a total of 3,423 groups in the account.

Log Group URL Utility and Alarm-to-Log-Group Mapping:

The `shared/log_groups.py` module was extracted into the shared utilities package, providing URL encoding, service classification from log group name patterns, and an alarm-to-log-group matching function used by the alarm email formatter. The `alarms-notifications` feature branch was merged to release, making the full alarm notification pipeline (EventBridge state publisher → SNS formatter → HTML email with log group deep-link) available in the DEV environment.

SRE Extended Review Sessions:

Two 60-minute SRE extended review sessions were held (Monday and Tuesday) with Kishor and the wider Innovation IT team. These sessions covered the full observability platform architecture, confirmed the resource inventory coverage, and reviewed the X-Ray implementation roadmap.

By the end of Week 19, the CloudWatch Logs layer was operational with saved queries for all platforms, the alarm-to-log-group pipeline was merged, and the SRE extended review confirmed the platform architecture was sound for the upcoming X-Ray tracing work.

Cross-Platform Variable Filter Fix:

A critical dashboard correctness bug was discovered and fixed: when a multi-select Grafana variable had an empty selection, `filterFieldsByName` was matching rows from all platforms rather than returning no results. The fix added `allValue=^$` with a `__no_resources__` sentinel value across all 13 multi-select variables, ensuring empty selections produced empty panels rather than misleading cross-platform data leakage. The inventory API project was renamed from `resources-inventory` to `infrastructure-inventory` to align

with the platform-agnostic naming convention adopted in Week 17, requiring updates across all Lambda names, CloudFormation stacks, API Gateway stages, documentation, and AWS Console bookmark links. The test suite reached 562 tests at 99% coverage (v1.5.1) after adding 50 Lambda API handler unit tests and 48 additional integration tests.

11.20 Week 20: Support Rotation #4 & X-Ray Tracing Specs (Jul 27–31)

Week 20 was the fourth RPA support rotation, combined with authoring the full X-Ray distributed tracing specification suite and the SRE Platform **v2.19.9** release.

RPA Production Support Rotation (Jul 27–31):

Tickets handled this week included: SD#15285857/15285410 FX Closeout retrigger for PICTET_RPA (checked UiPath folder, archived old Daily565 reports, triggered and notified Kartina on completion); SD#15285347 Auto Rec Download ASPECT client configuration (backed up gridview.py, added transaction and position columns with Horace verification); SD#15294806 MESO cash completeness for Johansen funds (verified DB state, obtained admin access from Horace to mark pending items as failed, retriggered to completion); SD#15286963 TUDOR3 daily reconciliation report (monitored Run ID progress, provided updates, and contacted Jay for an optimised query). Stale tickets SD#14238486, #14640319, and #14652654 were closed per the Production Support guide. Horace conducted KT sessions: 47 minutes on Monday, 43 minutes on Wednesday (group with Kishor and Edmund), and an extended 110-minute joint session with Jiannan covering full RPA KT for the incoming team member.

IISS-838–844: X-Ray Tracing Specification Suite:

Seven X-Ray tracing sub-task specifications were authored for the `ia_meso_process` repository: IISS-838 (Lambda Powertools Tracer with `@capture_method` decorators), IISS-839 (ECS ADOT sidecar with OpenTelemetry auto-instrumentation), IISS-840 (EKS DaemonSet collector), IISS-841 (API Gateway active tracing), IISS-842 (SQS message attribute trace propagation), IISS-843 (MSK record header trace propagation), and IISS-844 (EC2 X-Ray daemon). Each spec included implementation tasks, rollback procedures, and cost estimates.

SRE Platform v2.19.9 Release:

The log group discovery feature PR was merged and SRE Observability Platform **v2.19.9** was released, incorporating the saved queries, log group URL utility, and alarm deep-link pipeline developed in Weeks 18–19.

By the end of Week 20, the fourth support rotation was complete with five active tickets resolved, the X-Ray specification suite was published across seven IISS tickets, and v2.19.9 marked the completion of the CloudWatch Logs observability layer.

RPA Support Operational Detail:

The SD#15294806 MESO cash completeness incident for the Johansen fund illustrates the typical resolution workflow for stuck queue items. After verifying the database state and CloudWatch logs confirmed items were genuinely stuck (not still processing), the escalation to Horace for admin access took approximately 20 minutes. Items were marked failed using the admin UI, and the business user (Johansen) was asked to retrigger. The process ran to completion within 40 minutes of the retrigger. Johansen also requested standby assistance for downloading reports from S3 if they generated without saving to the expected location — this edge case was documented in the shift handover log for future rotations. The extended Jiannan RPA knowledge transfer session (110 minutes) transferred the full MESO retrigger and Corporate Actions onboarding procedures to the incoming support team member for the September continuation period.

11.21 Week 21: LLM Agent Major Refactor (Aug 3–7)

Week 21 was the delivery-densest week of the co-op placement's second half, focused on a comprehensive quality and reliability improvement programme for the **Ask Citco RPA** LLM agent (IISS-824).

Architectural Refactor and Structlog Migration:

The 1,140-line monolithic `event_service/function.py` was decomposed into 8 single-responsibility modules following the Model-View-Controller pattern. All print statements and root-logger calls across 11 Lambda functions were replaced with structlog structured key=value logging. Google-style docstrings with parameter and return types were added to all public functions, and all `sys.path.insert` hacks were removed in favour of a properly packaged module structure.

handler_utils Decorators and Test Milestone:

A shared `handler_utils` module was built providing three decorators: `@log_handler` (structured entry/exit logging with correlation IDs), `@scheduled_handler` (EventBridge schedule context extraction), and `@cfn_handler` (CloudFormation custom resource response handling). These decorators were applied to all eight Lambda handlers, eliminating repeated

boilerplate. The test suite grew from 60% to **100% coverage (233 tests)** through this week, with ruff reporting zero lint issues and Bandit clean on all security checks.

Structured CFN Nested Stack:

A parallel structured CloudFormation nested stack was built decomposing the flat 600-line template into six sub-stacks: `cfn.storage` (S3, DynamoDB), `cfn.iam` (all roles and policies), `cfn.lambda` (function definitions), `cfn.bedrock` (agent and knowledge base), `cfn.ecs` (container task definitions), and `cfn.monitoring` (alarms and dashboards). This structure enabled independent stack updates and brought the deployment model in line with the `ia-it-sre-infrastructure-monitoring-amg` nested stack pattern.

IISS-487 SRE Naming Convention and Metrics Live:

The SRE infrastructure repository service names, Lambda names, S3 bucket names, and IAM role names were renamed from the `meridian` prefix to the `ia-it-sre` convention. The inventory API was confirmed publishing 597 CloudWatch metrics live across all six platforms in DEV.

By the end of Week 21, the LLM agent had reached production-quality code standards (100% coverage, zero lint, zero security issues), the nested CFN architecture was established, and the SRE naming convention was applied platform-wide.

EventBridge vs APScheduler Decision:

A 59-minute architecture discussion with Horace on August 4 resolved the scheduling mechanism for the AExeo Report UI. APScheduler had been considered but had caused memory contention issues in a prior project. EventBridge Scheduler was selected for its managed lifecycle (~\$1/month), stateless invocation model, and architectural consistency with the 5-minute alarm re-notification pattern already running in IISS-487. An IAM assistance session (1 hour 50 minutes) followed, extracting the exact permissions required from the existing IISS-487 CloudFormation execution role: `scheduler:CreateSchedule`, `scheduler:UpdateSchedule`, `scheduler>DeleteSchedule`, and `iam:PassRole` with the `lambda.amazonaws.com` trust policy. The ECS task CPU and memory were aligned to production values (512 CPU units / 1,024 MB) based on CloudWatch utilization data from the 93-day analysis window, replacing the development-phase values that had been left in the template from initial scaffolding.

11.22 Week 22: KB Upload CLI & ECR Image Hardening (Aug 10–14)

Week 22 continued the IISS-824 quality programme with knowledge base upload tooling, Bedrock agent externalisation, and the FastAPI ECR image MVC refactor.

KB Content Upload CLI (`upload_kb_content.py`):

A CLI tool was built to upload NAV Checklist PDFs to the Bedrock Knowledge Base S3 bucket. The tool was refactored into a `_kb_utils/` package with four modules (`auth`, `s3`, `sync`, `bucket`), an `--env` flag with production confirmation prompt, a `status` subcommand that discovers Knowledge Base and Data Source IDs by name, and 78 tests (48 package tests + 30 CLI tests). KMS encryption was enforced consistently across all Athena queries, closing an SSE inconsistency identified in the Week 21 audit.

Bedrock Agent Instruction Externalisation:

The Bedrock agent instruction was externalised from an inline CloudFormation parameter to a plain-text file (`data/bedrock-agent-instruction.txt`) read from S3 via a CFN Custom Resource at deploy time. This eliminated the need to redeploy the full stack for instruction tuning iterations. A `status` subcommand was added to discover active Knowledge Base and Data Source IDs from AWS by name pattern, removing the need to hard-code resource IDs in deployment scripts.

FastAPI ECR Image MVC Refactor:

The `core-smart-rpa-llm-ecr-image` FastAPI container underwent a full MVC refactor: the application was restructured into an `app/` package (`routers`, `models`, `services`, `dependencies`), `structlog` logging replaced the custom logger, `pydantic-settings` replaced manual environment variable reads, LDAP TLS was properly configured, and CORS origins were locked to the Citco domain list. A hardened Docker container image was built and tested.

CodeBuild Pipeline Stabilisation:

A series of CodeBuild pipeline failures were resolved across the week: `AWS_ACCOUNT_ID` retrieval via STS lookup, `DATA_BUCKET_NAME` injection into the build environment, YAML parser colon-in-string errors in the `buildspec`, and account ID resolution from `parameters/{env}.json`.

By the end of Week 22, the KB upload CLI was tested and functional, the Bedrock agent instruction was decoupled from the CFN template, and the FastAPI ECR image was hardened and refactored

to production quality.

CodeBuild Pipeline Stabilisation Challenges:

The CodeBuild pipeline required six separate hotfix iterations before stabilising. Issues resolved in sequence: `AWS_ACCOUNT_ID` was not available as a built-in variable (fixed via STS lookup); the `DATA_BUCKET_NAME` environment variable was not injected into the build context (fixed by reading from `parameters/{env}.json`); YAML parser colon-in-string errors prevented the buildspec from parsing (fixed by quoting affected strings); the final solution read `CodeAccountId` from the parameters JSON file rather than querying it dynamically. An ad-hoc manual report run was completed for Kishor Deotale on August 11: five AExeo report requests were submitted via the `GET /arguments` API endpoint using `ReportID: 100502500`, `value date: 2026-08-10`, and `SourceSystem: RECON_EBINDER`, covering TWOSIGMA, SIGMATWO, and TSCONSOL clients. All five reports were verified as delivered to the production S3 bucket.

11.23 Week 23: LLM Agent Completion, X-Ray Instrumentation & AMG Alarm Emails (Aug 17–21)

Week 23 was one of the most productive delivery weeks of the co-op placement, completing the IISS-824 quality sprint, instrumenting X-Ray tracing across the LLM agent, delivering Citco-branded alarm emails for the AMG platform, and making initial progress on the IISS-919 Enterprise Dashboard.

IISS-824: 280 Tests at 100% Coverage and ircoekb Fixes:

The week opened on August 17 with the MVC refactor branch merged to release, bringing the test suite to **280 tests at 100% coverage**. The `ircoekb` Lambda timeout was increased from 300s to 900s based on the 93-day log analysis. An SSL hotfix was applied to skip certificate verification for internal Citco SDM endpoints (`nosec B501` annotations added for the known internal trust exception). Five dead scripts were archived into `_archive/`. On August 18, the Bedrock agent instruction was further externalised, usage report emails were completely redesigned as pivot-table HTML reports with Citco brand colours (`#002D72` navy), 32 `cfm-lint` warnings were resolved, and `eval()` calls were replaced with `json.loads()` across the codebase. On August 19, Pydantic validation models, tenacity exponential backoff, an SSM kill switch, and six CloudWatch alarms for all scheduled Lambda functions were implemented.

IISS-838/839/842: X-Ray Tracing Instrumentation:

Lambda Powertools `@tracer.capture_lambda_handler` and `@tracer.capture_method` decorators were applied to all Lambda functions on August 19. A cold-start fix was required for functions that load S3 data at module import time — these cannot be instrumented at the class-method level without triggering before the tracer initialises. A `tracing.md` file documenting every decorated and undecorated function with rationale was added. The ECS ADOT sidecar X-Ray tracing spec (IISS-839) and SQS trace header propagation spec (IISS-842) were also authored for the `ia_meso_process` BitBucket repository, and a new ECS ADOT sidecar instrumentation spec was written for the `core-smart-rpa-llm-ecr-image` container.

IISS-487: Citco-Branded HTML Alarm Emails and Notifications CLI:

On August 20–21, alarm email delivery was migrated from SES to Citco SMTP using the `TOIA-SMTP-Credentials` Secrets Manager secret. File-based HTML templates were introduced with Citco branding and per-platform digest format. Composite alarm re-notification with 5-minute cadence was implemented, with suppression logic preventing warning re-notifications when a critical sibling alarm is already active. IAM was corrected: missing `cloudwatch:DescribeAlarms` resource scope, stale SES permissions removed, and `kms:Decrypt` added for the SMTP credentials KMS key. Three feature branches (`mapper decommission`, `composite re-notification`, `SMTP emails`) were merged to release and then to main on August 21.

IISS-845 and IISS-919:

A dedicated `GET /api/log-groups` endpoint was added to the inventory API (IISS-845 Phase 2), enabling alarm emails to include per-resource log group deep-links. IISS-919 received its initial commit as a new `ia_infrastructure_dashboard` BitBucket repository, with an `@citco/core` + `@citco/theme` migration plan documented to align the Enterprise Dashboard with Citco's component library.

By the end of Week 23, IISS-824 reached full production quality, X-Ray tracing was instrumented end-to-end in the LLM agent, Citco SMTP alarm emails were live, and three major AMG features were merged to main.

11.24 Week 24: Support Rotation #5 & KT Handover (Aug 24–28)

Week 24 was the fifth and final RPA support rotation, combined with structured knowledge transfer to the team and planning for the Mitacs BSI extension.

RPA Production Support Rotation (Aug 24–28):

The final rotation followed the standard North America shift. Routine ticket monitoring covered MESO process completeness, FX Closeout status, and the VPL queue. An Auto Rec Download configuration update was completed (gridview.py backup and ASPECT client update). MESO cash completeness was investigated for a pending fund, with admin access coordinated through Horace to mark items failed and retrigger. A VPL IPV CheckpointException was escalated to the development team. Stale tickets were closed per the Production Support guide. End-of-rotation handover notes and a comprehensive weekly summary were posted to Confluence on Friday.

Knowledge Transfer Handover:

Structured handover sessions were conducted with Horace and Kri covering: the dashboard operating model (how to read alarm states, when to escalate vs self-heal), alarm runbooks (per-service threshold rationale and recommended response), inventory API deployment guide (stack ordering, environment promotion, S3 cache invalidation), and incident escalation procedures. Written handover notes were published to Confluence for the incoming support team.

Mitacs BSI Renewal Confirmation:

The Mitacs BSI renewal for Sep 1–Dec 31, 2026 was confirmed, providing continuity for four workstreams deferred from this engagement: X-Ray tracing implementation (IISS-837–844), log-group discovery (IISS-845–873), Enterprise UI Dashboards (IISS-509 and IISS-919), and IISS-824 LLM agent production hardening. IISS-509 frontend development was scaffolded: AG Grid tables rendering Inventory API responses with multi-dimensional filtering and export-to-Excel functionality. IISS-919 continued with drag-and-drop layout via react-grid-layout, threshold coloring for gauge panels, and responsive breakpoints for mobile views.

By the end of Week 24, the final support rotation was complete, all handover documentation was published to Confluence, and four workstreams were formally scoped for the Sep–Dec extension.

11.25 Week 25: Final Day & Co-op Placement Conclusion (Aug 31)

August 31 marked the final day of the co-op placement. A final knowledge transfer session was held with Horace and Kri to confirm operational handover of the infrastructure monitoring dashboard and RPA support rotation model. Incident escalation procedures and alarm runbook ownership

were formally transferred. The Mitacs BSI renewal scope was reviewed with Kishor, confirming the Sep–Dec extension focus on X-Ray tracing, log-group discovery, and the IISS-919 Enterprise Dashboard. IISS-919 component integration was reviewed with Soundarya and Sridhar to align on the BFF API contract and panel configuration schema before the handover.

12. CONCLUSIONS

Throughout my co-op placement I worked toward the technical and non-technical goals established at the outset in consultation with Kishor Deotale and Dr. Pawan Lingras. The following sections evaluate each goal against the delivered outcomes and reflect on the broader impact of the engagement.

12.1 Learning Goals Evaluation

Chapter 7 established technical and non-technical learning goals at the outset of my co-op placement. Table 12.1 evaluates each goal against evidence gathered from my delivered work.

Table 12.1: Learning Goals — Self-Evaluation at Co-op Placement Completion

Goal	Evidence	Status
AWS Cloud Services & Infrastructure	13 AWS service namespaces monitored; Lambda, ECS, EC2 inventoried; full DB spectrum covered	✓
Site Reliability Engineering Principles	Four golden signals applied; SLOs defined; MSK lag alarm reduces MTTD from 3.5h to <30s	✓
Infrastructure-as-Code	4 CloudFormation stacks; Lambda-backed provisioner; S3-staged templates	✓
Observability Platforms (AMG)	Multi-environment Grafana workspace; survived 10.4→12.4 upgrade	✓
Proactive vs. Reactive Reliability	Alarm-backed detection replaces business escalation; Meridian Incident proof point	✓
Python Development	Flask REST API; comprehensive tests at 99% coverage; Hypothesis PBT; structlog	✓
Software Engineering Practices	TDD with pytest; spec-driven development via Kiro IDE; PR-based merges	✓
AI-Augmented Engineering	Kiro IDE spec workflow; Amazon Q inline; Claude Sonnet for architecture	✓

continued on next page

Goal	Evidence	Status
Agile and Kanban Methodology	25+ JIRA tickets; story decomposition; work-log discipline; sprint rituals	✓
Technical Communication	RCA reports; Confluence pages; JIRA acceptance criteria; this report	✓
Production Incident Response	4 support rotations; 5 notable incidents resolved; IISS-706 hotfix within 7h	✓
Cross-Functional Collaboration	Halifax–Hyderabad coordination; Meridian team architecture sessions	✓
Financial Domain Knowledge	VPL queue, Corp Actions, MESO, CAIS familiarity; fund admin SLA awareness	✓

All my learning goals were met by my co-op placement completion. The deepest learning occurred at the intersection of engineering and operations: the direct experience of the Meridian Incident, followed immediately by building the instrumentation that would have detected it in under 30 seconds, compressed months of abstract SRE theory into a single, concrete engineering outcome.

12.2 Co-op Placement Hours Summary

Table 12.2: Co-op Placement Hours Summary

Work	Duration (hours)
Citco CTM Co-op Placement (Grafana dashboard, inventory API, RPA LLM agent, support rotations)	900
Report Writing (MCDA 5585 / 5586 Major Project Report)	120
Total	1020

The following chapters present the detailed evidence for each outcome: Chapter 13 covers deliverables and impact, Chapter 14 documents the engineering obstacles, and Chapter 15 outlines the planned next steps.

13. ACHIEVEMENTS & BUSINESS VALUE

Working on the IA-IT-SRE Infrastructure at Citco Technology Management, I achieved several significant milestones across observability platform development, API engineering, and production support operations. These accomplishments highlight my technical growth and the business value delivered during the co-op placement.

13.1 Key Achievements

- **CloudWatch Monitoring Platform:** Inventoried the complete AWS service fleet including Lambda functions, ECS clusters, and EC2 instances. Discovered and documented a critical tagging inconsistency affecting resource attribution. Developed a three-tier SQS fallback registry with comprehensive automated test coverage.
- **Grafana AMG Centralized Dashboard:** Designed and deployed a comprehensive observability workspace covering all six IA platforms. Implemented dynamic CloudWatch alarms via CloudFormation stacks with Lambda-backed provisioning. Successfully migrated the workspace through a breaking Grafana version upgrade.
- **Infrastructure Inventory REST API:** Built a production-ready Flask API with high test coverage. Implemented the Discoverer Registry pattern enabling extensible resource discovery. Integrated property-based testing with Hypothesis to verify tag disambiguation logic.
- **Container Insights Enablement:** Enabled ECS Container Insights across Meridian clusters, surfacing previously invisible task-level CPU and memory telemetry within hours of deployment.
- **Production Incident Response:** Completed multiple RPA support rotations, resolving notable production incidents. Delivered a critical CAIS Deadline hotfix within hours of initial alert, identifying a credential caching vulnerability during Secrets Manager rotation cycles.
- **Infrastructure-as-Code Proficiency:** Authored CloudFormation stacks with S3-staged templates and multi-environment reuse. Implemented a Lambda-backed dynamic provisioner that auto-discovers resources at deploy time.
- **Cross-Account IAM Architecture:** Designed a cross-account permission model enabling the

Grafana workspace to query CloudWatch metrics across all IA platform AWS accounts.

- **SRE Principles Application:** Mapped the four golden signals (latency, traffic, errors, saturation) to concrete CloudWatch metric namespaces. Defined SLO targets for Lambda, ECS, MSK, and SQS services.
- **Test-Driven Development:** Applied TDD practices across all deliverables using pytest, achieving high coverage on the inventory API and SQS registry components.
- **AI-Augmented Engineering:** Leveraged Kiro IDE's spec-driven workflow for API development. Used Amazon Q for inline code assistance and Claude Sonnet for architecture reasoning.
- **Technical Documentation:** Produced RCA reports for production incidents, Confluence knowledge base articles, and comprehensive acceptance criteria for all deliverables.

13.2 Business Value

The shift from a purely reactive to a proactively monitored infrastructure is the central business value proposition of this co-op placement's work.

13.2.1 Incident Detection: From Hours to Seconds

The Apr 17, 2026 Meridian incident—in which approximately 9 million Kafka messages queued, producing a 7.5-hour processing delay affecting multiple client funds—was detected only because a fund administrator noticed missing outputs. No automated alarm existed.

The EstimatedMaxTimeLag alarm deployed as part of IISS-487 directly addresses this gap:

- **Before:** Detection required human observation of missing downstream outputs, with latency of 2–3 hours when incidents began outside working hours.
- **After:** CloudWatch alarm fires within **30 seconds** of threshold breach, routing notification directly to the SRE team.

This improvement from multi-hour manual detection to sub-30-second automated alerting represents a qualitative step change in the team's ability to contain fund-impacting incidents before they reach business-level severity.

13.2.2 Operational Efficiency: Unified Fleet Visibility

Prior to this work, the IA-SRE team had no single view of all Lambda functions, ECS clusters, and EC2 instances under their operational mandate. Resource discovery was ad hoc and frequently lagged deployments by days or weeks.

The Grafana dashboard and inventory API together deliver:

- **Continuous visibility:** A single workspace replaces navigation across six separate AWS console tabs to assess platform health.
- **Machine-readable inventory:** The REST API enables downstream automation including tag compliance checks and dynamic alarm threshold recalibration.
- **Hidden issue discovery:** Container Insights immediately surfaced a previously invisible CPU saturation pattern on one of the Meridian Action Processor services.

13.2.3 Knowledge Transfer and Platform Resilience

Each support rotation produced documented incident summaries in the team's Confluence knowledge base. The CAIS hotfix analysis elevated a credential caching vulnerability to a platform-wide architectural concern, informing future RPA credential-handling guidelines and improving overall system resilience.

These achievements reflect my ability to deliver production-grade engineering solutions while contributing directly to the operational reliability of Citco's Intelligent Automation platform.

14. CHALLENGES & MITIGATIONS

Throughout this co-op placement, I encountered engineering challenges spanning AWS service limitations, Grafana platform constraints, corporate policy restrictions, and production incident responses. This chapter documents the most significant challenges that shaped delivery plans and produced lessons applicable to future SRE work.

14.1 AWS CloudWatch Limitations

CloudWatch imposes several hard limits encountered at modest operational scale (54 Lambda functions, 100+ dashboard panels):

- **Metrics Insights SQL 10 TPS limit:** With 90+ panels firing queries simultaneously, throttling caused intermittent panel errors. *Mitigation:* Collapsed rows by default, 5-minute refresh interval.
- **Log Insights 50 log group maximum:** The IA environment has 54+ Lambda log groups, exceeding the per-query limit. *Mitigation:* Filter to active log groups; separate saved query for full coverage.
- **SEARCH() visibility gap:** Functions without recent activity disappear from dashboards—the same root cause that made CAIS-Pricing invisible for 29 days. *Mitigation:* Replaced SEARCH() with Metric Insights SQL which enumerates all deployed functions.
- **No percentiles in GROUP BY mode:** P50/P99 statistics unavailable when grouping by dimension. *Mitigation:* Wildcard dimension queries for percentile panels; accepted tradeoff of missing idle functions.
- **51 KB CloudFormation dashboard limit:** Generated JSON exceeds inline CFN limits at scale. *Mitigation:* Python put_dashboard API deployment (no size limit).

14.2 Grafana Platform Constraints

Amazon Managed Grafana (AMG) with the CloudWatch plugin presented several limitations:

- **Mid-development version upgrade (10.4 → 12.4):** Breaking changes in query syntax required one full day of panel migration. *Lesson:* Design for portability across AMG versions.

- **Metrics Insights SQL no regex in WHERE clause:** The WHERE clause supports only LIKE patterns with simple wildcards, not regular expressions. Tag-based filtering (WHERE tag.AppName = 'MERIDIAN') is also unsupported. *Mitigation:* Implemented dashboard-level variables (\$platform, \$environment) with Infinity datasource backed by static inventory JSON files.
- **Dashboard freezing from excessive datapoints:** Before dashboard-level variables were introduced, panels rendered all resources simultaneously—54+ Lambda functions, 17+ ALBs, 30+ ECS services. The browser froze for 10–15 seconds on load as Grafana attempted to render thousands of data points across 113 panels. *Mitigation:* Platform variable filters reduced visible resources per dashboard load; collapsed rows defer queries until user expansion.
- **Single SQL query per panel:** Cannot combine multiple Metrics Insights queries on one panel—attempting to add a second query produces “Maximum number of queries exceeded” error. *Mitigation:* Split into separate panels (e.g., “Running Tasks” and “Desired Tasks” instead of one combined panel).
- **No percentiles in Metrics Insights SQL:** Aggregation functions like PERCENTILE(Duration, 50) are not supported. Percentile statistics (p50, p99) are only available via the standard Metric Search API, which lacks GROUP BY capability. *Tradeoff:* Accept either per-function grouping (Metrics Insights, no percentiles) or percentiles (Metric Search, only active functions visible).
- **No cross-panel filtering:** Clicking a resource in one panel does not filter other panels. *Mitigation:* Dashboard-level template variables with coordinated queries across all panels.
- **IAM credential rotation:** Monthly rotation breaks AMG access until trust policy is manually updated. *Mitigation:* Documented runbook with calendar reminders.

14.3 Infrastructure & Corporate Policy Challenges

Enterprise environment constraints required adaptation of standard AWS patterns:

- **Static metrics at deploy time:** Dashboard does not auto-discover new resources. *Mitigation:* Documented redeploy procedure with 3-tier fallback.

- **Lambda Insights layer stripped on redeploy:** Other teams' deployments remove the monitoring layer. *Mitigation:* Periodic enforcement script (`add_lambda_insights.py`).
- **ECS Fargate no free cluster metrics:** Requires Container Insights at \$0.30/metric/month. *Mitigation:* Manual per-cluster enablement with cost approval.
- **CloudFormation ResourceExistenceCheck hook:** Corporate guardrail blocked alarm stack deployment. *Mitigation:* Dynamic Lambda provisioner bypasses CFN constraints.
- **IAM permission boundaries:** Subscription filter approach required permissions outside team scope. *Mitigation:* Redesigned to use Metric Insights SQL instead.
- **S3 encryption policy:** Silent failures without `ServerSideEncryption=AES256`. *Mitigation:* Pre-commit hook to flag missing encryption headers.

14.4 Production Incidents

Production incidents provided valuable learning experiences:

14.4.1 CAIS Deadline Pipeline Hotfix (June 2026)

The CAIS data pipeline failed silently—no error notifications, no data delivered. I delivered a 3-part hotfix:

1. **UCM credential expiry:** SMB mount returned empty data on expired credentials without error. *Fix:* Explicit validation step verifying expected files exist.
2. **Empty list SQL error:** `WHERE id IN ( )` generated invalid SQL. *Fix:* Empty list guard before query construction.
3. **Module-level credential caching:** Stale credentials persisted across Secrets Manager rotation. *Fix:* Fresh credential retrieval before each use.

Key lesson:

Silent failures are worse than loud failures. Validate expected state, not just absence of exceptions.

14.4.2 RPA LLM Agent Lambda Memory Exhaustion (August 2026)

Business users reported the RPA Status Bot intermittently failing to respond to queries. CloudWatch logs revealed out-of-memory (OOM) errors on the `rpa_llm_agent_event_service_cicd` Lambda function, which handles Bedrock Agent API calls for process status lookups.

Root cause:

The Lambda was using CloudFormation’s implicit default of 128 MB memory. Functions using pandas, boto3, and JSON serialization together exceeded this allocation during peak query loads.

Fix:

Increased MemorySize from 128 MB to 256 MB in the CloudFormation template. Additionally standardized all nine Lambda functions in the RPA LLM stack to explicit 256 MB allocations, making memory settings visible and intentional in Infrastructure-as-Code rather than relying on implicit defaults.

Key lesson:

Never rely on CloudFormation default values for production workloads. Explicit resource allocations prevent silent degradation and make capacity constraints visible in code review.

14.5 Summary

Key themes from these challenges:

1. **AWS service limits are real constraints** encountered at modest scale—not edge cases. The 10 TPS Metrics Insights throttle affected dashboard load with only 90 panels.
2. **Silent failures are dangerous:** Multiple incidents involved operations that “succeeded” but produced no useful output.
3. **Enterprise environments add complexity:** Corporate policies break standard AWS patterns; budget time for adaptations.
4. **Managed services still require operational planning:** AMG’s “serverless” nature does not eliminate IAM credential lifecycle overhead or CloudWatch API throttling mitigation.
5. **Query language limitations shape architecture:** Metrics Insights SQL’s lack of regex filtering

and percentile aggregation required alternative approaches—dashboard variables backed by static inventory files and separate percentile panels using the legacy Metric Search API.

These themes inform the recommendations in Chapter 15.

15. FUTURE WORK

The work I delivered in this co-op placement establishes a monitoring foundation — not an endpoint. I have already defined several planned extensions in the JIRA backlog, scoped for implementation during the Mitacs BSI renewal (Sep 1 – Dec 31, 2026).

Distributed Tracing with X-Ray. The most technically compelling near-term extension is the integration of AWS X-Ray distributed tracing into the IA Lambda fleet using the `aws-lambda-powertools[tracer]` library. X-Ray would provide end-to-end request traces across Lambda invocations, SQS message dispatch, and downstream HTTP calls — the “service map” view that CloudWatch metrics alone cannot produce. At the 5% sampling rate recommended for production workloads, the additional cost is estimated at under \$2 per month for the current fleet size, making this an unusually cost-effective observability enhancement. A Grafana X-Ray data source plugin is available for AMG, meaning trace data would flow directly into the existing workspace without requiring a separate visualization tool.

Auto-Healing Agent

The alarm infrastructure I delivered closes the detection gap — the 2–3 hour window between failure onset and SRE awareness that the Meridian Incident exposed. But detection without automated remediation still leaves a response gap: a human must receive the SNS notification, triage the alarm, identify the correct recovery action, and execute it. For the most common and well-understood failure modes on the IA platform, this human step is unnecessary. The **auto-healing agent** eliminates it.

Concept. The auto-healing agent is an event-driven remediation layer that sits between the CloudWatch alarm infrastructure and the AWS services being monitored. Each alarm SNS message triggers a Lambda “healer” function. The healer evaluates the alarm payload, classifies the failure type, selects the appropriate recovery action from a configurable remediation catalogue, and executes it — logging the entire sequence via the IA Event Store with structured `PROCESS_EVENT` codes so that every automated intervention is fully auditable.

Architecture. The proposed architecture connects directly to the existing dashboard alarm infrastructure without requiring changes to the Grafana dashboard or CloudFormation alarm stacks:

1. A CloudWatch Alarm breaches its threshold (e.g., ECS task desired count $\neq$ running count on a Meridian service).
2. The alarm fires an SNS message to the `ia-sre-autohealing-topic` SNS topic (a new topic, parallel to the existing notification topic).
3. The SNS topic triggers the **Healer Lambda**, which receives the full alarm payload including the `AlarmName`, `NewStateValue`, `MetricName`, and any available dimensions (`ClusterName`, `ServiceName`, `FunctionName`).
4. The Healer Lambda looks up the alarm name in a **remediation catalogue** — a DynamoDB table mapping alarm patterns to recovery action functions.
5. The matched recovery action is invoked: an AWS API call, a Step Functions state machine, or a queued message to a downstream remediation worker.
6. The Healer Lambda logs the intervention to the IA Event Store with a `PROJ-BE-AUTOHEALING` code, including the alarm name, recovery action taken, and outcome status.
7. If the recovery action fails or the alarm re-fires within a configurable cool-down window, the Healer escalates by publishing to the existing notification SNS topic to alert the SRE team directly.

Remediation Catalogue (Initial Scope). The first phase of the auto-healing agent targets the failure modes observed or anticipated from the four support rotation weeks:

Design Principles. Three design principles constrain the auto-healing agent to avoid introducing new failure modes:

- **Idempotency.** Every recovery action must be safe to invoke multiple times. The DynamoDB remediation catalogue tracks the last invocation time per alarm; the Healer Lambda checks this before acting and skips duplicate invocations within the cool-down window.
- **Bounded autonomy.** The Healer acts on a whitelist of known-safe recovery actions. Any alarm with no matching catalogue entry falls through to standard SNS notification — the agent does not improvise. New recovery actions are added only after review and approval, following the same PR process as CloudFormation template changes.

Table 15.1: Auto-Healing Agent — Initial Remediation Catalogue

Alarm / Trigger	Recovery Action	Escalation if Fails
ECS task desired $\neq$ running	ecs:UpdateService (force new deployment)	SNS alert to SRE
MSK consumer lag > 10,000 messages	Scale ECS consumer service (+1 task)	SNS + JIRA ticket (auto-created via JIRA API)
Lambda error rate > 1% over 5 min	Trigger Step Functions diagnostic workflow; capture last 50 error log lines to S3	SNS alert with S3 log link
SQS queue age > 1 hour	Invoke queue-processor Lambda with force=true parameter	SNS alert
ECS CPU > 80% sustained 5 min	Trigger ECS service auto-scaling policy (RegisterScalableTarget)	SNS if already at max task count
RPA bot missing heartbeat (future)	Call UiPath Orchestrator API to restart robot job	SNS + support ticket

- **Full auditability.** Every automated intervention is logged to the IA Event Store with a structured event code (PROJ-I-AUTOHEALING), the alarm trigger name, the action executed, the outcome status, and a link to the AWS resource modified. The Event Store trail ensures that automated interventions are visible in the same audit log as manual ones.

Relation to the PoC. The auto-healing agent is the direct successor to the alarm-and-notify posture delivered by this co-op placement. The detection layer (CloudWatch alarms + SNS) was built first because a remediation layer without a reliable detection layer is fragile — automated actions triggered by false alarms cause more harm than the failures they prevent. Now that the detection infrastructure has been in production for several months, the signal quality of the alarms is validated, and the false-positive rate is well-characterized. The auto-healing agent can therefore be implemented against a tested alarm baseline rather than a hypothetical one.

This progression — detect first, then automate remediation — is the correct sequence for building reliable auto-healing in a regulated financial services environment, where an incorrectly triggered service restart during a VPL pricing window or a CAIS Deadline processing run would itself constitute a production incident.

RPA Bot CloudWatch Integration. Two follow-on stories in my JIRA backlog propose bridging the Blue Prism (BPM) and UiPath Cloud operational metrics into CloudWatch via a polling Lambda function. The Lambda would query the Blue Prism API and UiPath Orchestrator API on a configurable schedule, emit custom CloudWatch metrics for queue depth, bot session count, and process error rates, and feed those metrics into the existing Grafana dashboard infrastructure. This would close the last remaining visibility gap: the current dashboard covers the AWS infrastructure layer comprehensively but has no visibility into the RPA execution layer above it.

Error Budget Burn-Rate Alerting. The current alarm model fires on threshold breaches without distinguishing between a momentary spike and a sustained reliability regression. Multi-window burn-rate alerting — as described in the Google SRE Workbook — uses 1-hour and 6-hour Metric Math expressions to detect when the error budget is being consumed faster than the replenishment rate. Implementing this pattern in CloudWatch Metric Math would elevate the existing alarm infrastructure from threshold-based detection to SLO-aware alerting, enabling the team to distinguish incidents that require immediate response from those that are within the acceptable error budget envelope.

Tag Compliance Automation. The AppName tagging obligation I identified in Section 14.3 currently relies on a reactive CloudWatch alarm: the alarm fires after a non-compliant resource has already been deployed. A proactive alternative would query the inventory API on a schedule, compare the returned resource list against an expected tag manifest, and open a JIRA ticket automatically for any resource found to be non-compliant. This would shift tag compliance from a detection activity to a governance workflow.

Mitacs BSI Renewal Scope. The approved Mitacs BSI renewal (Sep 1 – Dec 31, 2026) will extend my industrial collaboration with a focus on three themes: X-Ray tracing integration, IaC hardening and drift detection for the CloudFormation stacks I delivered in this period, and capacity forecasting using CloudWatch anomaly detection models trained on the historical metric data now being accumulated by the deployed dashboards.

A. APPENDIX: WORK LOGS

The complete weekly work log documenting **900 hours** of internship work across 25 weeks (March 16 – August 31, 2026) is provided in the supplementary file:

BhavikBhagat_A00494758_WorkLogs.xlsx

The work log includes:

- Weekly task descriptions and hours logged per activity
- Weekly hour totals and cumulative running total
- Meeting and support shift hours tracked separately
- Work stream categorization (WS1–WS6)

An additional **120 hours** were dedicated to report writing and academic documentation (MCDA 5585 and 5586), bringing the total project effort to **1,020 hours**.

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